Cochrane Library

Cochrane Database of Systematic Reviews

Liver and spleen stiffness as assessed by vibration-controlled transient elastography for diagnosing clinically significant portal hypertension in comparison with other elastography-based techniques in adults with chronic liver disease (Review)

Vranić L, Nadarević T, Štimac D, Fraquelli M, Manzotti C, Casazza G, Colli A

Vranić L, Nadarević T, Štimac D, Fraquelli M, Manzotti C, Casazza G, Colli A.
Liver and spleen stiffness as assessed by vibration-controlled transient elastography for diagnosing clinically significant portal hypertension in comparison with other elastography-based techniques in adults with chronic liver disease.
Cochrane Database of Systematic Reviews 2026, Issue 1. Art. No.: CD015415.
DOI: 10.1002/14651858.CD015415.pub2.

www.cochranelibrary.com

WILEY

TABLE OF CONTENTS

[Diagnostic Test Accuracy Review]

Liver and spleen stiffness as assessed by vibration-controlled transient elastography for diagnosing clinically significant portal hypertension in comparison with other elastography-based techniques in adults with chronic liver disease

Luka Vranić[1,2a], Tin Nadarević[2,3], Davor Štimac[2,4], Mirella Fraquelli[5], Cristina Manzotti[6], Giovanni Casazza[7,8], Agostino Colli[9]

[1]Department of Gastroenterology, Clinical Hospital Centre Rijeka, Rijeka, Croatia. [2]Faculty of Medicine, University of Rijeka, Rijeka, Croatia. [3]Department of Diagnostic and Interventional Radiology, Clinical Hospital Centre Rijeka, Rijeka, Croatia. [4]Specialty Hospital Medico, Rijeka, Croatia. [5]Gastroenterology and Endoscopy Unit, Fondazione IRCCS Ca' Granda - Ospedale Maggiore Policlinico, Milano, Milan, Italy. [6]Gastroenterology and Digestive Endoscopy - Department of Oncology and Advanced Technologies, Azienda USL-IRCCS di Reggio Emilia, Reggio Emilia, Italy. [7]Dipartimento di Scienze Cliniche e di Comunità, Dipartimento di Eccellenza 2023-2027, Università degli Studi di Milano, Milan, Italy. [8]Fondazione IRCCS Ca' Granda Ospedale Maggiore Policlinico, Milan, Italy. [9]Cochrane Hepato-Biliary Group, Copenhagen Trial Unit, Centre for Clinical Intervention Research, The Capital Region, Copenhagen University Hospital – Rigshospitalet, Copenhagen, Denmark

[a]ORCID iD: https://orcid.org/0000-0003-2886-2668

Contact: Luka Vranić, luka-vranic@hotmail.com.

Editorial group: Cochrane Central Editorial Service.
Publication status and date: New, published in Issue 1, 2026.

Citation: Vranić L, Nadarević T, Štimac D, Fraquelli M, Manzotti C, Casazza G, Colli A. Liver and spleen stiffness as assessed by vibration-controlled transient elastography for diagnosing clinically significant portal hypertension in comparison with other elastography-based techniques in adults with chronic liver disease. *Cochrane Database of Systematic Reviews* 2026, Issue 1. Art. No.: CD015415. DOI: 10.1002/14651858.CD015415.pub2.

ABSTRACT

Background

Clinically significant portal hypertension (CSPH) in chronic liver disease (CLD) is a key driver of decompensation, with severe portal hypertension (SPH) leading to severe complications like oesophageal bleeding. Early detection is essential for timely treatment. The current gold standard for assessing portal pressure is hepatic venous pressure gradient (HVPG), an invasive and costly procedure with limited availability. Non-invasive alternatives are increasingly needed to estimate portal pressure (e.g. by liver and spleen stiffness measurement) and guide treatment. However, the diagnostic accuracy of vibration-controlled transient elastography (VCTE), point shear wave elastography (pSWE), two-dimensional shear wave elastography (2D-SWE), and magnetic resonance elastography (MRE) remains unknown.

Objectives

Primary objectives: to assess the diagnostic accuracy of liver stiffness measurement (LSM) and spleen stiffness measurement (SSM), as well as in combination, measured by any elastography technique (VCTE, pSWE, 2D-SWE, or MRE) in the detection of CSPH in adults with CLD; and to compare the diagnostic accuracies between the individual tests. We will regard a combination of tests as positive when at least one is positive.

Secondary objectives: to assess the diagnostic accuracy of LSM and SSM, as well as in combination, measured by any elastography technique (VCTE, pSWE, 2D-SWE, or MRE) in the detection of SPH in adults with CLD; and to investigate sources of heterogeneity in the results.

Search methods

We searched the Cochrane Hepato-Biliary Group Controlled Trials Register, the Cochrane Hepato-Biliary Group Diagnostic Test of Accuracy Studies Register, CENTRAL, MEDLINE ALL Ovid, Embase Ovid, LILACS, Science Citation Index – Expanded (Web of Science), and Conference Proceedings Citation Index – Science (Web of Science) until 8 April 2024. We applied no restrictions on language or document type.

Selection criteria

We included studies that evaluated the diagnostic accuracy of LSM and SSM either alone or in combination, as measured by different elastography techniques (VCTE, pSWE, 2D-SWE, or MRE), for the diagnosis of CSPH and SPH in adults with CLD. We only considered studies with cross-sectional design using HVPG measurement as the reference standard.

Data collection and analysis

Two review authors independently screened studies, extracted data, and assessed the risk of bias and applicability concerns using the QUADAS-C tool. In the case of different cut-off values, we used the hierarchical summary receiver operating characteristic (HSROC) model to meta-analyse data (sensitivities and specificities) and to estimate a summary ROC (SROC) curve. In the case of common cut-off values, we used the bivariate model. We presented uncertainty of the accuracy estimates using 95% confidence intervals (CIs).

Main results

We included 47 studies (7817 participants). The most evaluated index test was LSM by VCTE for CSPH (27 studies; 3818 participants).

We judged only two studies at low risk of bias for all domains. Most showed high risk in the index test domain due to lack of prespecified thresholds and high concern for applicability in patient selection. Overall, the certainty of evidence was very low.

Due to varying thresholds across studies, we used an HSROC model to obtain summary estimates. The main findings for detecting CSPH are below.

LSM by VCTE (27 studies, 3818 participants):
Sensitivity 72.6% (95% CI 60.0% to 82.5%) at fixed specificity of 90%
Specificity 75.9% (95% CI 63.5% to 85.1%) at fixed sensitivity of 90%

SSM by VCTE (6 studies, 391 participants):
Sensitivity 72.9% (95% CI 27.1% to 95.1%) at fixed specificity of 90%
Specificity 80.6% (95% CI 64.1% to 90.7%) at fixed sensitivity of 90%

SSM by pSWE (4 studies, 248 participants):
Sensitivity 84.1% (95% CI 8.4% to 99.7%) at fixed specificity of 90%
Specificity 86.1% (95% CI 40.4% to 98.3%) at fixed sensitivity of 90%

LSM by 2D-SWE (10 studies, 767 participants):
Sensitivity 73.5% (95% CI 52.6% to 87.4%) at fixed specificity of 90%
Specificity 83.4% (95% CI 68.2% to 92.2%) at fixed sensitivity of 80% (estimation at 90% not feasible)

SSM by 2D-SWE (6 studies, 353 participants):
Sensitivity 80.0% (95% CI 59.8% to 91.5%) at fixed specificity of 90%
Estimation at fixed sensitivity not feasible

HSROC analysis was not feasible for LSM by pSWE and LSM or SSM by MRE due to insufficient data.

For LSM by VCTE at 25 kPa (9 studies, 1553 participants), sensitivity was 62.3% (95% CI 53.0% to 70.7%) and specificity 94.1% (95% CI 87.2% to 97.3%). We explored heterogeneity and observed a prevalence effect: relative specificity 0.90 (95% CI 0.83 to 0.97). Other factors could not be assessed.

For SPH as secondary objective, LSM by VCTE was the most evaluated test (8 studies; 637 participants). Sensitivity corresponding to specificity of 90% was 67.2% (95% CI 48.8% to 81.4%), while specificity corresponding to sensitivity of 90% could not be calculated by the HSROC model.

Direct comparisons between two index tests were impossible due to inconsistent numbers of participants included for each index test. Due to variability in cut-off values reported across studies, we only performed indirect comparisons using SROC curves. Thus, no reliable results emerged from comparisons or combinations across techniques.

Authors' conclusions

Liver and spleen stiffness measurements may offer a non-invasive alternative to HVPG for detecting CSPH. However, the accuracy of individual techniques remains uncertain due to very low-certainty evidence and insufficient data for reliable comparisons. No test achieved both sensitivity and specificity ≥ 90%, limiting their utility for confidently ruling in or out CSPH.

For LSM by VCTE, the most commonly studied method, HSROC modelling yielded a sensitivity of 72.6% (95% CI 60.0% to 82.5%) at a fixed specificity of 90%, based on studies using various thresholds.

In a subgroup of nine studies (1553 participants) using the predefined 25 kPa cut-off, 38% of patients with CSPH would be missed, and 6% without CSPH would be incorrectly identified.

The certainty of the evidence is very low, mainly due to high risk of bias, heterogeneity, and imprecision.

High-quality research is needed with predefined thresholds, standardised methodology, and improved reporting. Future studies should target key subpopulations (e.g. compensated CLD, specific aetiologies) and assess combinations of non-invasive tools to enhance diagnostic accuracy and clinical usefulness.

Funding

No internal or external sources of support.

Registration

https://doi.org/10.1002/14651858.CD015415

PLAIN LANGUAGE SUMMARY

How accurate is liver and spleen stiffness measured by different elastography techniques in diagnosing clinically significant portal hypertension?

Key messages

- Liver stiffness measurement measured by vibration-controlled transient elastography with cut-off of 25 kPa may detect clinically significant portal hypertension (CSPH) as previously suggested by guidelines.
- The reliability of results is low due to problems with the methods of the currently available studies.
- More high-quality studies are needed before elastography techniques can be used routinely to assess portal hypertension.

Why is improving the diagnosis of clinically significant portal hypertension (CSPH) important?

Portal hypertension is high blood pressure in the veins of the liver, often seen in people with chronic liver disease. This condition can lead to serious complications like fluid build-up in the abdomen (ascites), yellowing of the skin (jaundice), kidney failure, confusion, and dangerous bleeding from veins in the oesophagus. Early diagnosis is important to start treatment and to prevent these complications. The standard reliable test, called the hepatic venous pressure gradient, is invasive (involves inserting a tube into a vein), expensive, and not widely available. Therefore, we need non-invasive tests, like elastography, a type of imaging test that can measure how stiff the liver or spleen is, to indirectly assess the portal pressure.

What did we want to find out?

We wanted to know how accurate different elastography techniques and their combinations are for diagnosing CSPH in people with chronic liver disease.

What did we do?

We searched for studies of various elastography techniques that measure liver or spleen stiffness, using either specialised machines such as vibration-controlled transient elastography, or standard ultrasound and magnetic resonance machines like point shear wave elastography, two-dimensional shear wave elastography, and magnetic resonance elastography, to find out how well these tests can detect CSPH at different thresholds (cut-off values).

Main results

We included 47 studies with a total of 7817 participants. On average, about 63 out of every 100 people in these studies had CSPH. The most studied test was liver stiffness measurement by vibration-controlled transient elastography. There was a lack of studies looking at other elastography techniques or combinations of tests. There were also few studies on spleen stiffness.

Liver stiffness measurement by vibration-controlled transient elastography

We looked at 27 studies involving 3818 people. These studies used different thresholds to decide when liver stiffness measurement results suggest the presence of CSPH.

When the test is used to catch nearly everyone who has CSPH (90 out of 100 people), it may also correctly suggest that 76 out of 100 people actually do not have CSPH.

On the other hand, if the test is used in a way that avoids false results in people who do not have CSPH (correctly ruling out 90 out of 100 without the condition), it will correctly find CSPH in about 73 out of 100 people who actually do have it.

Using a threshold of 25 kPa for liver stiffness (based on 9 studies with 1553 participants), the test correctly identified 62 out of 100 people who had CSPH. However, it also correctly ruled out CSPH in almost everyone (94 out of 100 people) who did not have the condition.

While liver stiffness measurement by vibration-controlled transient elastography with a 25 kPa threshold might detect CSPH, the overall low quality of the studies means that these results are not very reliable. More high-quality research is needed before elastography can be recommended for assessing portal hypertension.

What are the limitations of the evidence?

Most studies had limitations, such as problems with how people were selected for inclusion in the study and how thresholds were defined. Only 2 out of 47 studies were assessed as having no major concerns. This lowers our confidence in the accuracy of the results.

How up-to-date is this evidence?

The evidence is current to 8 April 2024.

SUMMARY OF FINDINGS

Summary of findings 1. Diagnostic accuracy of LSM and SSM by VCTE, pSWE, 2D-SWE, and MRE for the diagnosis of CSPH

Review question: what is the diagnostic accuracy of liver stiffness measurement (LSM) and spleen stiffness measurement (SSM) by vibration-controlled transient elastography (VCTE), point shear wave elastography (pSWE), 2D-shear wave elastography (2D-SWE), or magnetic resonance elastography (MRE) for the diagnosis of clinically significant portal hypertension (CSPH) and severe portal hypertension (SPH) in adults with chronic liver disease (CLD)?

Population: adults with CLD

Setting: clinical setting (secondary or tertiary centres)

Study design: prospective and retrospective cross-sectional cohort type accuracy studies

Index tests:

- LSM by VCTE
- LSM by pSWE
- LSM by 2D-SWE
- LSM by MRE
- SSM by VCTE
- SSM by pSWE
- SSM by 2D-SWE
- SSM by MRE

Target condition:

- CSPH (HVPG > 10 mmHg)
- SPH (HVPG > 12 mmHg)

Reference standards: hepatic venous pressure gradient (HVPG)

Risk of bias/concerns about applicability:

- Patient selection: high/unclear risk of bias in 26 studies (55%), unclear/high concern in 37 studies (79%)
- Index tests: high/unclear risk of bias in 44 studies (94%), unclear/high concern in 1 study (2%)
- Reference standard: high/unclear risk of bias in 27 studies (57%), no studies with applicability concerns
- Flow and timing: high risk of bias in 30 studies (64%)

Clinical implications in a hypothetical cohort of 1000 people

Index test	Number of participants	Sensitivity	Specificity	Assumed preva-	True positives will	False negatives will	True negatives will	False positives will	Certainty of evidence

for CSPH	(number of studies)	(95% CI)	(95% CI)	lence of CSPH[a]	receive appropriate treatment with non-selective beta-blockers	be misdiagnosed and will miss the opportunity to receive appropriate treatment timely	appropriately avoid unnecessary further testing or inappropriate treatment	be inappropriately treated with beta-blockers	
LSM by VCTE (HSROC; specificity 90%)	3818 (27 studies)	72.6%(60.0% to 82.5%)	90%	62.5%	454	171	338	38	Very low[b] ⊕◯◯◯
LSM by VCTE (HSROC; sensitivity 90%)	3818 (27 studies)	90%	75.9% (63.5% to 85.1%)	62.5%	563	63	285	90	Very low[b] ⊕◯◯◯
LSM by VCTE (cut-off 25 kPa)	1553 (9 studies)	62.3% (53% to 70.7%)	94.1% (87.2% to 97.3%)	62.5%	389	236	353	22	Very low[b] ⊕◯◯◯ for both sensitivity and specificty
SSM by VCTE (HSROC; specificity 90%)	391 (6 studies)	72.9% (27.1% to 95.1%)	90%	62.5%	456	169	338	38	Very low[c] ⊕◯◯◯
SSM by VCTE (HSROC; sensitivity 90%)	391 (6 studies)	90%	80.6% (64.1% to 90.7%)	62.5%	563	63	302	73	Very low[c] ⊕◯◯◯
SSM by pSWE (HSROC; specificity 90%)	248 (4 studies)	84.1% (8.4% to 99.7%)	90%	62.5%	526	99	338	38	Very low[c] ⊕◯◯◯
SSM by pSWE (HSROC; sensitivity 90%)	248 (4 studies)	90%	86.1% (40.4% to 98.3%)	62.5%	563	63	323	52	Very low[c] ⊕◯◯◯

LSM by 2D-SWE (HSROC; specificity 90%)	767 (10 studies)	73.5% (52.6% to 87.4%)	90%	62.5%	459	166	338	38	Very low[c] ⊕◯◯◯
LSM by 2D-SWE (HSROC; sensitivity 90%)	767 (10 studies)	90%	Cannot be calculated	62.5%	N/A	N/A	N/A	N/A	Very low[c] ⊕◯◯◯
SSM by 2D-SWE (HSROC; specificity 90%)	353 (6 studies)	80.0% (59.8% to 91.5%)	90%	62.5%	500	125	338	38	Very low[c] ⊕◯◯◯
SSM by 2D-SWE (HSROC; sensitivity 90%)	353 (6 studies)	90%	Cannot be calculated	62.5%	N/A	N/A	N/A	N/A	N/A

Index test for SPH	Number of participants (number of studies)	Sensitivity (95% CI)	Specificity (95% CI)	Assumed prevalence of SPH[d]	True positives will receive appropriate treatment with non-selective beta-blockers	False negatives will be misdiagnosed and will miss the opportunity to receive appropriate treatment timely	True negatives will appropriately avoid unnecessary further testing or inappropriate treatment	False positives will be inappropriately treated with beta-blockers	Certainty of evidence
LSM by VCTE (HSROC; specificity 90%)	637 (8 studies)	67.2% (48.8% to 81.4%)	90%	62.5%	420	205	338	38	Very low[c] ⊕◯◯◯
LSM by VCTE (HSROC; sensitivity 90%)	637 (8 studies)	90%	Cannot be calculated	62.5%	N/A	N/A	N/A	N/A	N/A
LSM by 2D-SWE (HSROC; specificity 90%)	492 (7 studies)	59.8% (19.4% to 90.2%)	90%	62.5%	374	251	338	38	Very low[c] ⊕◯◯◯

LSM by 2D-SWE (HSROC; sensitivity 90%)	492 (7 studies)	90%	Cannot be calculated	62.5%	N/A	N/A	N/A	N/A	N/A

Abbreviations: CI: confidence interval; **HSROC:** hierarchical summary receiver operating characteristic

GRADE certainty of the evidence

High: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

Comparisons between different index tests

Direct (within-study) comparisons between different index tests were impossible due to the inconsistent numbers of participants included for each index test in the included studies.

Indirect comparisons were conducted only using summary receiver operating characteristic curves, due to the variability in cut-off values reported across included studies. Indirect comparisons show no overall difference between summary receiver operating characteristic curves.

[a]We chose the prevalence of CSPH of 62.5%, as this was the median prevalence observed in our review and may reflect the true prevalence in routine clinical practice.
[b]We deemed these estimates to be at a very low level of certainty, downgrading them by three levels due to risk of bias, inconsistency, and imprecision. Risk of bias was downgraded by one level because almost all studies were judged at high risk of bias; inconsistency was downgraded by one level due to wide range of cut-off values or different definitions of populations in studies; imprecision was downgraded by one level due to wide CIs of estimates.
[c]We deemed these estimates to be at a very low level of certainty, downgrading them by one level due to risk of bias and by two levels due to imprecision. Risk of bias was downgraded by one level because all studies were judged at high risk of bias; imprecision was downgraded by two levels due to generally low number of participants per study, low number of studies, and wide CIs of estimates.
[d]We chose the prevalence of SPH of 62.5%, as this was the median prevalence observed in our review and may reflect the true prevalence in routine clinical practice.

BACKGROUND

Target condition being diagnosed

Portal hypertension manifests with a set of clinical symptoms caused by increased pressure in the portal venous system. Chronic liver disease (CLD), which is the spectrum of (advanced) liver fibrosis and cirrhosis, is the most common cause of increased portal pressure. Portal hypertension due to CLD results from structural changes in liver parenchyma such as regenerative nodule formation, sinusoidal remodelling, and occlusion with consequent increased vascular (sinusoidal) resistance caused by chronic inflammation and increased collagen deposition. Increased peripheral vascular tone due to endothelial dysfunction and increased vasoconstrictor production are also contributing factors to portal hypertension. Furthermore, vasodilatation occurring in systemic circulation leads to activation of neurohumoral and vasoconstrictive systems (renin-angiotensin-aldosterone system), which causes hyperdynamic circulatory state and further increase in portal venous flow and portal pressure [1, 2].

Hepatic venous pressure gradient (HVPG) measurement is currently considered the gold standard for the evaluation of portal hypertension in advanced CLD. HVPG is measured by retrograde insertion of a balloon-tipped central vein catheter into the main hepatic vein, which enables measurement of the gradient between the pressure in the hepatic sinusoidal capillary network and the free hepatic (systemic) pressure [3]. HVPG measurement is thus an invasive technique associated with possible complications. Moreover, it is not widely available in clinical practice and is restricted to specialist centres, which limits its feasibility. Given its potential disadvantages, HVPG measurement is usually employed only for research purposes. Upper endoscopy allows an indirect assessment of portal hypertension and serves as diagnosis and surveillance of oesophageal varices, a condition that is associated with HVPG ≥ 10 mmHg, but which can be present at lower levels than 10 mmHg [2, 4]. Moreover, upper endoscopy is a costly procedure that may require sedation and is associated with potentially serious complications [5, 6]. Non-invasive alternative techniques with acceptable diagnostic accuracy and easy implementation in routine practice are therefore needed.

It is important to distinguish between mild (or subclinical) portal hypertension, defined by HVPG between 6 mmHg and 9 mmHg, and clinically significant portal hypertension (CSPH) and severe portal hypertension (SPH), defined by HVPG ≥ 10 mmHg and 12 mmHg, respectively [1, 2]. CSPH is the main driver of decompensation, with a direct impact on the median survival of people with CLD: in compensated CLD, survival is more than 10 years, while it is approximately two to four years in people with decompensated state [2, 7, 8, 9]. CSPH is also associated with an increased risk of CLD complications such as oesophageal varices [10], ascites, hepatic encephalopathy [11], and hepatocellular carcinoma [12]. On the other hand, an HVPG ≥ 12 mmHg, defined as SPH, is associated with a greatly increased risk of oesophageal bleeding, the most fatal consequence of portal hypertension, with six-week mortality between 15% and 25% [1, 13, 14, 15, 16]. A large cross-sectional study on compensated CLD patients showed that CSPH, any-size varices, and high-risk varices are present in 66%, 42%, and 13% of patients, respectively [17]. Many people with CSPH remain asymptomatic and undiagnosed for years. However, the overall risk of decompensation is 7% to 10% per year [10, 18]. Also, a decrease in HVPG to less than 12 mmHg or a reduction greater than 20% from baseline leads to significant risk reduction of recurrent haemorrhage, encephalopathy, ascites, and death [19, 20]. Moreover, in compensated CLD patients, a decrease by 10% from baseline reduces the risk for varices development [10], first variceal bleeding, and death [21, 22]. In routine clinical practice, CSPH is still diagnosed indirectly with endoscopy based on the presence of oesophageal varices. Even though varices usually appear when HVPG > 10 mmHg, they can also appear at lower levels [23, 24]. On the other hand, only 50% to 60% of people with CSPH have oesophageal varices [2, 25]. Thus, endoscopy is not a precise method to diagnose CSPH, but is much more available than HVPG. Timely assessment of portal hypertension, and detection of CSPH and SPH in populations at risk, are therefore important for patient prognosis and to ensure appropriate and early management. Consequently, the 'pre-primary prophylaxis' has been proposed for people with CSPH but without varices, to prevent variceal development. However, a large, multicentre, randomised, placebo-controlled trial showed no difference between non-selective beta-blockers (NSBBs) and placebo in variceal development prevention [10]. More recently, Villanueva and colleagues showed that people with subclinical portal hypertension (< 10 mmHg) have less hyperdynamic circulation and lower portal pressure reduction with intravenous propranolol than people with CSPH, indicating that NSBB therapy has greater potential to prevent decompensation in CSPH than in earlier stages [10, 26]. Following the PREDESCI trial, the paradigm has shifted from treating and prevention of only varices and variceal bleeding, to actually treating CSPH when diagnosed to prevent any type of decompensation. The PREDESCI trial showed that long-term NSBB therapy could increase decompensation-free survival, mainly by decreasing the incidence of ascites, but without significant reduction of the incidence of high-risk varices [27]. Thus, contrary to previous guidelines, the recent Baveno VII consensus recommends NSBB not only in people with any oesophageal varices or high-risk varices as primary or secondary prophylaxis of variceal bleeding, but also in those with CSPH due to their higher risk of developing decompensation [28]. Thus, widely available, cost-effective non-invasive tests with acceptable diagnostic accuracy are needed for timely diagnosis and early preventive treatment.

A list of abbreviations is provided in Table 1.

Index test(s)

Elastography-based techniques are the most widely used non-invasive methods for estimation of portal pressure. These include vibration-controlled transient elastography (VCTE), two-dimensional shear wave elastography (2D-SWE), point shear wave elastography (pSWE), and magnetic resonance elastography (MRE), which by assessing liver stiffness and spleen stiffness allow indirect estimation of portal pressure [29]. Liver stiffness is a marker of the amount of liver fibrosis and can be used in a diagnostic approach and follow-up in people with CLD. Liver stiffness by VCTE is the most used non-invasive test and shows a strong association with portal pressure values [30]. Liver stiffness measurement (LSM) by VCTE is thus recommended in the latest Baveno VII guidelines for people with CLD for the assessment of CSPH, but only in certain aetiologies of CLD. There are no recommendations for LSM by other elastography techniques, thus their role remains unknown for clinical practice [28]. Spleen stiffness reflects the effects of portal hypertension on the spleen parenchyma, making it a potential marker of portal pressure. Current guidelines recommend spleen stiffness measurement (SSM) by VCTE only as an additional diagnostic tool in cases of viral aetiology. The role and utility of SSM in other aetiologies, as well as when measured by other elastography techniques, remain unclear [28]. It also remains unclear whether the diagnostic accuracy of SSM by VCTE is comparable or superior to LSM in the detection of CSPH, and if a combination of these two measurements can improve their single diagnostic accuracy. Furthermore, recent guidelines have not addressed specific cut-off values for SPH for either LSM or SSM [2, 4, 28].

Vibration-controlled transient elastography (VCTE) (Fibroscan; Echosens, Paris, France) is a non-invasive diagnostic modality that evaluates both liver and spleen stiffness. Fibroscan is a one-dimensional ultrasound VCTE that measures the velocity of a low-frequency (50 Hz) elastic shear wave propagating through the liver or spleen parenchyma. The velocity of the wave is directly related to tissue stiffness: the stiffer the tissue, the faster the wave. However, the spleen parenchyma is stiffer than liver parenchyma, thus liver 50 Hz probe might lead to overestimation of SSM. Recently, a novel spleen-dedicated 100 Hz probe was introduced for SSM [31]. Since it represents new technology, it has not yet been widely evaluated. VCTE is the most widely used non-invasive technique, and is user-friendly (rapid, easy to learn, performed at bedside) with high reproducibility [32]. However, it has the following disadvantages: impaired applicability due to obesity and ascites; inability to choose the scanned region; false-positive results in the case of elevated transaminases, acute hepatitis, extra-hepatic cholestasis, liver congestion, recent food intake, excessive alcohol consumption, and inaccuracy in discriminating intermediate stages of fibrosis [33].

• Liver stiffness as measured by VCTE

Liver stiffness is a marker of the amount of liver fibrosis. It is a consequence of the pathologic excessive accumulation of collagen at the expense of normal liver parenchyma that occurs in CLD. In adult patients, liver stiffness can be measured with VCTE by using M or XL probe (according to body mass index (BMI)) in the supine position. The result is expressed in kilopascals, which range from 1.5 to 75 kPa, with normal values around 5 kPa. Higher values are expected in people with low or high BMI (U-shaped distribution) [33]. In 7261 patients and over 13,000 examinations, measurement failure was observed in 3% and unreliable results in 15.8% of patients, mostly due to obesity and operator experience. VCTE-LSM applicability was thus 81% [34]. However, the introduction of the XL probe improved applicability (to more than 95%) in people with non-alcoholic fatty liver disease [35, 36]. An excellent interobserver reproducibility has also been reported for VCTE with intraclass correlation coefficient (ICC) of 0.98 [32]. A cut-off of 8 kPa is recommended to rule out advanced fibrosis in high-risk populations, irrespective of CLD aetiology. Moreover, a cut-off of 12 kPa to 15 kPa has been proposed to rule in advanced fibrosis for people with alcoholic liver disease based on evidence including one Cochrane review [37, 38]. According to the Baveno consensus, a cut-off of 10 kPa can rule out CLD; values 10 kPa to 15 kPa are considered suggestive of CLD; while values above 15 kPa are highly suggestive of CLD [4, 28]. In addition to estimating liver fibrosis, liver stiffness appears to be a reliable surrogate marker for assessing portal hypertension [30]. The Baveno VII consensus recommends a cut-off value of liver stiffness by VCTE of ≥ 25 kPa to confirm CSPH in cases of viral, alcohol-related, and non-obese non-alcoholic steatohepatitis-related CLD. The diagnostic value of LSM by VCTE in other aetiologies remains unknown [2, 4, 28]. Furthermore, LSM by VCTE has been employed in combination with other non-invasive tests, such as platelet count, to assess the risk of CSPH and high-risk varices [2, 4, 17, 28].

• Spleen stiffness as measured by VCTE

Spleen stiffness reflects the effects of portal hypertension on the spleen parenchyma, making it a marker of portal pressure values. The pathogenesis of spleen enlargement in cirrhotic patients is not fully understood; however, it is not simply due to passive spleen congestion, but also to tissue hyperplasia due to the combination of fibrogenesis, angiogenesis, and enlargement of splenic lymphoid tissue [39]. The working principle of VCTE is the same as for the liver; however, since the spleen is a stiffer organ, a higher shear wave frequency is needed (100 Hz), resulting in higher measured values (6 kPa to 100 kPa) [37]. However, since 100 Hz spleen-dedicated probe has only recently been introduced [31], in most studies and in clinical practice, liver 50 Hz probe is still being used and evaluated for SSM. High inter- and intraobserver agreements have been reported (0.89 and 0.94, respectively), with a failure rate of 15% to 20% [40]. A study of 100 people with hepatitis C virus-related cirrhosis showed that a cut-off value of 40 kPa is highly sensitive for ruling out CSPH, and a cut-off of 52 kPa is highly specific for ruling in CSPH. These results were comparable to LSM and showed a strong correlation with HVPG [39]. Another study confirmed these findings [41]. The latest guidelines recommend SSM by VCTE in viral aetiology to rule out and rule in CSPH using SSM less than 21 kPa and SSM more than 50 kPa. The same consensus highlighted the need for validation of the best cut-off value using a novel spleen-dedicated 100 Hz probe [28].

Shear wave elastography is based on real-time imaging and shear wave propagation and detection. These techniques include point shear wave elastography (pSWE) (Virtual touch tissue quantification, Siemens; elastography point quantification, ElastPQ, Philips; Samsung TS80A, Samsung) and two-dimensional shear wave elastography (2D-SWE) (Aixplorer Supersonic Imagine, ACUSON Sequoia, Siemens; ElastQ, Philips; 2D-SWE.GE, General Electric), both of which use acoustic radiation force impulse imaging technology. These two techniques are 'hybrid' methods that allow both B-mode visualisation and the assessment of liver stiffness/spleen stiffness. They aim to overcome the inherent limitation of VCTE, which lacks the ability to select the specific region to be scanned. By providing targeted imaging, these hybrid techniques can reduce the failure rate, particularly in obese patients and those with ascites [33]. In a study comparing the reliability of pSWE and 2D-SWE for LSM, the failure rate was low for both methods (1% for pSWE and 5% for 2D-SWE). However, the intraobserver agreement was significantly better for pSWE than for 2D-SWE (ICC 0.915 versus 0.829) [42]. For SSM, the overall feasibility of pSWE ranges from 85% to 100%; however, low inter- and intraobserver agreements have been reported, unlike LSM. It appears that reproducibility is highly dependent on the operator's experience [43, 44]. On the other hand, SSM by 2D-SWE is less feasible (up to 29% failure rate) but demonstrates a higher intra- and interoperator reproducibility [45]. A recent meta-analysis evaluating LSM by shear wave elastography found a diagnostic performance comparable to VCTE [46]. However, cut-offs differ between techniques and devices, which underlines the need for device-specific cut-off values. Moreover, shear wave elastography is also feasible in people with ascites [47], and most studies include a high proportion of patients with ascites in whom the presence of portal hypertension is already known [48]. Notably, shear wave elastography techniques, combined with B-mode imaging, are theoretically ideal for SSM.

Two-dimensional shear wave elastography (2D-SWE) is a newly developed elastography technique that can be implemented in most ultrasound machines. It is based on shear wave propagation in tissues by focused ultrasonic beams while performing real-time ultrasound imaging and assessing LSM. The size of the region of interest is approximately 2 x 2 cm. The velocity of shear wave propagation can be measured at varying locations within the region of interest and presented as liver stiffness/spleen stiffness results in kPa [33, 49]. The most evaluated device is Aixplorer Supersonic Imagine. Liver stiffness cut-off values for ruling in CSPH vary substantially between studies (15.2 kPa to 24.6 kPa) [50, 51, 52, 53, 54]. This variation may be partially attributed to differences in evaluated populations [48]. SSM cut-offs to rule in CSPH also differ between studies (ranging from 25.3 kPa to 34.7 kPa) [50, 51, 53, 54]. While applicability and results of liver stiffness are considered comparable between 2D-SWE and VCTE, a study found that SSM by 2D-SWE was feasible in only 66% of patients and demonstrated a lower area under the receiver operating characteristic (AUROC) for CSPH compared to LSM [53]. Algorithms that combine both techniques appear to improve both applicability and diagnostic accuracy [51, 54]. However, a subsequent validation study found that these combination algorithms did not enhance the diagnostic accuracy of liver stiffness measured by 2D-SWE [55].

Point shear wave elastography (pSWE) is based on the mechanical excitation of liver tissue using acoustic radiation force impulses that cause shear wave propagation and consequently localised displacements in tissue. The shear wave propagation velocity measured by ultrasonographic waves (in metres per second) correlates with tissue stiffness. The region of interest is smaller than VCTE and 2D-SWE (approximately 1.0 cm x 0.5 cm), but it can be selected by the operator under B-mode visualisation. The major advantage of pSWE is that it can be implemented in most commercial ultrasound machines. Like VCTE, LSM is influenced by elevated aminotransferases and food intake. However, the failure rate is lower than VCTE [49, 56]. Studies that used pSWE to measure liver stiffness proposed cut-offs of 2.17 m/s and 2.58 m/s for CSPH [47, 57], and 2.54 m/s for SPH [57]. SSM by pSWE showed higher AUROC than LSM. Moreover, pSWE seems to be applicable even in people with ascites, obesity, or both [57, 58].

Magnetic resonance elastography (MRE) utilises a modified phase-contrast method to evaluate the velocity of shear wave propagation in the liver on 3D imaging. Shear waves are generated by a passive driver that produces continuous acoustic vibrations transmitted through the entire abdomen, including the liver, at a frequency of 60 Hz. MRE's advantage lies in its ability to evaluate the entire liver tissue, and its applicability in people with ascites or obesity, or both [59]. The failure rate of MRE-LSM is low (5.6%) [60], and test-retest repeatability is high (reported ICC is 0.95) [61]. On the other hand, SSM using MRE is 100% feasible and shows a strong linear association between liver stiffness and splenic stiffness (r^2 = 0.75) [62, 63]. However, due to its limited availability, high cost, and the time-consuming nature of the procedure, it has limited clinical value and is generally more suited for research purposes [33, 59]. While MRE has proven accurate in diagnosing and staging liver fibrosis [64], data on CSPH and SPH are limited. Spleen-derived parameters appear to correlate better than liver-derived parameters with HVPG and can identify people with CSPH, SPH, and high-risk varices [65, 66].

The possible advantages and disadvantages of different index tests are summarised in Table 2 [32, 33, 37, 40, 42, 45, 49, 61].

Each of the available techniques uses different cut-off values for the target condition. We planned to assess the diagnostic accuracy estimates overall and for the most used cut-off values.

Clinical pathway

Given that CSPH is the main driver of decompensation in people with CLD, and that SPH can lead to oesophageal bleeding, it is crucial to detect these conditions at the time of patient presentation. Early detection may facilitate timely intervention and treatment [28]. Based on the estimated portal pressure, people with CLD, independently of aetiologies, can be divided into those with mild portal hypertension (HVPG 5 mmHg to 9 mmHg), CSPH (≥ 10 mmHg), and SPH (≥ 12 mmHg). The reference standard to assess portal pressure is HVPG measurement. However, its invasiveness, high costs, and requirement for specific expertise limit its use in routine practice, and it is not available in most centres [2, 4, 28]. Current practice is thus more oriented to detecting complications than CSPH and SPH. Upper endoscopy is frequently used in clinical practice in people with CLD as an indirect assessment of portal hypertension detecting the presence of oesophageal varices, a condition that is associated with HVPG ≥ 10 mmHg [2, 4]. However, only 50% to 60% of people with CSPH have oesophageal varices [2, 25], thus many patients with CSPH might miss a chance for early treatment. Elastography-based methods may non-invasively detect people with CSPH or SPH without using HVPG or upper endoscopy, and allow the identification of people at high risk of decompensation and complications for whom preventive strategies could be applied. Thus, the ideal target population are patients with compensated advanced chronic liver disease (cACLD) (LSM > 10 kPa) or with compensated histologically proven cirrhosis without prior decompensation event. Since CSPH is the main driver of decompensation, decompensated patients, or those currently compensated but with previous decompensation episode, most likely already have CSPH, which questions the utility of non-invasive techniques to evaluate portal hypertension. Current guidelines recommend preventing CSPH by eliminating the underlying cause of liver disease (i.e. aetiological treatment) and treating comorbidities [1, 2, 4, 28, 67]. CSPH can be ruled out if LSM by VCTE is < 15 kPa and platelet count is > 150,000/mm^3. Current guidelines recommend liver stiffness measured by VCTE to rule in CSPH, using a cut-off value of ≥ 25 kPa, in people with viral, alcohol-related, and non-obese non-alcoholic steatohepatitis-related CLD. However, the diagnostic accuracy of liver stiffness by VCTE in other aetiologies or by any other elastography technique in any aetiology is currently unknown [28]. American Association for the Study of Liver Diseases (AASLD) practice guidelines recommend that LSM by VCTE < 15 kPa alone can be used to rule out CSPH [68]. In the case of LSM < 25 kPa, the ANTICIPATE model, which combines LSM and platelet count (and BMI in metabolic dysfunction-associated steatotic liver disease (MASLD) aetiology), should be used for CSPH prediction [17, 28]. Regarding SSM, SSM by VCTE can be used in viral aetiology to both rule out and rule in CSPH with SSM < 21 kPa and SSM > 50 kPa. SSM in other aetiologies cannot be recommended, nor are there recommendations for novel spleen-dedicated 100 Hz probe [28]. AASLD guidelines recommend considering SSM by VCTE > 40 kPa to rule in CSPH, irrespective of aetiology [68]. If CSPH is present, NSBB therapy (preferably carvedilol) should be initiated to prevent decompensation [28]. In the case of NSBB contraindication or intolerance, patients should undergo endoscopy if LSM by VCTE ≥ 20 kPa, or platelet count ≤ 150,000/mm^3 [4, 28]. Additionally, in people who are not candidates for NSBBs and in whom endoscopy would be required according to the Baveno VI criteria (LSM by VCTE ≥ 20 kPa or platelet count ≤ 150,000/mm^3), SSM by VCTE ≤ 40 kPa can identify people with a low probability of high-risk oesophageal varices, preventing the need for screening endoscopy. Cut-offs for SPH have not been addressed for LSM and SSM in any recent guidelines [2, 4, 28].

Prior test(s)

The index tests for diagnosing conditions related to cACLD are typically performed after the initial clinical and laboratory diagnosis and often following a preliminary ultrasound examination. This sequence ensures that the elastography tests are conducted in a well-defined clinical context, following a broad assessment that includes the following.

1. Clinical diagnosis: this involves a thorough clinical evaluation of the patient, including history taking, and physical examination to suspect cACLD.
2. Laboratory diagnosis: blood tests and other laboratory assessments are used to confirm or support the clinical suspicion of cACLD and to evaluate liver function, fibrosis, and other related markers.
3. Ultrasound examination: an initial abdominal ultrasound may be performed to assess liver morphology, detect any focal lesions, and evaluate the presence of ascites or signs of portal hypertension. This imaging technique helps in identifying any immediate structural abnormalities and guides the selection of further diagnostic tests.
4. Index tests: only after these preliminary steps, are non-invasive elastography tests, such as LSM by VCTE or SSM, conducted. These tests might provide additional diagnostic value by quantifying stiffness and potentially identifying portal hypertension or other complications associated with cACLD.

Role of index test(s)

In the diagnostic pathway for chronic advanced liver disease, non-invasive tests such as LSM and SSM, performed using various elastography techniques, may serve as replacements for the HVPG measurement. While HVPG is considered the gold standard for estimating portal pressure, it is an invasive procedure associated with potential complications and is typically limited to specialised centres. Conversely, LSM and SSM offer safer, more accessible options for assessing portal hypertension in clinical practice.

Alternative test(s)

There are other non-invasive tests employing various techniques that have been assessed and used for screening populations at risk of CSPH. These non-invasive tests include imaging and functional tests and blood-based biomarkers that evaluate different features of portal hypertension syndrome [48]. These non-invasive tests include:

1. platelet count [69];
2. spleen length [69];
3. extracellular matrix protein serum levels [70];
4. plasma Von Willebrand factor antigen (vWF-Ag) [71, 72];
5. vWF-Ag/platelet ratio [73];
6. enhanced liver fibrosis (ELF) test [74, 75];
7. indocyanine green 15-minute retention (ICG-r15) [76, 77];
8. contrast-enhanced ultrasound-based methods [78, 79, 80, 81];
9. magnetic resonance-based techniques besides elastography [82, 83, 84];
10. computed tomography-based techniques [85, 86, 87];
11. non-patented serum biomarkers (aspartate aminotransferase:platelet ratio index (APRI), Lok Index, Fibrosis-4 (FIB-4), FibroTest) [88, 89, 90, 91]; and
12. combinations of unrelated non-invasive tests: liver stiffness-spleen diameter-to-platelet ratio score [92], portal hypertension risk score [74], and platelet count-to-spleen length ratio [69].

However, a recent review questioned the precision of the estimated accuracy of studies assessing these non-invasive tests, and their role in clinical practice remains unknown [1, 48].

Rationale

This review aims to assess the accuracy of LSM and SSM measured with different elastography techniques in diagnosing CSPH and SPH in adults with CLD.

Previous studies show a correlation of both LSM and SSM, evaluated by different elastography techniques, with HVPG measurement [39, 41, 51, 54, 57, 58, 93, 94]. These techniques are increasingly recognised for their non-invasiveness, applicability, and relatively short learning curve. Different cut-off values and diagnostic accuracies have been established, depending on the specific technique or aetiology of CLD. Therefore, a comprehensive assessment of the diagnostic performance of both LSM and SSM assessed by VCTE, pSWE, 2D-SWE, and MRE in detecting CSPH and SPH is needed.

Previously published systematic reviews with meta-analysis on LSM and SSM accuracy to evaluate CSPH and SPH have yielded inconsistent and conflicting results [30, 46, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106]. Most of these reviews focused either on LSM or SSM, analysing a single index test or including participants with a single aetiology of CLD. As a result, they lack comparative data between the two measurements (LSM and SSM), or insights into the relative superiority and applicability of one elastography technique over another. The only exception is the recent systematic review with meta analysis by Dajti and colleagues, in which diagnostic accuracies of SSM and SSM-based algorithms as measured by different elastography techniques were evaluated and compared to current recommendations (mainly LSM and LSM-based composite scores and algorithms measured with VCTE) [100]. However, we believe that, with the Cochrane structural systematic review process, we can provide more reliable and updated evidence and offer comparative data, indirect or direct (if feasible), on the diagnostic accuracy of LSM and SSM, as well as on different elastography techniques across various CLD aetiologies.

We found no related Cochrane review, protocol, or registered title that assessed or planned to assess the diagnostic accuracy of liver and spleen stiffness to evaluate portal hypertension in people with CLD. This review will complement another Cochrane review that we have registered: 'Liver and spleen stiffness for the diagnosis of oesophageal varices in people with chronic liver disease'. Moreover, our review will also complement two earlier published Cochrane reviews that assessed the role of capsule endoscopy [107], and the role of platelet count, spleen length, and platelet count-to-spleen length ratio [69], for the diagnosis of oesophageal varices. As previously mentioned, oesophageal varices are a direct consequence of CSPH. Therefore, these reviews will collectively provide evidence for clinical practice for people with CLD.

OBJECTIVES

- To assess the diagnostic accuracy of LSM and SSM, as well as in combination, measured by any elastography technique (VCTE, pSWE, 2D-SWE, or MRE) in the detection of CSPH in adults with CLD.
- To compare the diagnostic accuracies between the individual tests. We will regard a combination of tests as positive when at least one is positive.

Secondary objectives

- To assess the diagnostic accuracy of LSM and SSM, as well as in combination, measured by any elastography technique (VCTE, pSWE, 2D-SWE, or MRE) in the detection of SPH in adults with CLD.
- To investigate sources of heterogeneity in the results.

Potential sources of heterogeneity

- Different populations based on CLD aetiology, in particular viral compared to non-viral aetiology: studies including more than 80% of participants with viral aetiology compared to those including less than 80% of participants with viral aetiology. We have chosen a threshold of 80%, as in North America and Europe the prevalence of chronic viral hepatitis is usually lower, while in Asia and Africa the prevalence is higher [108].
- Different prevalences of CSPH: studies with prevalence of CSPH > 50% compared to studies with prevalence of CSPH ≤ 50%. We have chosen a threshold of 50% according to a median prevalence of any size oesophageal varices in adults at the diagnosis of cirrhosis [107].
- Different prevalences of SPH: studies with prevalence of SPH > 25% compared to studies with prevalence of SPH ≤ 25%. We have chosen a threshold of 25% according to a median prevalence of high-risk oesophageal varices in adults at the diagnosis of cirrhosis [107].
- Different populations based on CLD severity: studies including more than 75% of participants with cirrhosis (Child-Pugh class A) compared to studies including less than 75% of participants with cirrhosis (Child-Pugh class A). We have chosen a threshold of 75% according to a median prevalence of adults with cirrhosis in Child-Pugh class A of 74% in a large cohort study [8].
- Type of probe used (liver, LSM@50Hz versus spleen, SSM@100Hz) for SSM by VCTE.
- Type of probe used (M versus XL) when liver, LSM@50Hz probe is used for SSM by VCTE.
- Number of unsuccessful or indeterminate results to obtain any LSM or SSM: studies with more than 20% compared to studies with less than 20% of unsuccessful or indeterminate results. We have chosen a threshold of 20%, as the test applicability estimated in a large series was 81% [34].

METHODS

The protocol was published in August 2022 in the Cochrane Database of Systematic Reviews. The reference citation is as follows.

Vranić L, Nadarevic T, Štimac D, Fraquelli M, Manzotti C, Casazza G, et al. Liver and spleen stiffness as assessed by vibration controlled transient elastography for diagnosing clinically significant portal hypertension in comparison with other elastography-based techniques in adults with chronic liver disease. Cochrane Database of Systematic Reviews 2022, Issue 8. Art. No.: CD015415. DOI: 10.1002/14651858.CD015415

Criteria for considering studies for this review

Types of studies

We included studies that, irrespective of publication status and language, evaluated the diagnostic accuracy of LSM and SSM either alone or in combination, as measured by different techniques (VCTE, pSWE, 2D-SWE, MRE), for the diagnosis of CSPH and SPH in adults with CLD. The studies should have used HVPG measurement as the reference standard.

We considered studies of cross-sectional design. We excluded case-control (two-gate) studies that compared adults with known portal hypertension to matched controls, as these are at higher risk of bias. Such designs often include individuals at the extremes of the disease spectrum (e.g. definite cases and definite non-cases), which can lead to inflated diagnostic accuracy estimates due to spectrum bias. [109].

Participants

We included adults of any age and sex with CLD, irrespective of duration of illness and aetiology. The diagnosis of CLD was based on histology or on a combination of physical, laboratory, endoscopic, and radiologic findings, as defined by the trialists.

We planned to exclude participants with other different conditions associated with raised LSM or SSM values, such as acute hepatitis, cholestatic liver disease, past or current hepatocellular carcinoma, multiple focal liver lesions, past or current portal vein thrombosis, and secondary portal hypertension (e.g. right-sided heart failure, Budd-Chiari syndrome). We also planned to exclude participants with a history of treatments for portal hypertension (splenectomy, partial splenic embolisation, transjugular intrahepatic portosystemic shunt, balloon-occluded retrograde transvenous obliteration, NSBB therapy, or endoscopic therapies).

We planned to exclude studies including participants with already-known CSPH and SPH, unless investigators presented data in such a way that allowed for extraction of the data on the remaining participants. However, unlike in our protocol, we included studies with less than 20% of participants with hepatocellular carcinoma, cholestatic liver disease, history of NSBB use (not current), and previously known CSPH/SPH or oesophageal varices to avoid excluding larger studies with eligible participants. Nevertheless, we categorised these studies at high risk of bias and concern for applicability according to Quality Assessment of Diagnostic Accuracy Studies (QUADAS-2) [110]. We thus decided to perform a sensitivity analysis by excluding such studies (see Supplementary material 7).

Index tests

1. Liver stiffness as measured by VCTE
2. Spleen stiffness as measured by VCTE
3. Liver and spleen stiffness as measured by 2D-SWE
4. Liver and spleen stiffness as measured by pSWE
5. Liver and spleen stiffness as measured by MRE

Target conditions

The target conditions in this review were:

1. CSPH, defined as HPVG ≥ 10 mmHg;
2. SPH, defined as HVPG ≥ 12 mmHg (see Secondary objectives).

Reference standards

HVPG measurement is the only acceptable reference standard for assessing portal hypertension. HVPG represents the gradient between the pressure in the portal vein and the intra-abdominal portion of the inferior vena cava. It is determined by measuring the difference between the wedged hepatic venous pressure (WHVP) and the free hepatic venous pressure. This is a moderately invasive procedure performed by right jugular, left jugular, or brachial vein catheterisation. A balloon-tipped catheter is advanced under fluoroscopic control into the inferior vena cava and hepatic vein. When blood flow is stopped by occluding the hepatic vein by the wedged catheter, the registered pressure reflects the pressure from the preceding vascular territory, specifically the hepatic sinusoids. Thus, the wedged hepatic venous pressure indicates hepatic sinusoidal pressure, rather than the portal pressure itself. In a normal liver, wedged hepatic venous pressure is slightly lower than the portal pressure due to pressure equilibration through interconnected sinusoids. In cirrhosis, fibrous septa disrupt this pressure equilibration, allowing WHVP to accurately estimate portal pressure. Free hepatic venous pressure measures the pressure in a non-occluded hepatic vein. The normal HVPG value ranges from 1 mmHg to 5 mmHg. Values above 5 mmHg define portal hypertension, but only HVPG values above 10 mmHg (termed CSPH) are associated with the development of CLD complications. HVPG values above 12 mmHg (termed SPH) represent the threshold pressure for variceal rupture and bleeding [3]. However, HVPG has its own limitations, especially in cholestatic diseases and MASLD. In fact, in MASLD the pattern of liver injury often includes perisinusoidal fibrosis, contributing to presinusoidal resistance which leads to underestimation of portal pressure. Thus, there might be significant discordance between HVPG measurements and clinical signs of portal hypertension and decompensation [111, 112]. In cholestatic liver disease, especially primary biliary cholangitis, veno-venous collaterals (or inter-sinusoidal channels) form in early stages of disease which cause partial decompression of the hepatic venous system, which results in lower WHVP and consequently underestimation of HVPG [113]. Thus, HVPG might be imprecise in highly prevalent MASLD aetiology among CLD patients.

The interval between index tests and HVPG measurements must be less than a week. If a study reported a longer interval period, we included the study but considered it at high risk of bias due to the potential evolution of the target condition.

Search methods for identification of studies

To minimise bias in our search results, we followed the guidance in Chapter 6 of the *Cochrane Handbook for Diagnostic Test of Accuracy Reviews* [114] and in PRISMA-S (PRISMA-S Checklist; [115]) to describe the search process for the review. The Cochrane Hepato-Biliary Group Information Specialist assisted with the development of the search strategies and performed the electronic searches.

Electronic searches

We searched the following electronic databases:

1. Cochrane Hepato-Biliary Group (CHBG) Controlled Trials Register and the CHBG Diagnostic Test of Accuracy Studies Register via the Cochrane Register of Studies Web (searched 8 April 2024);
2. the Cochrane Central Register of Controlled Trials (CENTRAL) in the Cochrane Library (2024, Issue 3);
3. MEDLINE ALL Ovid (1946 to 8 April 2024);
4. Embase Ovid (Excerpta Medica Database) (1974 to 8 April 2024);
5. LILACS (VHL Regional Portal) (1982 to 8 April 2024);
6. Science Citation Index – Expanded (Web of Science) (1900 to 8 April 2024); and
7. Conference Proceedings Citation Index – Science (Web of Science) (1990 to 8 April 2024).

The last two were searched simultaneously through the Web of Science.

We applied no restrictions on language or document type.

The search strategies with the date range of the searches for the respective databases are provided in Supplementary material 1.

Searching other resources

We also searched ClinicalTrials.gov (clinicaltrials.gov/), the European Medicines Agency (EMA) (www.ema.europa.eu/ema/), the World Health Organization International Clinical Trials Registry Platform (WHO ICTRP) (www.who.int/ictrp), and the US Food and Drug Administration (FDA) (www.fda.gov). The searches were conducted between 1 May 2023 and 1 May 2024. The search strategy for each resource is presented in Supplementary material 8. We found no additional studies for inclusion.

We attempted to identify other potentially eligible studies not identified by the electronic searches by manually searching the reference lists of the included studies and checking review articles, other systematic reviews, and/or meta-analyses. We also searched in abstract books from meetings of the European Association for the Study of the Liver (EASL), Asian Pacific Association for the Study of the Liver (APASL), and the American Association for the Study of Liver Diseases (AASLD) held 10 years prior to the search date (2014 to 2024). We manually screened the abstract books of the relevant conferences year by year using study-by-study inspection. We manually searched the conference books each by each year for eligible studies. We found no additional studies for inclusion.

We searched Google Scholar (scholar.google.com/) for relevant grey literature sources, including reports, dissertations, theses, and conference abstracts. The searches were conducted between 1 May 2023 and 1 May 2024. The search strategy is presented in Supplementary material 8. We searched Google Scholar manually using the specified terms. We found no additional studies for inclusion.

We used the PubMed/MEDLINE “similar articles search” tool on all included studies. The searches were conducted between May 2023 and May 2024. We found no additional studies for inclusion.

We searched for and examined any relevant retraction statements using the Retraction Watch Database (retractionwatch.com/retraction-watch-database-user-guide/). To identify errata related to the included studies, we also searched PubMed/MEDLINE. Identifying errata was important, as they can reveal critical limitations or corrections that may affect study validity [116]. These searches were conducted between 1 May 2023 and 1 May 2024. The search strategy for Retraction Watch Database is presented in Supplementary material 8. We found no additional studies for inclusion.

We attempted to contact authors of the included studies, experts, and researchers in the field, as well as relevant pharmaceutical companies, as needed, to enquire about potentially missed, unpublished, or ongoing studies. We contacted authors of included studies or studies with insufficient data for additional data as needed. Those authors who were also experts in the field referred us to other studies, but in all cases they had already been encountered and judged for inclusion or exclusion. We attempted to contact authors, experts, and researchers between 1 May 2023 and 1 May 2024.

We used items from the PRISMA-S checklist that were relevant to our review to ensure that we reported and documented our searches as advised [115].

Data collection and analysis

We followed the guidelines provided in the *Cochrane Handbook for Diagnostic Test of Accuracy Reviews* [117].

Selection of studies

Two review authors (LV and CM) independently reviewed the titles or abstracts of the retrieved records. After our first selection of records of possible interest, we obtained the full-text records. LV and CM independently screened the full-text records against the inclusion criteria and made final decisions regarding study inclusion. Any disagreements were resolved by discussion, or if a disagreement persisted, a third review author (TN or AC) arbitrated. We listed all excluded studies in the 'Characteristics of excluded studies' table with the reasons for their exclusion (see Supplementary material 3). When a single study had multiple reports, we listed all publication reports under the main reference, defined as reference with the most complete data. We illustrated the study selection process graphically using a PRISMA-S flow diagram (Figure 1) [118, 119]. AC and GC were co-authors of the Colecchia 2012 [120] study and were thus not involved in the eligibility assessment of that particular study.

Figure 1. Study flow diagram [118, 119]. Date of last search 8 April 2024.

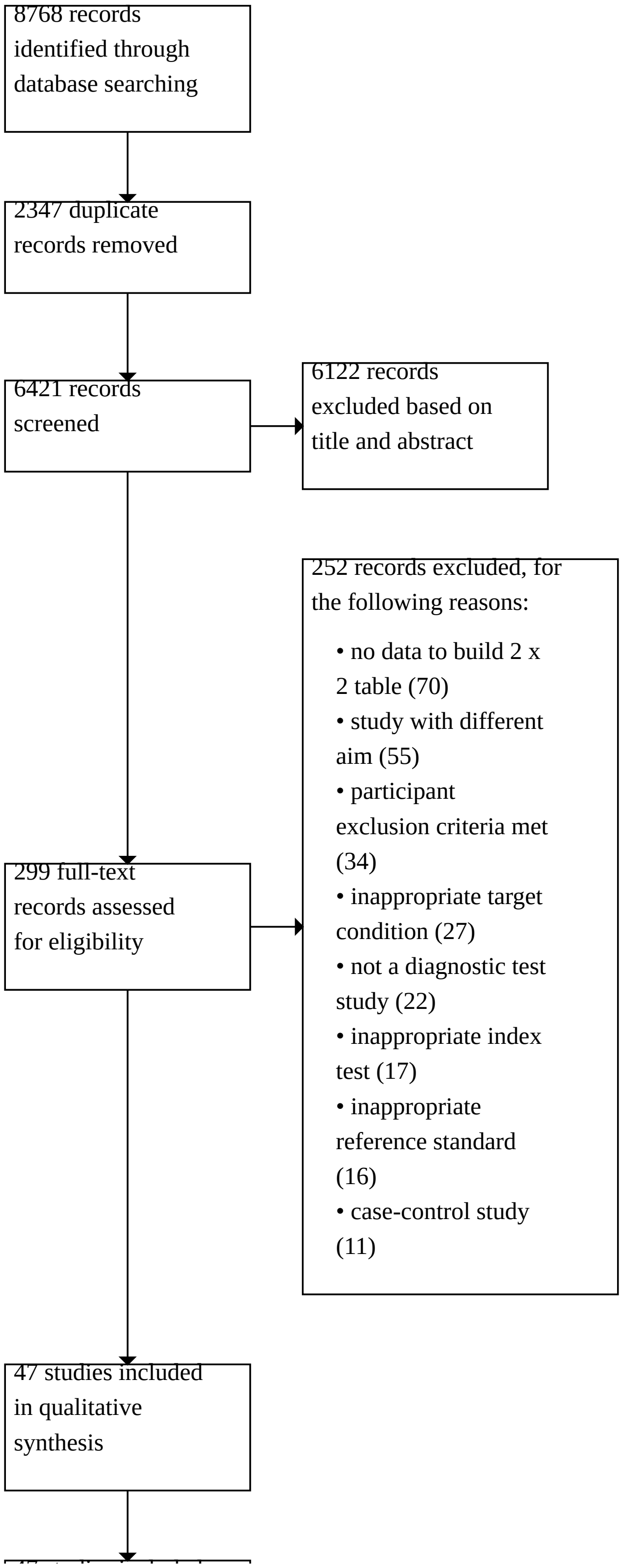
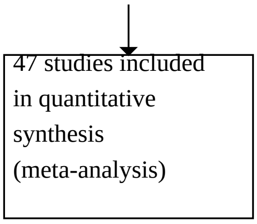


Figure 1. (Continued)



Data extraction and management

Two review authors (LV and TN) independently extracted study data from each included study using a piloted data extraction form developed specifically for this review. Any discrepancies were resolved through discussion, or if a discrepancy persisted, a third review author (CM or MF) arbitrated. If multiple reports of a single study occurred, we examined each report separately, and then combined the reports into a single study report, using the piloted data extraction form. We extracted all necessary data to calculate true-positive, false-positive, true-negative, and false-negative values, using the reference standard of HVPG. We summarised these data in 2 x 2 tables, according to the index tests, and entered the data into RevMan [121]. We contacted the authors of the included studies in the case of missing information or for clarification.

We extracted the following data from the included studies.

1. General characteristics: type of study report (e.g. abstract, full-text article, letter, or comment, etc.), year of publication, language of publication, country in which the participants were recruited, study duration (start date and completion date), setting (hospital, outpatient clinic, etc.), study design (prospective compared to retrospective, randomised compared to non-randomised), funding, inclusion and exclusion criteria, contact for correspondence
2. Sample size: number of participants meeting the criteria and total number of participants screened
3. Baseline characteristics: age (mean/median, standard deviation, range), sex, ethnicity, disease severity (cirrhosis versus non-cirrhosis), disease aetiology
4. Index test used with predefined positivity criteria (cut-off values used)
5. Blinding the ultrasound operators for the results of HVPG
6. Blinding the HVPG operators for the results of LSM or SSM, or both
7. Time interval between the index test and the reference standard
8. Study results: true positives for each presented cut-off value, true negatives for each presented cut-off value, false positives for each presented cut-off value, false negatives for each presented cut-off value. We extracted these data for each index test and for the two target conditions (CSPH and SPH).
9. The frequency of invaluable or indeterminate results due to technical failure
10. Missing data
11. Other relevant findings (e.g. presence of oesophageal varices)

AC and GC were co-authors of the Colecchia 2012 study and were thus not involved in the data extraction and management of that particular study.

Assessment of methodological quality

Two review authors (LV and TN) independently assessed the risk of bias in the included studies and applicability of their results using Quality Assessment of Diagnostic Accuracy Studies – Comparative (QUADAS-C), a tool for assessing risk of bias in comparative diagnostic accuracy studies, which retains the same four-domain structure of QUADAS-2 (patient selection, index test, reference standard, and flow and timing) and comprises additional questions to each QUADAS-2 domain specific to the test comparisons [122]. There are signalling questions for each domain used to make judgements regarding the risk of bias. The proposed answers are 'yes', 'no', or 'unclear'. The first three domains are also assessed in terms of concerns regarding applicability, and the proposed answers are 'high', 'low', or 'unclear' [110]. We resolved any disagreements by discussion, or through arbitration with a third review author (CM or MF) if necessary.

AC and GC were co-authors of the Colecchia 2012 and were thus not involved in the risk of bias assessment of that particular study.

The signalling questions addressed to our review subject, along with the criteria used to assess the methodological quality of the included studies, are provided in Supplementary material 6. We considered blinding to be relevant for both the index tests and the reference standard, as elastography and HVPG are operator-dependent – not only in obtaining the numerical results, but also in determining whether the results are valid. Therefore, knowledge of either test's results could bias the outcome. We considered an appropriate time interval being one week between the index test and the reference standard test, or between different index tests.

If all domains relating to bias and applicability were judged as 'low', then we considered the overall judgement of that particular study to be 'at low risk of bias' and 'low concern regarding applicability'. If at least one domain was judged as 'high' or 'unclear', then we considered the overall judgement to be 'at risk of bias' or as having 'concerns regarding applicability', or both. Guidance to assess the risk of bias and applicability of the results is provided in Supplementary material 6.

We provided the judgements regarding risk of bias and applicability in the 'Characteristics of included studies' table and graphically per domain for each included study (see Supplementary material 2, Figure 2, and Figure 3).

Figure 2. Risk of bias and applicability concerns graph: review authors' judgements by each domain for each study for both risk of bias and applicability concerns.

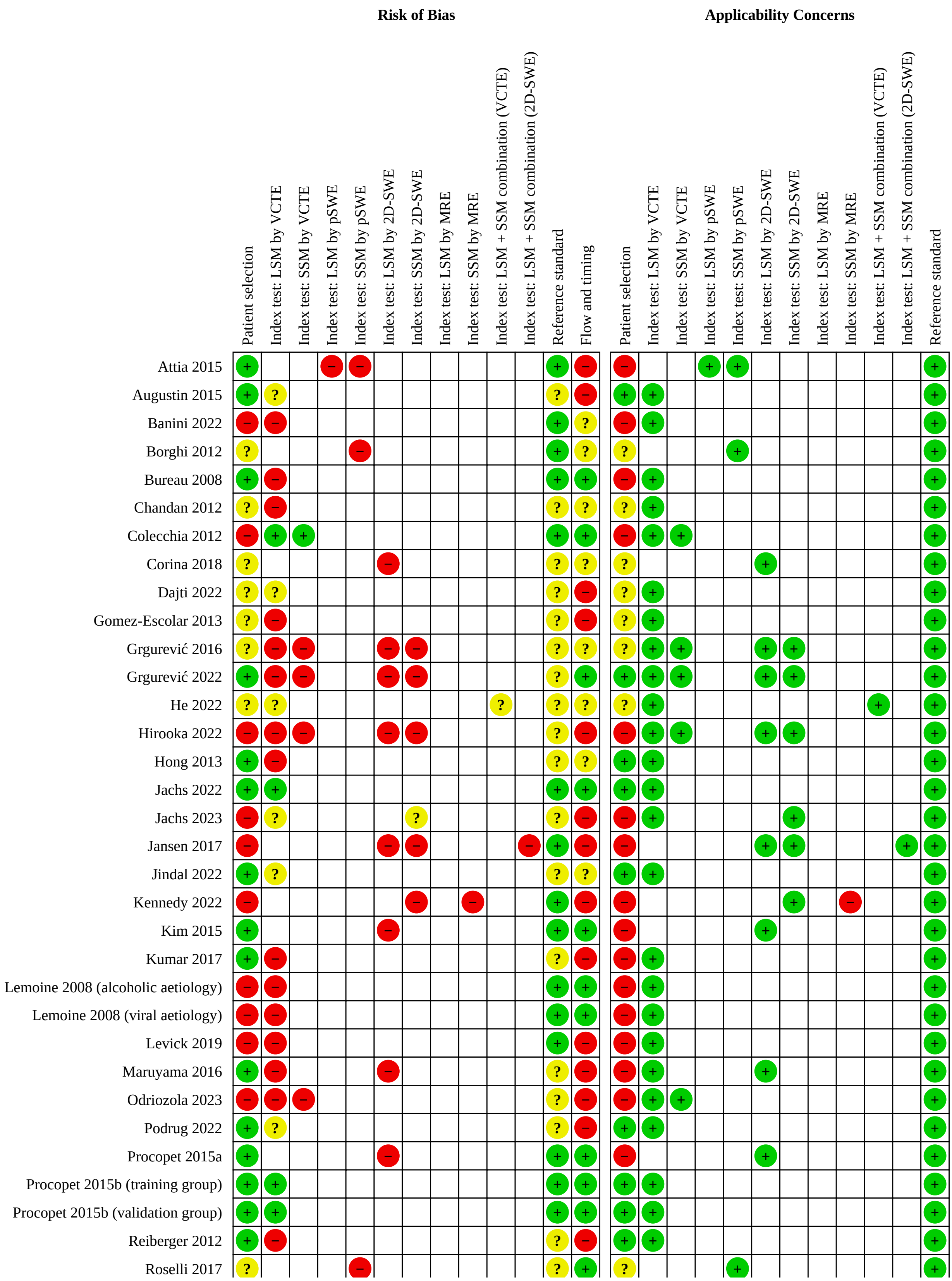


Figure 2. (Continued)

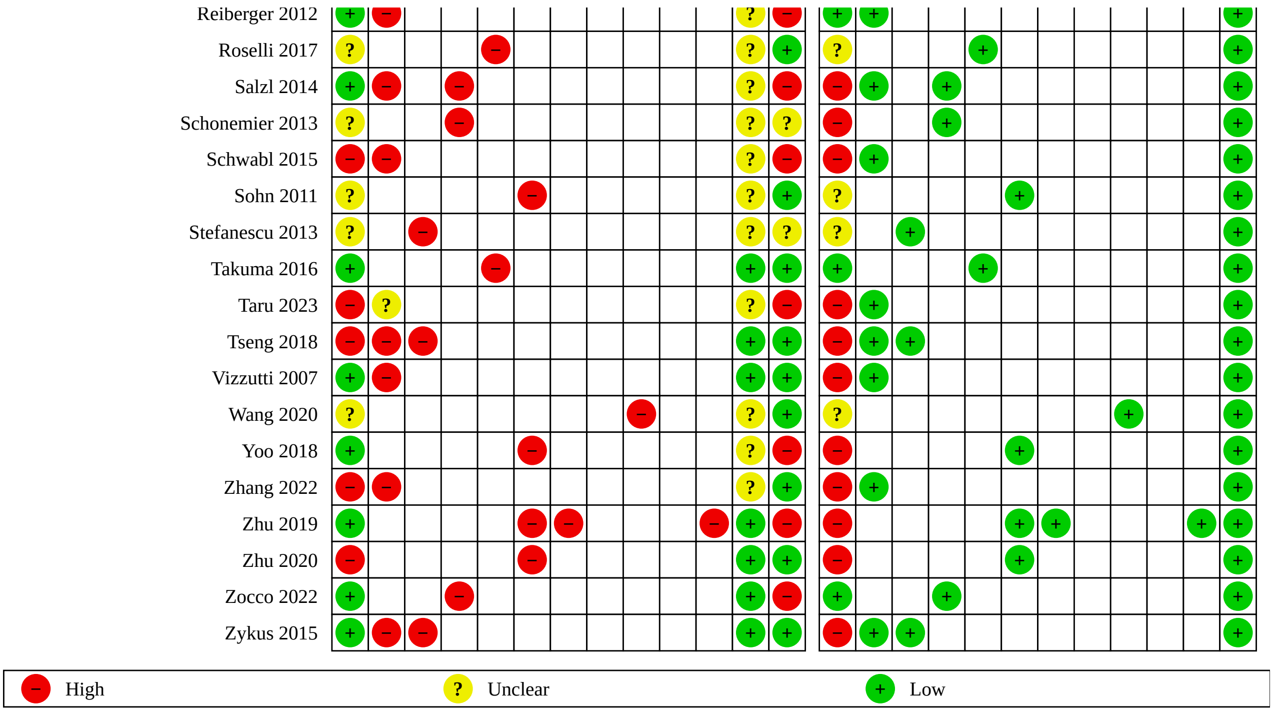
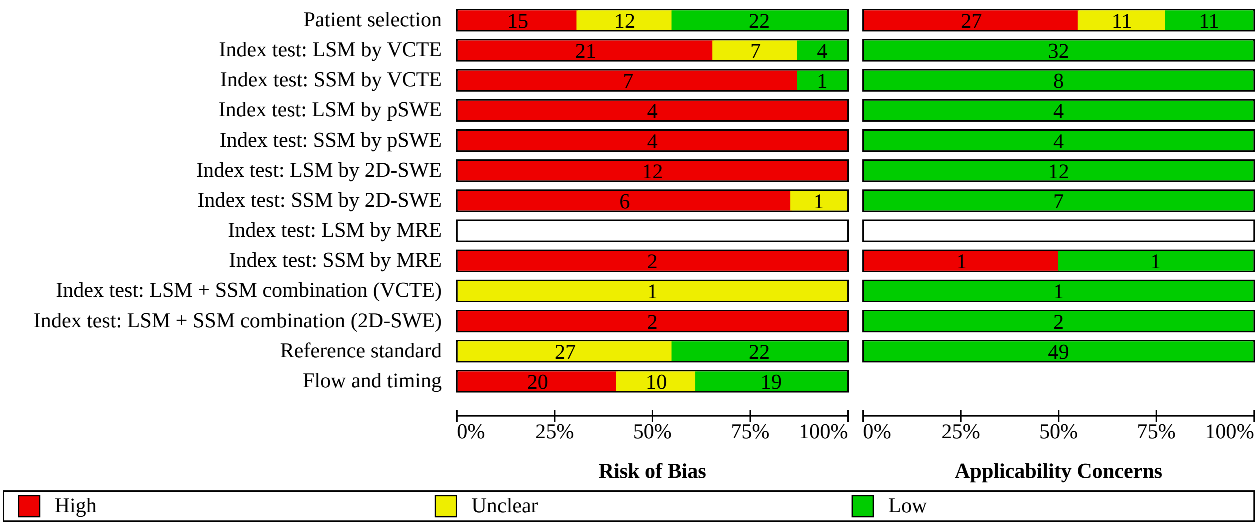


Figure 3. Risk of bias and applicability concerns graph: review authors' judgements about each domain presented as percentages across included studies.



Statistical analysis and data synthesis

We carried out statistical analyses according to the recommendations provided in the *Cochrane Handbook for Systematic Reviews of Diagnostic Test Accuracy* [117]. We designed 2 x 2 tables (see Data extraction and management) for each primary study for the index tests. We undertook the following strategy of analysis.

Index tests

1. Liver stiffness and spleen stiffness as measured by VCTE
2. Liver stiffness and spleen stiffness as measured by 2D-SWE
3. Liver stiffness and spleen stiffness as measured by pSWE
4. Liver stiffness and spleen stiffness as measured by MRE

Target conditions: CSPH and SPH.

We considered liver stiffness and spleen stiffness, measured by VCTE, 2D-SWE, pSWE, and MRE, to be positive when they exceed a defined cut-off (threshold) value. Firstly, we performed a graphical descriptive analysis of the included studies. We reported forest plots (sensitivity and specificity separately, with their 95% confidence intervals (CIs)), and provided a graphical representation of studies in the receiver operating characteristic (ROC) space (sensitivity against 1 − specificity). Secondly, we performed a meta-analysis. In the case that primary studies reported accuracy estimates of liver and spleen stiffness using different cut-off values, we used the hierarchical summary ROC model (HSROC) to pool data (sensitivities and specificities) and to estimate a summary ROC (SROC) curve [117]. In the case of studies reporting more than one cut-off value, we extracted the results associated with the most commonly used cut-off. We reported results using graphical representations (plotting HSROC curves) and reported combinations of sensitivity and specificity at different points on the curves. We selected four points on the HSROC curve according to the following strategy: we obtained specificity values corresponding to the two points at a prespecified high sensitivity (sensitivity 80% and 90%); we obtained sensitivity values corresponding to the two points at a prespecified high specificity (specificity 80% and 90%). When considering studies with a common cut-off value, we used the bivariate model. We provided estimates of summary sensitivity and specificity corresponding to a common cut-off value, when at least three studies were identified with a common cut-off value. We used the pooled estimates obtained from the fitted models to calculate summary estimates of positive and negative likelihood ratios (LR+ and LR-, respectively). As we identified a wide variation of cut-off values among the included studies assessing the different index tests, we grouped these cut-offs into the narrowest possible ranges. We then applied the bivariate model to provide estimates of summary sensitivity and specificity, provided that at least three studies were identified within a given cut-off range.

For comparisons between tests, we conducted a direct comparison by considering only the studies that compared index tests in the same participants. We performed pair-wise comparisons between the tests by adding a covariate for the index test to the bivariate or HSROC model, depending on the chosen model. We assessed differences in test accuracy by using the log-likelihood ratio test for comparison of models with and without the index test covariate term. We calculated the relative sensitivity (i.e. ratio between the sensitivities of the two tests) and the relative specificity (i.e. ratio between the two specificities). In the case of insufficient data for direct comparisons, we performed indirect comparisons by including in the analysis all the studies that assessed at least one of the considered tests. However, we are aware of the bias that arises from indirect comparison of diagnostic test accuracies, and we highlighted this as a limitation in the Discussion section of the review [123].

We performed all the statistical analyses using SAS statistical software, and macro METADAS [117, 124].

Investigations of heterogeneity

Potential sources of heterogeneity are listed in Secondary objectives.

We estimated the effects by adding covariates to the bivariate or HSROC model, depending on the model chosen (Statistical analysis and data synthesis). We assessed the statistical significance of the covariates on sensitivity and specificity by using the log-likelihood ratio test for comparison of models with and without the covariate term. We considered P values of less than 0.05, two-sided, as statistically significant.

Unlike the original protocol, and based on the available data, we decided to investigate heterogeneity only in studies that used a common predefined cut-off value (Supplementary material 7).

Sensitivity analyses

We performed sensitivity analysis by excluding studies judged to be at high risk of bias, specifically those where at least one domain of QUADAS-2 was deemed at high risk of bias (Supplementary material 6). In addition, we defined the questions below as the most relevant, and conducted a sensitivity analysis by excluding studies with answers 'no' or 'unclear'.

- Were the results of the index tests interpreted without knowledge of the reference standard results, in the case of the reference standard being done before the index test?
- Was there an appropriate time interval between the index test and the reference standard?

Moreover, we performed sensitivity analysis by excluding studies published only in abstract or letter form.

We decided to perform sensitivity analyses only in studies that used a common predefined cut-off value. Moreover, we decided to perform a sensitivity analysis by excluding studies that included up to 20% of participants that met exclusion criteria (hepatocellular carcinoma, cholestatic liver disease, history of NSBB use, and previously known CSPH/SPH or oesophageal varices) or did not report enough data. These criteria were not originally planned in the protocol and were added post hoc (Supplementary material 7).

Assessment of reporting bias

We did not plan to test for publication bias due to the lack of validated methods for diagnostic test accuracy reviews.

Summary of findings

We prepared a summary of findings table to present the main results and key information regarding the certainty of evidence assessed using the GRADE approach [125, 126, 127, 128]. As recommended, we rated the certainty of evidence as high (not downgraded), moderate (downgraded by one level), low (downgraded by two levels), or very low (downgraded by more than two levels) based on five GRADE domains: risk of bias, indirectness, inconsistency, imprecision, and publication bias. For sensitivity and specificity, the certainty of evidence starts as high when there are high-quality observational studies (cross-sectional cohort type accuracy studies) that enrolled participants with diagnostic uncertainty. When there was a reason for downgrading, we used our judgement to classify the reason as either serious (downgraded by one level) or very serious (downgraded by two levels) and recorded this information in the footnotes. We applied the GRADE judgements for the GRADE domains as follows.

1. Risk of bias: we used QUADAS-2 to assess risk of bias.
2. Indirectness: we used QUADAS-2 for concerns of applicability and looked for important differences between the populations

studied (e.g. the spectrum of disease), the setting, and the index test.

3. Inconsistency: we carried out prespecified analyses to investigate potential sources of heterogeneity and downgraded when we could not explain inconsistency in the accuracy estimates.
4. Imprecision: we looked at the CIs of sensitivity and specificity estimates and at the unexplained heterogeneity of the results.
5. Publication bias: we did not evaluate publication bias due to the lack of validated methods for diagnostic test accuracy reviews.

AC and GC were co-authors of the Colecchia 2012 study and were thus not involved in certainty of evidence assessment using the GRADE approach of that particular study.

RESULTS

Results of the search

We identified 8768 records overall by searching the following databases on 8 April 2024: CHBG Controlled Trials Register (15 records), CHBG Diagnostic Test of Accuracy Studies Register (19 records), CENTRAL (2024, Issue 3) in the Cochrane Library (271 records), MEDLINE Ovid (1417 records), Embase Ovid (5167 records), LILACS (VHL Regional Portal) (5 records), and Science Citation Index Expanded and Conference Proceedings Citation Index – Science (Web of Science) (1874 records). The search of all other sources excluding database searches took place between May 2023 and May 2024. We identified no additional studies by searching other sources. However, we e-mailed the corresponding authors of 46 potential eligible studies to retrieve additional data; the results are discussed below. After the exclusion of 2347 duplicates, 6421 records remained for eligibility screening. We excluded 6122 records based on title and abstract screening. Thus, for the full-text analysis, 299 records remained to assess for eligibility. We excluded 252 studies for various reasons (see Supplementary material 3). Hence, we included 47 studies (overall 49 cohorts) for the final analysis (Figure 1), including 7817 participants (Attia 2015 [129]; Augustin 2015 [130]; Banini 2022 [131]; Borghi 2012 [132]; Bureau 2008 [133]; Chandan 2012 [134]; Colecchia 2012; Corina 2018 [135]; Dajti 2022 [136]; Gomez-Escolar 2013 [137]; Grgurević 2016 [138]; Grgurević 2022 [139]; He 2022 [140]; Hirooka 2022 [141]; Hong 2013 [142]; Jachs 2022 [143]; Jachs 2023 [144]; Jansen 2017 [145]; Jindal 2022 [146]; Kennedy 2022 [147]; Kim 2015 [148]; Kumar 2017 [149]; Lemoine 2008 (alcoholic aetiology) [150]; Lemoine 2008 (viral aetiology) [151]; Levick 2019 [152]; Maruyama 2016 [153]; Odriozola 2023 [154]; Podrug 2022 [155]; Procopet 2015a [156]; Procopet 2015b (training group) [157]; Procopet 2015b (validation group) [158]; Reiberger 2012 [159]; Roselli 2017 [160]; Salzl 2014 [161]; Schonemier 2013 [162]; Schwabl 2015 [163]; Sohn 2011 [164]; Stefanescu 2013 [165]; Takuma 2016 [166]; Taru 2023 [167]; Tseng 2018 [168]; Vizzutti 2007 [169]; Wang 2020 [170]; Yoo 2018 [171]; Zhang 2022 [172]; Zhu 2019 [173]; Zhu 2020 [174]; Zocco 2022 [175]; Zykus 2015 [176]).

We applied no language restrictions in our searches, which resulted in the retrieval of full-text articles of three studies published in non-English languages (Liusina 2017 [177]; Perez 2019 [178]; Zhu 2020), of which one, fulfilling the inclusion criteria, was included in the review (Zhu 2020). We obtained the full-text translations through Google Chrome Engine.

We e-mailed the corresponding authors of 46 studies either because data could not be extracted due to missing 2 x 2 tables, or to request additional information. If no response was received within one month, we followed up with a second request. We did not receive replies from the authors of 34 studies from whom we had requested 2 x 2 table data (Baik 2014 [179]; Benmassaoud 2021 [180]; Bota 2015 [181]; Buck 2014 [182]; Chang 2019 [183]; Cho 2013 [184]; De Filippi 2011 [185]; Hametner 2016 [186]; Han X 2024 [187]; Herran 2022 [188]; Hirooka 2011 [189]; Hirooka 2012 [190]; Iacob 2019 [191]; Ilan 2017 [192]; Jachs 2023 (3) [193]; Kedarisetty 2014 [194]; Liu C 2021 [195]; Liu F 2018 [196]; Ortiz 2020 [197]; Ozkaya 2023 [198]; Park SH 2009 [199]; Park SY 2012 [200]; Radu 2018 [201]; Reiberger 2010 [202]; Ryu 2021 [203]; Sav 2018 [204]; Semmler 2024 [205]; Sharma 2013 [206]; Simbrunner 2020 [207]; Stefanescu 2015 [208]; Vonghia 2015 [209]; Vuppalanchi 2017 [210]; Wang 2023 [211]; Zhu 2020 (2) [212]). The corresponding authors of Berzigotti 2009 [213] and Berzigotti 2013 [214] reported that they no longer have the requested data. The authors of Sperl 2019 [215] provided data, but the study did not fulfil the inclusion criteria. We requested additional data for several studies that fulfilled the inclusion criteria (Colecchia 2012; Grgurević 2016; Grgurević 2022; Procopet 2015a; Procopet 2015b (training group); Procopet 2015b (validation group); Reiberger 2012; Roselli 2017; Wang 2020; Zocco 2022). Of these, only the authors of Colecchia 2012 and Zocco 2022 provided the requested data. The authors of Reiberger 2012 indicated that they no longer have the requested data.

We reported the main characteristics of the 47 studies in the 'Overview of Syntheses and Included Studies' table (Table 3) and 'Characteristics of included studies' table (Supplementary material 2). Eleven studies were published only as abstracts (Borghi 2012; Chandan 2012; Corina 2018; Gomez-Escolar 2013; Grgurević 2016; Roselli 2017; Schonemier 2013; Sohn 2011; Taru 2023; Wang 2020; Yoo 2018), and one study was a letter (Stefanescu 2013).

The included studies were conducted from 2007 to 2023.

No studies are awaiting classification.

The reference flow is shown in Figure 1.

Methodological quality of included studies

We reported the results of the quality assessment of the included studies in detail in the 'Characteristics of included studies' table (Supplementary material 2), and summarised the results in Figure 2 and Figure 3. As stated in Assessment of methodological quality, we used the QUADAS-2 tool for assessment of the methodological quality of studies, and the QUADAS-C tool for studies that compared two or more index tests. The results of the comparative diagnostic accuracy study methodology assessment are summarised in Figure 4.

Figure 4. QUADAS-C methodology assessment of studies with multiple index tests; the results of comparative diagnostic accuracy study methodology assessment by each study and each domain. PS = patient selection; IT = index tests; RS = reference standard; FT = flow and timing

Author	Year	C PS 1.1	C PS 1.2	C PS 1.3	C PS 1.4	C PS 1.5	C IT 2.1	C IT 2.2	C IT 2.3	C IT 2.4	C IT 2.5	C RS 3.1	C RS 3.2	C RS 3.3	C FT 4.1	C FT 4.2	C FT 4.3	C FT 4.4	C FT 4.5
Attia	2015	yes	yes	not applicable	not applicable	low risk	yes	no	not applicable	yes	high risk	yes	yes	low risk	no	yes	yes	no	high risk
Colecchia	2012	no	yes	not applicable	not applicable	high risk	no	yes	not applicable	yes	high risk	yes	yes	low risk	no	yes	yes	yes	high risk
Grgurević (1)	2016	no	yes	not applicable	not applicable	high risk	no	unclear	not applicable	yes	high risk	no	yes	high risk	no	yes	yes	unclear	high risk
Grgurević (2)	2022	yes	yes	not applicable	not applicable	low risk	no	unclear	not applicable	yes	high risk	no	yes	high risk	yes	yes	yes	yes	low risk
He	2022	no	yes	not applicable	not applicable	high risk	no	unclear	not applicable	yes	high risk	no	yes	high risk	no	yes	yes	unclear	high risk
Hirooka	2022	no	yes	not applicable	not applicable	high risk	no	no	not applicable	yes	high risk	no	yes	high risk	no	yes	yes	no	high risk
Jachs	2023	no	yes	not applicable	not applicable	high risk	no	yes	not applicable	yes	high risk	no	yes	high risk	no	yes	yes	yes	high risk
Jansen	2017	no	yes	not applicable	not applicable	high risk	no	unclear	not applicable	yes	high risk	no	yes	high risk	no	yes	no	no	high risk
Kennedy	2022	no	yes	not applicable	not applicable	high risk	no	unclear	not applicable	yes	high risk	yes	yes	low risk	no	yes	yes	no	high risk
Maruyama	2016	yes	yes	not applicable	not applicable	low risk	no	no	not applicable	yes	high risk	no	yes	high risk	no	yes	yes	no	high risk
Odriozola	2023	no	yes	not applicable	not applicable	high risk	no	unclear	not applicable	yes	high risk	no	yes	high risk	no	yes	yes	no	high risk
Salzl	2014	yes	yes	not applicable	not applicable	low risk	no	unclear	not applicable	yes	high risk	no	yes	high risk	no	yes	yes	no	high risk
Tseng	2018	no	yes	not applicable	not applicable	high risk	no	unclear	not applicable	yes	high risk	yes	yes	low risk	yes	yes	yes	yes	low risk
Zhu	2019	yes	yes	not applicable	not applicable	low risk	no	unclear	not applicable	yes	high risk	yes	yes	low risk	no	yes	yes	no	high risk
Zykus	2015	yes	yes	not applicable	not applicable	low risk	yes	unclear	not applicable	yes	high risk	yes	yes	low risk	yes	yes	yes	yes	low risk

QUADAS-2 methodology quality assessment

Patient selection domain

Risk of bias

We judged 21 studies (including 22 cohorts) as low risk for the patient selection domain (Attia 2015; Augustin 2015; Bureau 2008; Grgurević 2022; Hong 2013; Jachs 2022; Jindal 2022; Kim 2015; Kumar 2017; Maruyama 2016; Podrug 2022; Procopet 2015a; Procopet 2015b (training group); Procopet 2015b (validation group); Reiberger 2012; Salzl 2014; Takuma 2016; Vizzutti 2007; Yoo 2018; Zhu 2019; Zocco 2022; Zykus 2015), and 12 studies as unclear due to insufficient information provided (Borghi 2012; Chandan 2012; Corina 2018; Dajti 2022; Gomez-Escolar 2013; Grgurević 2016; He 2022; Roselli 2017; Schonemier 2013; Sohn 2011; Stefanescu 2013; Wang 2020). The remaining 14 studies (including 15 cohorts) were at high risk for this domain, mainly because of inappropriate inclusion/exclusion criteria (Banini 2022; Colecchia 2012; Hirooka 2022; Jachs 2023; Jansen 2017; Kennedy 2022; Lemoine 2008 (alcoholic aetiology); Lemoine 2008 (viral aetiology); Levick 2019; Odriozola 2023; Schwabl 2015; Taru 2023; Tseng 2018; Zhang 2022; Zhu 2020). In a change from the protocol, we included studies enrolling less than 20% of participants with hepatocellular carcinoma, cholestatic liver disease, history of NSBB use (not current), and previously known CSPH/SPH or oesophageal varices to avoid missing data from large studies with eligible participants. We judged these studies at high risk of bias for this domain because the inclusion of participants with known portal hypertension or prior treatment could affect HVPG measurements or disease progression, potentially biasing the reference standard. Furthermore, in conditions such as hepatocellular carcinoma and cholestatic liver disease, HVPG is known to be less reliable in reflecting portal pressure, which could compromise the accuracy assessment of the index tests. We decided to perform sensitivity analysis by excluding such studies and studies that did not report sufficient data regarding population characteristics (Supplementary material 7).

Applicability

We judged 10 studies (including 11 cohorts) at low concern regarding applicability (Augustin 2015; Grgurević 2022; Hong 2013; Jachs 2022; Jindal 2022; Podrug 2022; Procopet 2015b (training group); Procopet 2015b (validation group); Reiberger 2012; Takuma 2016; Zocco 2022). We judged 11 studies at unclear concern regarding applicability due to insufficient information (Borghi 2012; Chandan 2012; Corina 2018; Dajti 2022; Gomez-Escolar 2013; Grgurević 2016; He 2022; Roselli 2017; Sohn 2011; Stefanescu 2013; Wang 2020). We judged 26 studies at high concern for applicability as they included predominately participants with decompensated liver cirrhosis (Attia 2015; Hirooka 2022; Jansen 2017; Kennedy 2022; Kim 2015; Kumar 2017; Lemoine 2008 (alcoholic aetiology); Lemoine 2008 (viral aetiology); Levick 2019; Maruyama 2016; Odriozola 2023; Procopet 2015a; Salzl 2014; Schwabl 2015; Taru 2023; Tseng 2018; Vizzutti 2007; Yoo 2018; Zhang 2022; Zhu 2019; Zhu 2020; Zykus 2015) or participants with single aetiology (Colecchia 2012; Jachs 2023; Lemoine 2008 (alcoholic aetiology); Lemoine 2008 (viral aetiology); Vizzutti 2007; Zhu 2019; Zhu 2020) or participants with hepatocellular carcinoma, cholestatic liver disease, history of NSBB use (not current), and previously known CSPH/SPH or oesophageal varices in less than 20% of total cohort (Banini 2022; Bureau 2008; Kennedy 2022; Odriozola 2023; Schonemier 2013; Tseng 2018; Zhang 2022).

Index test domain

Risk of bias

We judged only three studies (including four cohorts) at low risk of bias for the index test domain (Colecchia 2012; Jachs 2022; Procopet 2015b (training group); Procopet 2015b (validation group)). Seven studies were at unclear risk of bias because the authors did not report if the index tests were interpreted without the knowledge of the results of the reference standard (Augustin 2015; Dajti 2022; He 2022; Jachs 2023; Jindal 2022; Podrug 2022; Taru 2023). Although elastography provides numerical results, its acquisition and validation are operator-dependent. Factors such as the selection of valid measurements and exclusion of unreliable readings may introduce subjectivity; therefore, lack of blinding could still influence the reported results. Most of the studies were judged at high risk of bias due to the lack of predefined cut-off values (Attia 2015; Banini 2022; Borghi 2012; Bureau 2008; Chandan 2012; Colecchia 2012; Corina 2018; Gomez-Escolar 2013; Grgurević 2016; Grgurević 2022; Hirooka 2022; Hong 2013; Jansen 2017; Kennedy 2022; Kim 2015; Kumar 2017; Lemoine 2008 (alcoholic aetiology); Lemoine 2008 (viral aetiology); Levick 2019; Maruyama

2016; Odriozola 2023; Procopet 2015a; Reiberger 2012; Roselli 2017; Salzl 2014; Schonemier 2013; Schwabl 2015; Sohn 2011; Stefanescu 2013; Takuma 2016; Tseng 2018; Vizzutti 2007; Wang 2020; Yoo 2018; Zhang 2022; Zhu 2019; Zhu 2020; Zocco 2022; Zykus 2015). Only studies that assessed LSM by VCTE cut-off of 25 kPa had predefined cut-off criteria (Augustin 2015; Dajti 2022; He 2022; Jachs 2022; Jachs 2023; Jindal 2022; Podrug 2022; Procopet 2015b (training group); Procopet 2015b (validation group); Taru 2023).

Applicability

All studies except Kennedy 2022 were judged as having low concern regarding applicability for the index test domain. Kennedy 2022 is a pilot study assessing the clinical utility of MRE for CSPH/SPH using both 1.5T and 3T magnetic resonance systems and evaluating both 2D-MRE and 3D-MRE. The lack of standardisation in this study raises high concerns regarding applicability for this domain. However, they also evaluated the diagnostic accuracy of SSM by 2D-SWE, which was used in a standardised way, thus this index test is considered as low concern regarding applicability for the index test domain. In all other studies, already-validated and known devices were used in a standardised way for each test as recommended by guidelines [216].

Reference standard domain

Risk of bias

All included studies used the appropriate reference standard, HVPG, which was an inclusion criterion. Twenty-seven studies did not report whether investigators performing HVPG were blinded to the results of index tests (Augustin 2015; Chandan 2012; Corina 2018; Dajti 2022; Gomez-Escolar 2013; Grgurević 2016; Grgurević 2022; He 2022; Hirooka 2022; Hong 2013; Jachs 2023; Jindal 2022; Kumar 2017; Maruyama 2016; Odriozola 2023; Podrug 2022; Reiberger 2012; Roselli 2017; Salzl 2014; Schonemier 2013; Schwabl 2015; Sohn 2011; Stefanescu 2013; Taru 2023; Wang 2020; Yoo 2018; Zhang 2022). Although HVPG provides numerical results, its acquisition and interpretation are also operator-dependent. Decisions such as obtaining and interpreting pressure waveforms and selecting representative readings may introduce subjectivity; therefore, lack of blinding could still influence the obtained results. Other studies reported that investigators were blinded to the results of index tests and vice versa, and were thus judged at low risk of bias for the reference standard domain. All studies performed HVPG in a standardised way.

Applicability

We judged all studies at low concern regarding applicability for this domain.

Flow and timing domain

Risk of bias

Overall, we judged 17 studies (including 19 cohorts) at low risk of bias for the flow and timing domain (Bureau 2008; Colecchia 2012; Grgurević 2022; Jachs 2022; Kim 2015; Lemoine 2008 (alcoholic aetiology); Lemoine 2008 (viral aetiology); Procopet 2015a; Procopet 2015b (training group); Procopet 2015b (validation group); Roselli 2017; Sohn 2011; Takuma 2016; Tseng 2018; Vizzutti 2007; Wang 2020; Zhang 2022; Zhu 2020; Zykus 2015).

We judged 20 studies at high risk of bias for the flow and timing domain. Five studies had an inappropriate time interval (> 1 week) between the index test and reference standard (Dajti 2022; Jachs 2023; Odriozola 2023; Podrug 2022; Yoo 2018); 11 studies had missing data (Attia 2015; Gomez-Escolar 2013; Hirooka 2022; Kumar 2017; Levick 2019; Reiberger 2012; Salzl 2014; Schwabl 2015; Taru 2023; Zhu 2019; Zocco 2022); and four studies had both (Augustin 2015; Jansen 2017; Kennedy 2022; Maruyama 2016). In all studies, all participants received the same reference standard. Of those judged at high risk of bias, Gomez-Escolar 2013, Levick 2019, Salzl 2014, and Taru 2023 did not report the time interval between the index test and the reference standard.

We judged 10 studies at unclear risk of bias because they did not report the time interval between the index test and the reference standard (Banini 2022; Borghi 2012; Chandan 2012; Corina 2018; Grgurević 2016; He 2022; Hong 2013; Jindal 2022; Schonemier 2013; Stefanescu 2013). Moreover, Chandan 2012, Corina 2018, and He 2022 did not report sufficient data to conclude whether there were some missing data.

QUADAS-2 overall assessment

Overall, we judged only two studies, including three cohorts, at low risk of bias for all domains (Jachs 2022; Procopet 2015b (training group); Procopet 2015b (validation group)). We judged the other studies at high risk of bias for at least one domain. Most studies were deemed at high risk of bias for the index test domain, predominantly due to the lack of predefined cut-off values.

Regarding applicability, we judged 10 studies to have low concern for applicability for all three domains (Augustin 2015; Grgurević 2022; Hong 2013; Jachs 2022; Jindal 2022; Podrug 2022; Procopet 2015b (training group); Procopet 2015b (validation group); Reiberger 2012; Takuma 2016; Zocco 2022). All studies had low concern for applicability for the index test domain, except for Kennedy 2022. All studies had low concern for applicability for the reference standard domain. Applicability issues were identified primarily in the patient selection domain. This was due to the inclusion of participants not matching the clinical question, such as those with decompensated cirrhosis, single aetiology of CLD, hepatocellular carcinoma, cholestatic liver disease, history of NSBB use (not current), and previously known CSPH/SPH or oesophageal varices, although these conditions were present in less than 20% of the total cohort.

The overall assessment by each domain and by each index test is provided graphically in Figure 3.

QUADAS-C methodology quality assessment

We identified 15 comparative diagnostic accuracy studies that matched our inclusion/exclusion criteria (Attia 2015; Colecchia 2012; Grgurević 2016; Grgurević 2022; He 2022; Hirooka 2022; Jachs 2022; Jansen 2017; Kennedy 2022; Maruyama 2016; Odriozola 2023; Salzl 2014; Tseng 2018; Zhu 2019; Zykus 2015). The results of QUADAS-C methodology quality assessment are summarised in Figure 4.

Patient selection domain

We judged nine comparative diagnostic accuracy studies at high risk of bias for the patient selection domain (Colecchia 2012; Grgurević 2016; He 2022; Hirooka 2022; Jachs 2023; Jansen 2017; Kennedy 2022; Odriozola 2023; Tseng 2018). Since all included studies included participants in a non-randomised manner, questions 1.3 and 1.4 regarding allocation sequence in randomisation were not applicable (Figure 4). Moreover, all studies intended for all participants to receive all index tests evaluated in each study (question 1.2; Figure 4). Thus, those studies at high risk of bias for this domain are the ones that were judged at high risk or unclear risk of bias for this domain by QUADAS-2. The other six comparative studies were judged at low risk of bias for the patient selection domain (as in QUADAS-2 assessment).

Index test domain

We judged all 15 comparative diagnostic accuracy studies at high risk of bias for the index test domain, mainly because in most studies, excluding Colecchia 2012 and Jachs 2023, the results of index tests were not interpreted without the knowledge of the results of another index test(s), or no pertinent information was reported (question 2.2; Figure 4). Mostly, the same investigators performed multiple index tests. Moreover, almost all studies, excluding Attia 2015 and Zykus 2015, were judged at unclear or high risk for this domain in the QUADAS-2 assessment, which automatically defines these studies at high risk for this domain in the QUADAS-C assessment (question 2.1; Figure 4).

Reference standard domain

We judged six comparative studies at low risk of bias for the reference standard domain, as in the QUADAS-2 assessment (Attia 2015; Colecchia 2012; Kennedy 2022; Tseng 2018; Zhu 2019; Zykus 2015). The other nine studies deemed at unclear or high risk of bias for the reference standard domain at QUADAS-2 assessment were also judged as having a high risk of bias for this domain at QUADAS-C assessment.

Flow and timing domain

We judged three studies at low risk of bias for the flow and timing domain (Grgurević 2022; Tseng 2018; Zykus 2015). All other studies were deemed at high risk of bias for this domain by QUADAS-2 assessment. Additionally, most studies had disproportionate proportions and reasons for missing data across index tests (Attia 2015; Grgurević 2016; He 2022; Hirooka 2022; Jansen 2017; Kennedy 2022; Maruyama 2016; Odriozola 2023; Salzl 2014; Zhu 2019). In all studies, the interval between index tests was appropriate.

QUADAS-C overall assessment

None of the 15 comparative studies were judged at low risk of bias across all domains. The results of QUADAS-C methodology quality assessment are summarised in Figure 4.

Findings

The main results of individual index tests are shown in Summary of findings 1. The planned summary of findings table for the combination of tests could not be done due to insufficient data.

Descriptive analysis of the included studies

We included overall 47 studies encompassing 49 cohorts, as two studies each assessed two independent cohorts. Overall, 7817 participants were included. The prevalence of CSPH was 62.5% (interquartile range (IQR) 51% to 76.7%; reported in 43 studies), and of SPH 62.5% (IQR 54% to 71.7%; reported in 19 studies). Twenty-nine studies were conducted in Europe. Thirty-four studies were published as full-text articles. Regarding study design, all included studies were cross-sectional, according to the inclusion criteria. The median number of analysed participants per study was 76 (IQR 46 to 107; reported in all studies). The median age was 55 (IQR 52 to 59; reported in 39 studies). The proportion of males was 68.7% (IQR 63.8% to 72.8%; reported in 38 studies). The median proportion of participants with liver cirrhosis was 100% (IQR 69.4 to 100%; reported in 32 studies), hepatitis C virus (HCV) aetiology 16.2% (IQR 3.4% to 47.5%; reported in 19 studies), hepatitis B virus (HBV) aetiology 17.2% (IQR 1.4% to 67.9%; reported in 18 studies), alcoholic aetiology 18.8% (IQR 6% to 36.8%; reported in 34 studies), non-alcoholic fatty liver disease (NAFLD) aetiology 17% (IQR 0% to 31.4%; reported in 20 studies). The median Model For End-Stage Liver Disease (MELD) score was 9 (IQR 8 to 10; reported in 14 studies); the median of participants with compensated cirrhosis as Child-Pugh score A was 68% (IQR 59.9% to 97.6%; reported in 16 studies); and the median of participants with decompensated cirrhosis as Child-Pugh score B or C was 31.2% (IQR 2.4% to 37.5%; reported in 16 studies).

Liver stiffness measurement (LSM) by vibration-controlled transient elastography (VCTE) for clinically significant portal hypertension (CSPH)

Description of included studies

We identified 27 studies with 3818 participants that evaluated the diagnostic accuracy of LSM by VCTE for CSPH (Augustin 2015; Banini 2022; Bureau 2008; Colecchia 2012; Dajti 2022; Gomez-Escolar 2013; Grgurević 2016; Grgurević 2022; He 2022; Hong 2013; Jachs 2022; Jachs 2023; Jindal 2022; Kumar 2017; Lemoine 2008 (alcoholic aetiology); Lemoine 2008 (viral aetiology); Levick 2019; Maruyama 2016; Odriozola 2023; Podrug 2022; Procopet 2015b (training group); Procopet 2015b (validation group); Reiberger 2012; Salzl 2014; Schwabl 2015; Taru 2023; Vizzutti 2007; Zhang 2022; Zykus 2015). The cut-off values ranged from 8.5 kPa to 34.9 kPa, the most evaluated cut-off being 25 kPa (nine studies). Only 11 studies reported the frequency of invaluable or indeterminate results due to technical failure; the failure rate ranged from 0% to 28.4% (median 5.9%) (Bureau 2008; Colecchia 2012; Grgurević 2022; Kumar 2017; Maruyama 2016; Odriozola 2023; Podrug 2022; Procopet 2015b (training group); Salzl 2014; Schwabl 2015; Zykus 2015).

Meta-analysed results

Figure 5 shows a forest plot of sensitivity and specificity with their 95% CIs, and a graphical representation of studies in the ROC space (sensitivity against 1 − specificity). We performed a meta-analysis using the HSROC, as the primary studies reported diagnostic accuracy estimates of LSM by VCTE using different cut-off values (Figure 6). In 11 studies, more than one cut-off value was evaluated (Augustin 2015; Banini 2022; Bureau 2008; Colecchia 2012; Grgurević 2022; Jachs 2023; Jansen 2017; Odriozola 2023; Procopet 2015a; Procopet 2015b (training group); Procopet 2015b (validation group); Zykus 2015). According to the SROC curve, the specificities corresponding to sensitivities of 80% and 90% were 86.4% (95% CI 80.1% to 90.9%) and 75.9% (95% CI 63.5% to 85.1%), respectively. On the other hand, sensitivities corresponding to specificities of 80% and 90% were 87.2% (95% CI 78.8% to 92.6%) and 72.6% (95% CI 60.0% to 82.5%), respectively. We deemed these estimates of specificity and sensitivity at a very low level of certainty, downgrading them by three levels due to risk of bias, inconsistency, and imprecision. We considered the wide inconsistency in the results to be primarily due to differences in the definition of cut-off values, which ranged from 8.5 kPa to 34.9 kPa.

Figure 5. Forest plot of sensitivity and specificity of liver stiffness measurement (LSM) by vibration-controlled transient elastography (VCTE) at any cut-off against hepatic venous pressure gradient (HVPG) reference standard > 10 mmHg (clinically significant portal hypertension; CSPH) in 27 studies ordered alphabetically. TP = true positive; FP = false positive; FN = false negative; TN = true negative. Values between brackets are the 95% confidence intervals (CIs) of sensitivity and specificity. The figure shows the estimated sensitivity and specificity of the study (blue square) and its 95% CI (black horizontal line).

Study	TP	FP	FN	TN	LS cut-off value (CSPH)	Sensitivity (95% CI)	Specificity (95% CI)
Augustin 2015	17	1	9	13	25	0.65 [0.44, 0.83]	0.93 [0.66, 1.00]
Banini 2022	9	33	7	93	21.3	0.56 [0.30, 0.80]	0.74 [0.65, 0.81]
Bureau 2008	68	5	8	69	21	0.89 [0.80, 0.95]	0.93 [0.85, 0.98]
Colecchia 2012	30	1	35	34	25	0.46 [0.34, 0.59]	0.97 [0.85, 1.00]
Dajti 2022	30	7	30	128	25	0.50 [0.37, 0.63]	0.95 [0.90, 0.98]
Gomez-Escolar 2013	40	1	4	6	17.15	0.91 [0.78, 0.97]	0.86 [0.42, 1.00]
Grgurević 2016	10	1	1	8	10.4	0.91 [0.59, 1.00]	0.89 [0.52, 1.00]
Grgurević 2022	31	5	9	31	20.2	0.78 [0.62, 0.89]	0.86 [0.71, 0.95]
He 2022	45	0	5	51	25	0.90 [0.78, 0.97]	1.00 [0.93, 1.00]
Hong 2013	30	6	6	17	21.95	0.83 [0.67, 0.94]	0.74 [0.52, 0.90]
Jachs 2022	119	11	69	103	25	0.63 [0.56, 0.70]	0.90 [0.83, 0.95]
Jachs 2023	19	3	13	16	25	0.59 [0.41, 0.76]	0.84 [0.60, 0.97]
Jindal 2022	258	30	222	116	25	0.54 [0.49, 0.58]	0.79 [0.72, 0.86]
Kumar 2017	219	16	59	32	21.6	0.79 [0.73, 0.83]	0.67 [0.52, 0.80]
Lemoine 2008 (alcoholic aetiology)	36	1	4	7	34.9	0.90 [0.76, 0.97]	0.88 [0.47, 1.00]
Lemoine 2008 (viral aetiology)	21	3	13	7	20.5	0.62 [0.44, 0.78]	0.70 [0.35, 0.93]
Levick 2019	3	0	1	9	28.9	0.75 [0.19, 0.99]	1.00 [0.66, 1.00]
Maruyama 2016	9	0	7	6	31.2	0.56 [0.30, 0.80]	1.00 [0.54, 1.00]
Odriozola 2023	13	2	16	53	27.6	0.45 [0.26, 0.64]	0.96 [0.87, 1.00]
Podrug 2022	27	4	13	32	25	0.68 [0.51, 0.81]	0.89 [0.74, 0.97]
Procopet 2015b (training group)	55	25	3	75	13.6	0.95 [0.86, 0.99]	0.75 [0.65, 0.83]
Procopet 2015b (validation group)	13	9	2	20	13.6	0.87 [0.60, 0.98]	0.69 [0.49, 0.85]
Reiberger 2012	227	38	49	188	18	0.82 [0.77, 0.87]	0.83 [0.78, 0.88]
Salzl 2014	38	4	4	13	16.8	0.90 [0.77, 0.97]	0.76 [0.50, 0.93]
Schwabl 2015	44	15	3	100	18.1	0.94 [0.82, 0.99]	0.87 [0.79, 0.93]
Taru 2023	14	0	11	37	24	0.56 [0.35, 0.76]	1.00 [0.91, 1.00]
Vizzutti 2007	46	1	1	13	13.6	0.98 [0.89, 1.00]	0.93 [0.66, 1.00]
Zhang 2022	78	9	29	21	14.1	0.73 [0.63, 0.81]	0.70 [0.51, 0.85]
Zykus 2015	69	4	9	25	17.4	0.88 [0.79, 0.95]	0.86 [0.68, 0.96]

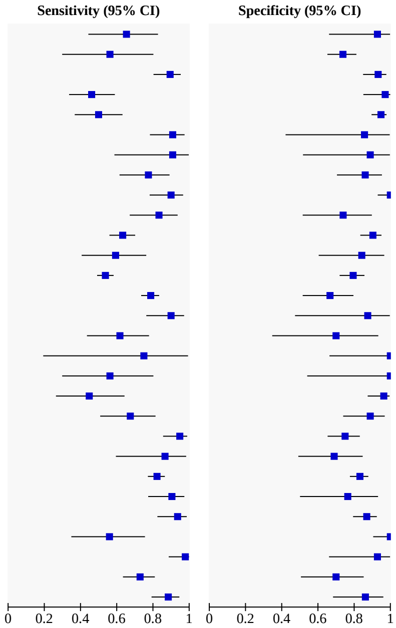
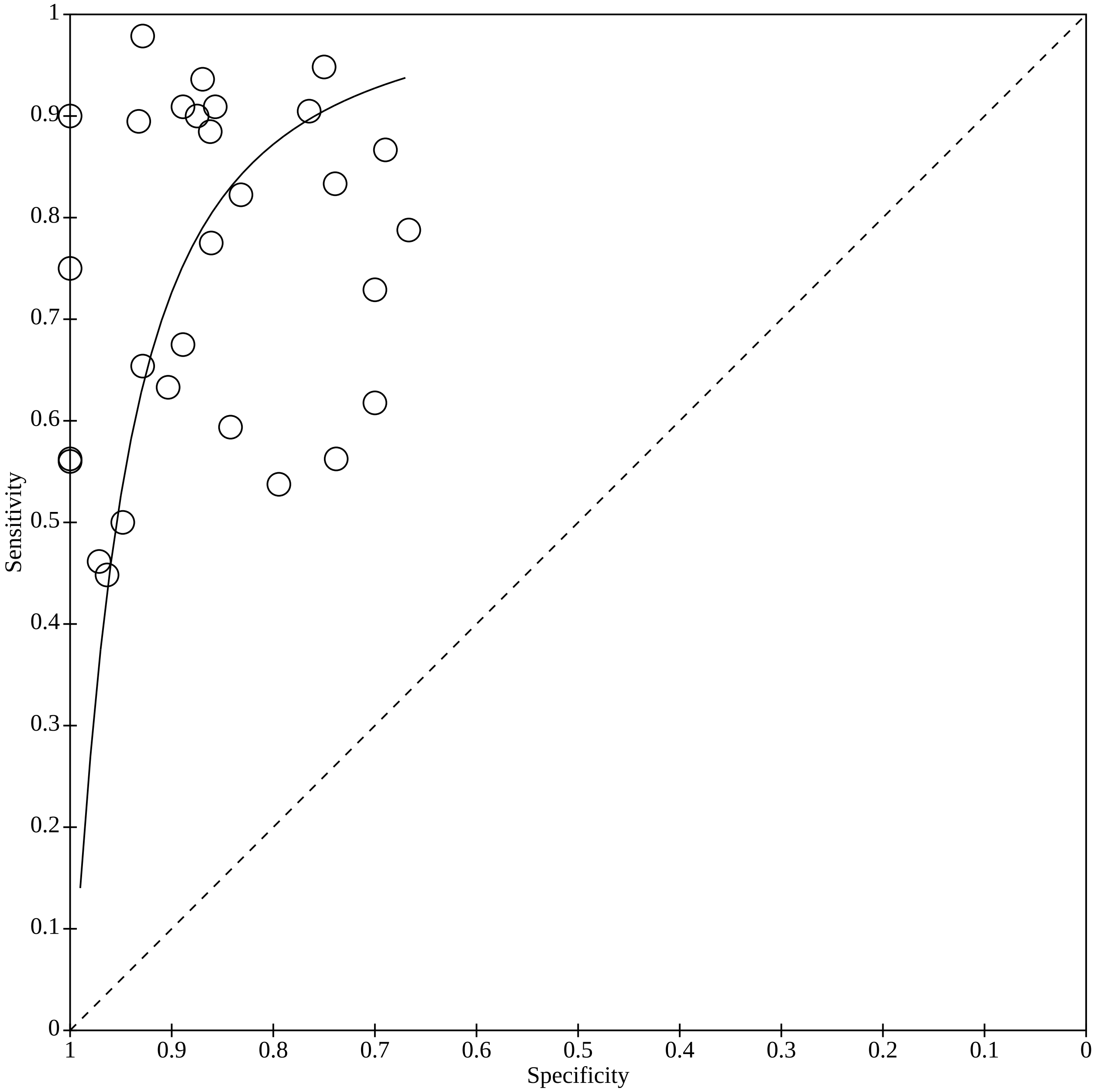


Figure 6. Summary receiver operating characteristic (ROC) comparing liver stiffness measurement (LSM) by vibration-controlled transient elastography (VCTE) with any cut-off value and hepatic venous pressure gradient (HVPG) reference standard > 10 mmHg (clinically significant portal hypertension; CSPH) in 27 studies.



Based on cut-off values used in the included studies, we performed a meta-analysis with investigations of heterogeneity and sensitivity analysis that included only studies that evaluated a cut-off of 25 kPa as the only predefined cut-off value.

LSM cut-off value of 25 kPa

Description of included studies

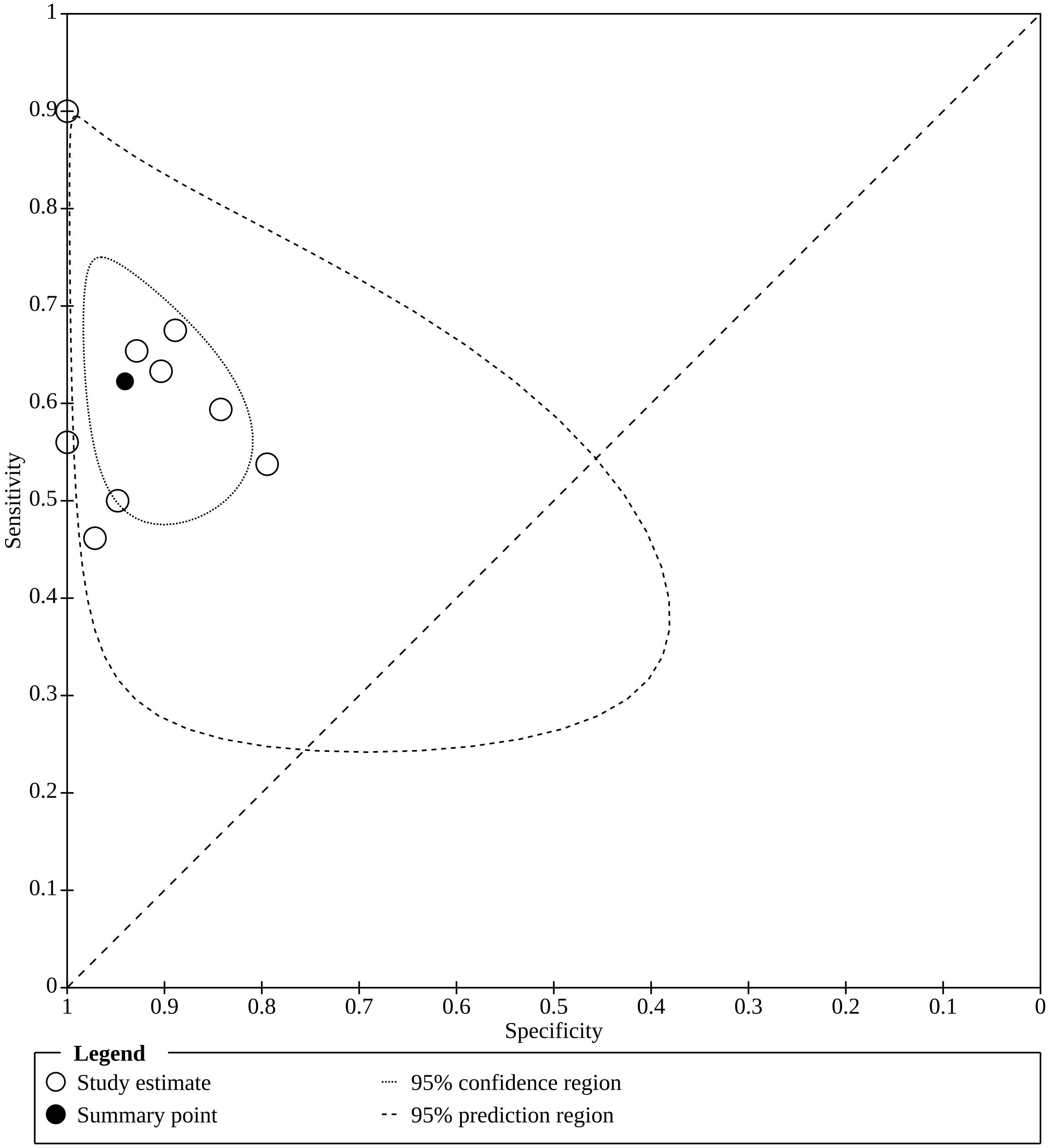
We identified nine studies with 1553 participants that evaluated the diagnostic accuracy of LSM by VCTE for CSPH with a cut-off value of 25 kPa (Augustin 2015; Colecchia 2012; Dajti 2022; He 2022; Jachs 2022; Jachs 2023; Jindal 2022; Podrug 2022; Taru 2023). The prevalence of CSPH was 62% (IQR 50% to 65%). Seven studies were conducted in Europe, and seven were published as full-text articles (Augustin 2015; Colecchia 2012; Dajti 2022; Jachs 2022; Jachs 2023; Jindal 2022; Podrug 2022); two studies were identified only as abstracts (He 2022; Taru 2023). The median number of analysed participants per study was 100 (IQR 62 to 195). The median age was 54 years (IQR 50.8 to 59; reported in 7 studies), and the proportion of males was 70% (IQR 68.7% to 74.3%; reported in 7 studies). The median proportion of participants with liver cirrhosis was 100% (IQR 100% to 100%; reported in 5 studies), with HCV aetiology 16.6% (IQR 11.8% to 100%; reported in 3 studies), HBV aetiology 10.2% (IQR 0% to 100%; reported in 3 studies), alcoholic aetiology

17.9% (IQR 15.2% to 30.3%; reported in 5 studies), NAFLD aetiology 26.4% (IQR 13.2% to 30.3%; reported in 5 studies); median MELD score was 9 (IQR 8 to 9; reported in 3 studies), median participants with compensated cirrhosis as Child-Pugh score A was 100% (IQR 68% to 100%; reported in 3 studies); and median participants with decompensated cirrhosis as Child-Pugh score B or C was 0% (IQR 0% to 32%; reported in 3 studies).

Meta-analysed results

By using the bivariate model, we obtained the following meta-analysed estimates in nine studies with 1553 participants (Figure 7): sensitivity 62.3% (95% CI 53.0% to 70.7%), specificity 94.1% (95% CI 87.2% to 97.3%), LR+ 10.48 (95% CI 4.43 to 24.78), LR- 0.4 (95% CI 0.31 to 0.52). These estimates from the bivariate model agree with the results from the HSROC model. We deemed these estimates at a very low level of certainty, downgrading them by three levels due to risk of bias, inconsistency, and imprecision.

Figure 7. Plot of studies in receiver operating characteristic (ROC) space comparing liver stiffness measurement (LSM) by vibration-controlled transient elastography (VCTE) with cut-off 25 kPa and hepatic venous pressure gradient (HVPG) reference standard > 10 mmHg (clinically significant portal hypertension; CSPH) in 9 studies.



Heterogeneity analysis

We investigated heterogeneity in studies assessing LSM cut-off value of 25 kPa. Table 4 shows comparisons across different predefined subgroups. Due to the low number of studies, we could only investigate heterogeneity in a subgroup analysis comparing studies with CSPH prevalence > 50% and those with less than 50%, and we found an overall effect on the prevalence. In particular, the sensitivity was 59.6% (49.2% to 69.2%) in studies with a prevalence higher than 50% and 68.1% (52.8% to 80.3%) in studies with a prevalence lower than 50% (relative sensitivity 0.87, 95% CI 0.67 to 1.14, P = 0.322); the specificity was 88.3% (79.8% to 93.6%) in studies with a prevalence higher than 50% and 98.2% (92.8%

to 99.6%) in studies with a prevalence lower than 50% (relative specificity 0.90, 95% CI 0.83 to 0.97, P = 0.009).

Sensitivity analysis

Sensitivity analysis considering studies judged at low risk of bias was precluded due to an insufficient number of studies (only one study, Jachs 2022). When considering seven studies published as full text, we obtained a sensitivity of 56.8% (95% CI 51.3% to 62.1%) and specificity of 90.3% (95% CI 84.4% to 94.2%; Table 4). When considering studies without any participants that met our exclusion criteria, we obtained a sensitivity of 55.2% (95% CI 49% to 61.3%) and specificity of 91.4% (95% CI 83.7% to 95.7%; Table 4). Sensitivity analysis considering studies that had an appropriate interval between the index test and the reference standard (less than one week) was precluded due to an insufficient number of studies (only two studies (Colecchia 2012; Jachs 2022)). Sensitivity analysis considering studies where index tests were interpreted without knowledge of the reference standard results was precluded due to an insufficient number of studies (only two studies (Colecchia 2012; Jachs 2022)).

Other cut-off values

Based on cut-off values used in the included studies, we performed meta-analysis including only studies that evaluated the most used cut-off values in ranges of 3 kPa:

- 6 studies (7 cohorts) reporting data with cut-off values ranging from 11.4 to 14.4 kPa;
- 9 studies reporting data with cut-off values ranging from 15 to 18 kPa;
- 11 studies reporting data with cut-off values ranging from 20 to 23 kPa;
- 12 studies reporting data with cut-off values ranging from 25 to 28 kPa.

The results of the bivariate analyses for these four subgroups are shown in Table 5.

LSM by VCTE for severe portal hypertension (SPH)

Description of included studies

We identified eight studies with 637 participants that evaluated the diagnostic accuracy of LSM by VCTE for SPH (Chandan 2012; Colecchia 2012; Hirooka 2022; Hong 2013; Maruyama 2016; Tseng 2018; Vizzutti 2007; Zykus 2015). The cut-off values ranged from 12.1 kPa to 35 kPa. Only four studies reported the frequency of invaluable or indeterminate results due to technical failure; the failure rate ranged from 1.8% to 12.0% (Colecchia 2012; Maruyama 2016; Tseng 2018; Zykus 2015).

Meta-analysed results

Figure 8 shows a forest plot of sensitivity and specificity with their 95% CIs, and Figure 9 shows a graphical representation of studies in the ROC space (sensitivity against 1 − specificity). We performed a meta-analysis using the HSROC model, as the primary studies reported diagnostic accuracy estimates of LSM by VCTE using different cut-off values (Figure 9). Two studies evaluated more than one cut-off value (Colecchia 2012; Zykus 2015). In such cases, we extracted the results obtained with the cut-off most frequently used. According to the SROC curve, specificity corresponding to sensitivity of 80% was 74.0% (95% CI 51.8% to 88.2%), while specificity corresponding to sensitivity of 90% could not be calculated by the HSROC model. On the other hand, sensitivities corresponding to specificities of 80% and 90% were 76.6% (95% CI 67.0% to 84.1%) and 67.2% (95% CI 48.8% to 81.4%), respectively. We deemed these estimates at a very low level of certainty, downgrading them by three levels due to risk of bias, inconsistency, and imprecision.

Figure 8. Forest plot of sensitivity and specificity of liver stiffness measurement (LSM) by vibration-controlled transient elastography (VCTE) at any cut-off against hepatic venous pressure gradient (HVPG) reference standard > 12 mmHg (severe portal hypertension; SPH) in 8 studies ordered alphabetically. TP = true positive; FP = false positive; FN = false negative; TN = true negative. Values between brackets are the 95% confidence intervals (CIs) of sensitivity and specificity. The figure shows the estimated sensitivity and specificity of the study (blue square) and its 95% CI (black horizontal line).

Study	TP	FP	FN	TN	LS cut-off value (SPH)	Sensitivity (95% CI)	Specificity (95% CI)
Chandan 2012	50	14	17	27	21.2	0.75 [0.63, 0.84]	0.66 [0.49, 0.80]
Colecchia 2012	30	1	24	45	25	0.56 [0.41, 0.69]	0.98 [0.88, 1.00]
Hirooka 2022	22	9	9	41	24.5	0.71 [0.52, 0.86]	0.82 [0.69, 0.91]
Hong 2013	26	8	5	20	24.25	0.84 [0.66, 0.95]	0.71 [0.51, 0.87]
Maruyama 2016	9	0	7	6	31.2	0.56 [0.30, 0.80]	1.00 [0.54, 1.00]
Tseng 2018	67	6	19	7	14	0.78 [0.68, 0.86]	0.54 [0.25, 0.81]
Vizzutti 2007	33	5	2	21	17.6	0.94 [0.81, 0.99]	0.81 [0.61, 0.93]
Zykus 2015	55	8	12	32	20.6	0.82 [0.71, 0.90]	0.80 [0.64, 0.91]

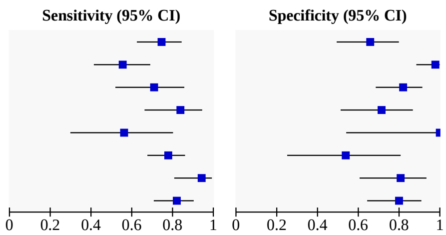
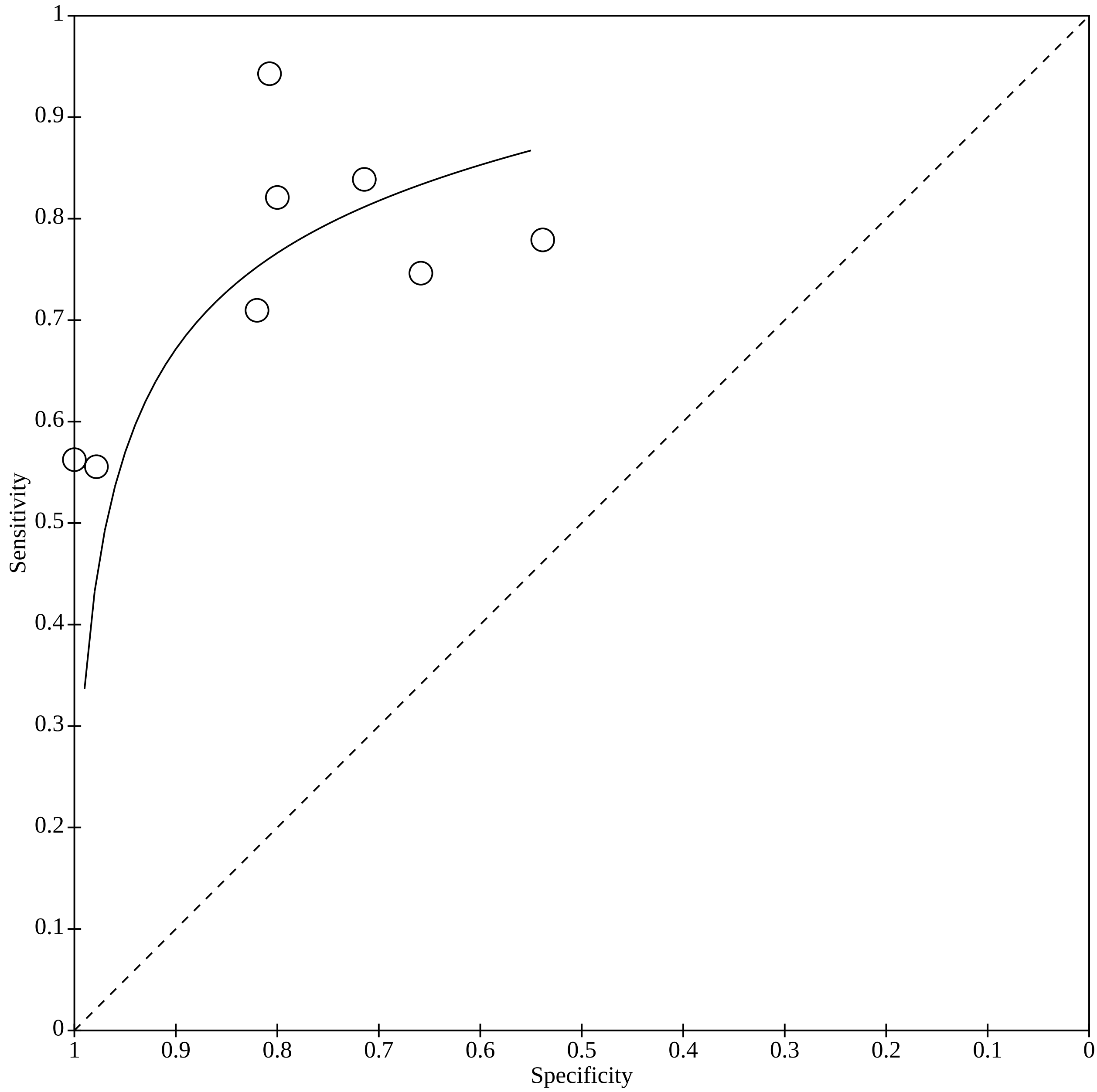


Figure 9. Summary receiver operating characteristic (ROC) comparing liver stiffness measurement (LSM) by vibration-controlled transient elastography (VCTE) with any cut-off value and hepatic venous pressure gradient (HVPG) reference standard > 12 mmHg (severe portal hypertension; SPH) in 8 studies.



Based on cut-off values used in the included studies, we grouped them into three groups based on a cut-off value range of 2 kPa (16 to 18 kPa and 20 to 22 kPa, and 24 to 26 kPa). The results of the bivariate analyses for these three subgroups are shown in Table 5.

Spleen stiffness measurement (SSM) by VCTE for CSPH

Description of included studies

We identified six studies with 391 participants that evaluated the diagnostic accuracy of SSM by VCTE for CSPH (Colecchia 2012; Grgurević 2016; Grgurević 2022; Odriozola 2023; Stefanescu 2013; Zykus 2015). The cut-off values ranged from 29.3 kPa to 52.8 kPa. Four studies reported the frequency of invaluable or indeterminate results due to technical failure; the failure rate ranged from 7.5% to 19.7% (Colecchia 2012; Grgurević 2022; Odriozola 2023; Zykus 2015).

Meta-analysed results

Figure 10 shows a forest plot of sensitivity and specificity with their 95% CIs, and Figure 11 shows a graphical representation of studies in the ROC space (sensitivity against 1 − specificity). We performed a meta-analysis using the HSROC model, as the primary studies reported diagnostic accuracy estimates of SSM by VCTE using different cut-off values (Figure 11). Three studies evaluated more than one cut-off value (Colecchia 2012; Grgurević 2022; Odriozola 2023). In such cases, we extracted the results obtained with the cut-off most frequently used. According to the SROC curve, the specificities corresponding to sensitivities of 80% and 90% were 87.5% (95% CI 72.7% to 94.8%) and 80.6% (95% CI 64.1% to 90.7%), respectively. On the other hand, sensitivities corresponding to 80% and 90% specificities were 90.6% (95% CI 70.9% to 97.4%) and 72.9% (95% CI 27.1% to 95.1%), respectively. We deemed these estimates at a very low level of certainty, downgrading them by three levels due to risk of bias, inconsistency, and imprecision.

Figure 10. Forest plot of sensitivity and specificity of spleen stiffness measurement (SSM) by vibration-controlled transient elastography (VCTE) at any cut-off against hepatic venous pressure gradient (HVPG) reference standard > 10 mmHg (clinically significant portal hypertension; CSPH) in 6 studies ordered alphabetically. TP = true positive; FP = false positive; FN = false negative; TN = true negative. Values between brackets are the 95% confidence intervals (CIs) of sensitivity and specificity. The figure shows the estimated sensitivity and specificity of the study (blue square) and its 95% CI (black horizontal line).

Study	TP	FP	FN	TN	SS cut-off value (CSPH)	Sensitivity (95% CI)	Specificity (95% CI)
Colecchia 2012	64	9	1	26	40	0.98 [0.92, 1.00]	0.74 [0.57, 0.88]
Grgurević 2016	7	0	4	9	41.8	0.64 [0.31, 0.89]	1.00 [0.66, 1.00]
Grgurević 2022	31	8	4	18	43.5	0.89 [0.73, 0.97]	0.69 [0.48, 0.86]
Odriozola 2023	20	2	9	43	49.9	0.69 [0.49, 0.85]	0.96 [0.85, 0.99]
Stefanescu 2013	13	4	1	19	42.7	0.93 [0.66, 1.00]	0.83 [0.61, 0.95]
Zykus 2015	59	5	17	18	47.6	0.78 [0.67, 0.86]	0.78 [0.56, 0.93]

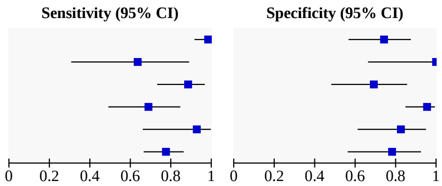
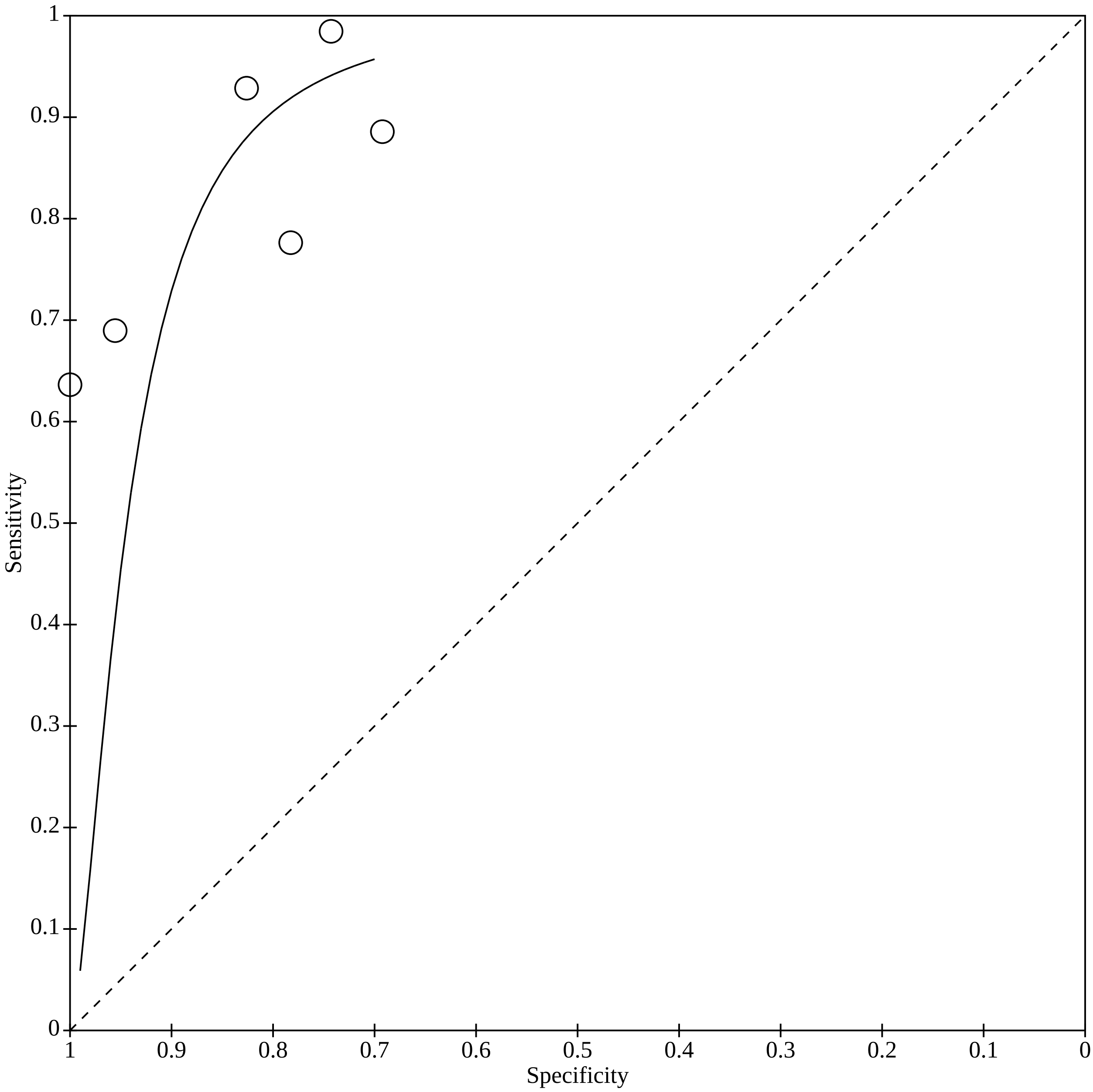


Figure 11. Summary receiver operating characteristic (ROC) comparing spleen stiffness measurement (SSM) by vibration-controlled transient elastography (VCTE) with any cut-off value and hepatic venous pressure gradient (HVPG) reference standard > 10 mmHg (clinically significant portal hypertension; CSPH) in 6 studies.



Based on cut-off values used in the included studies, we performed meta-analysis including only studies that evaluated the most used cut-off values in ranges of 4 kPa (40 to 44 kPa and 46 to 50 kPa). The results of the bivariate analyses for these two subgroups are shown in Table 5.

SSM by VCTE for SPH

Description of included studies

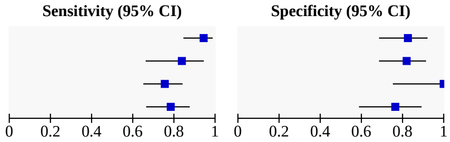
We identified four studies with 379 participants that evaluated the diagnostic accuracy of SSM by VCTE for SPH (Colecchia 2012; Hirooka 2022; Tseng 2018; Zykus 2015). The cut-off values ranged from 41.3 kPa to 55 kPa. Colecchia 2012 reported more than one cut-off value. Only three studies reported the frequency of invaluable or indeterminate results due to technical failure; the failure rate ranged from 7.5% to 11.5% (Colecchia 2012; Tseng 2018; Zykus 2015).

Meta-analysed results

Figure 12 shows a forest plot of sensitivity and specificity with their 95% CIs. Due to model convergence issues, we did not perform a meta-analysis using the HSROC model.

Figure 12. Forest plot of sensitivity and specificity of spleen stiffness measurement (SSM) by vibration-controlled transient elastography (VCTE) at any cut-off against hepatic venous pressure gradient (HVPG) reference standard > 12 mmHg (severe portal hypertension; SPH) in 4 studies ordered alphabetically. TP = true positive; FP = false positive; FN = false negative; TN = true negative. Values between brackets are the 95% confidence intervals (CIs) of sensitivity and specificity. The figure shows the estimated sensitivity and specificity of the study (blue square) and its 95% CI (black horizontal line).

Study	TP	FP	FN	TN	SS cut-off value (SPH)	Sensitivity (95% CI)	Specificity (95% CI)
Colecchia 2012	51	8	3	38	49	0.94 [0.85, 0.99]	0.83 [0.69, 0.92]
Hirooka 2022	26	9	5	41	47.2	0.84 [0.66, 0.95]	0.82 [0.69, 0.91]
Tseng 2018	65	0	21	13	48.9	0.76 [0.65, 0.84]	1.00 [0.75, 1.00]
Zykus 2015	51	8	14	26	50.7	0.78 [0.67, 0.88]	0.76 [0.59, 0.89]



LSM by point shear wave elastography (pSWE) for CSPH

Description of included studies

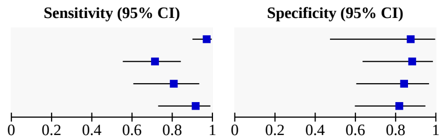
We identified four studies with 228 participants that evaluated the diagnostic accuracy of LSM by pSWE for CSPH (Attia 2015; Salzl 2014; Schonemier 2013; Zocco 2022). The cut-off values ranged from 2.17 to 2.58 m/s, and one study evaluated cut-off value in kilopascals (24.2 kPa) (Zocco 2022). Two studies reported the frequency of invaluable or indeterminate results due to technical failure; the failure rates ranged from 0% to 1% (Attia 2015; Salzl 2014).

Meta-analysed results

Figure 13 shows a forest plot of sensitivity and specificity with their 95% CIs. Due to model convergence issues, we did not perform a meta-analysis using the HSROC model.

Figure 13. Forest plot of sensitivity and specificity of liver stiffness measurement (LSM) by point shear wave elastography (pSWE) at any cut-off against hepatic venous pressure gradient (HVPG) reference standard > 10 mmHg (clinically significant portal hypertension; CSPH) in 4 studies ordered alphabetically. TP = true positive; FP = false positive; FN = false negative; TN = true negative. Values between brackets are the 95% confidence intervals (CIs) of sensitivity and specificity. The figure shows the estimated sensitivity and specificity of the study (blue square) and its 95% CI (black horizontal line).

Study	TP	FP	FN	TN	LS cut-off value (CSPH)	Sensitivity (95% CI)	Specificity (95% CI)
Attia 2015	68	1	2	7	2.17	0.97 [0.90, 1.00]	0.88 [0.47, 1.00]
Salzl 2014	30	2	12	15	2.58	0.71 [0.55, 0.84]	0.88 [0.64, 0.99]
Schonemier 2013	21	3	5	16	2.5	0.81 [0.61, 0.93]	0.84 [0.60, 0.97]
Zocco 2022	22	4	2	18	24.2	0.92 [0.73, 0.99]	0.82 [0.60, 0.95]



LSM by pSWE for SPH

Description of included studies

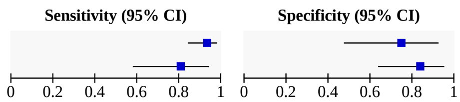
We identified two studies with 124 participants that evaluated the diagnostic accuracy of LSM by pSWE for SPH (Attia 2015; Zocco 2022). Attia 2015 evaluated the cut-off of 2.54 m/s, while Zocco 2022 evaluated the cut-off in kilopascals (25.7 kPa). One study reported no invaluable or indeterminate results due to technical failure (Attia 2015).

Meta-analysed results

Figure 14 shows a forest plot of sensitivity and specificity with their 95% CIs. We could not perform a meta-analysis using the HSROC model, as the primary studies were incomparable due to different units of measurement.

Figure 14. Forest plot of sensitivity and specificity of liver stiffness measurement (LSM) by point shear wave elastography (pSWE) at any cut-off against hepatic venous pressure gradient (HVPG) reference standard > 12 mmHg (severe portal hypertension; SPH) in 2 studies ordered alphabetically. TP = true positive; FP = false positive; FN = false negative; TN = true negative. Values between brackets are the 95% confidence intervals (CIs) of sensitivity and specificity. The figure shows the estimated sensitivity and specificity of the study (blue square) and its 95% CI (black horizontal line).

Study	TP	FP	FN	TN	LS cut-off value (SPH)	Sensitivity (95% CI)	Specificity (95% CI)
Attia 2015	58	4	4	12	2.54	0.94 [0.84, 0.98]	0.75 [0.48, 0.93]
Zocco 2022	17	4	4	21	25.7	0.81 [0.58, 0.95]	0.84 [0.64, 0.95]



SSM by pSWE for CSPH

Description of included studies

We identified four studies with 248 participants that evaluated the diagnostic accuracy of SSM by pSWE for CSPH (Attia 2015; Borghi 2012; Roselli 2017; Takuma 2016). The cut-off values ranged from 2.32 to 3.26 m/s, while Roselli 2017 evaluated the cut-off in kilopascals (42.7 kPa). Three studies reported the frequency of invaluable or indeterminate results due to technical failure; the failure rate ranged from 0% to 3.3% (Attia 2015; Borghi 2012; Takuma 2016).

Meta-analysed results

Figure 15 shows a forest plot of sensitivity and specificity with their 95% CIs, and Figure 16 shows a graphical representation of studies in the ROC space (sensitivity against 1 − specificity). We performed a meta-analysis using the HSROC model, as the primary studies reported diagnostic accuracy estimates of LSM by pSWE using different cut-off values (Figure 16). According to the SROC curve, specificities corresponding to sensitivities of 80% and 90% were 91.6% (95% CI 29.7% to 99.6%) and 86.1% (95% CI 40.4% to 98.3%), respectively. On the other hand, sensitivities corresponding to specificities of 80% and 90% were 94.3% (95% CI 52.6% to 99.6%) and 84.1% (95% CI 8.4% to 99.7%), respectively. We deemed these estimates at a very low level of certainty, downgrading them by one level due to risk of bias and by two levels due to imprecision.

Figure 15. Forest plot of sensitivity and specificity of spleen stiffness measurement (SSM) by point shear wave elastography (pSWE) at any cut-off against hepatic venous pressure gradient (HVPG) reference standard > 10 mmHg (clinically significant portal hypertension; CSPH) in 4 studies ordered alphabetically. TP = true positive; FP = false positive; FN = false negative; TN = true negative. Values between brackets are the 95% confidence intervals (CIs) of sensitivity and specificity. The figure shows the estimated sensitivity and specificity of the study (blue square) and its 95% CI (black horizontal line).

Study	TP	FP	FN	TN	SS cut-off value (CSPH)	Sensitivity (95% CI)	Specificity (95% CI)
Attia 2015	67	1	3	7	2.32	0.96 [0.88, 0.99]	0.88 [0.47, 1.00]
Borghi 2012	20	1	8	11	3.26	0.71 [0.51, 0.87]	0.92 [0.62, 1.00]
Roselli 2017	25	7	1	37	42.7	0.96 [0.80, 1.00]	0.84 [0.70, 0.93]
Takuma 2016	33	11	1	15	3.1	0.97 [0.85, 1.00]	0.58 [0.37, 0.77]

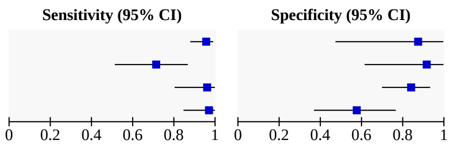
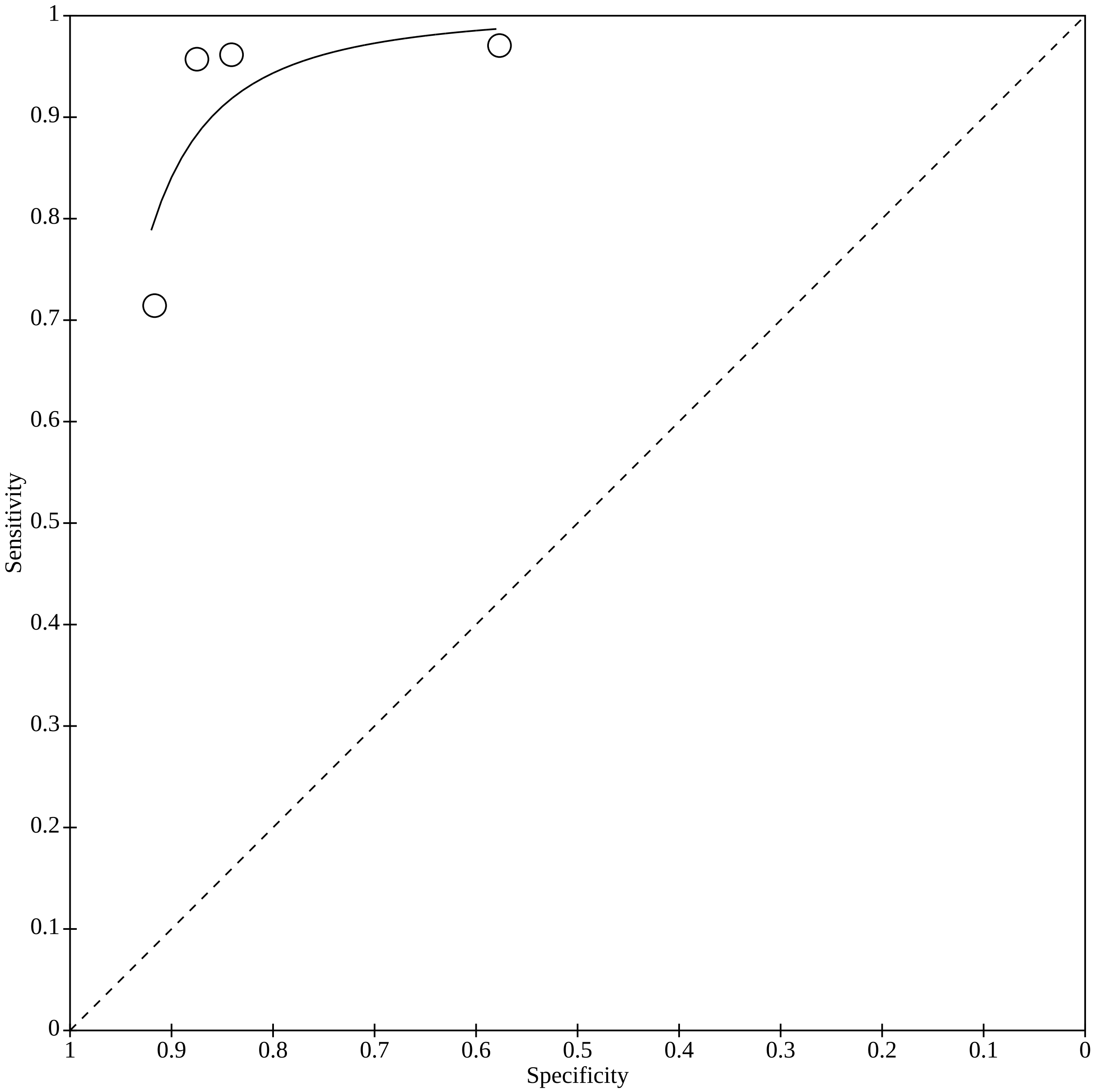


Figure 16. Summary receiver operating characteristic (ROC) comparing spleen stiffness measurement (SSM) by point shear wave elastography (pSWE) with any cut-off value and hepatic venous pressure gradient (HVPG) reference standard > 10 mmHg (clinically significant portal hypertension; CSPH) in 4 studies.



SSM by pSWE for SPH

Description of included studies

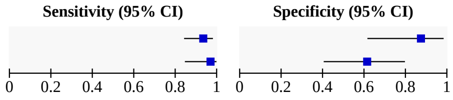
We identified two studies with 138 participants that evaluated the diagnostic accuracy of SSM by pSWE for SPH (Attia 2015; Takuma 2016). These two studies reported the frequency of invaluable or indeterminate results due to technical failure; the failure rates were 0% and 3.3%, respectively.

Meta-analysed results

Figure 17 shows a forest plot of sensitivity and specificity with their 95% CIs. We could not perform a meta-analysis using the HSROC model due to an insufficient number of studies. Attia 2015 reported that the cut-off value of 2.53 m/s had sensitivity of 94% (95% CI 84% to 98%) and specificity of 88% (95% CI 62% to 98%) (78 participants), while Takuma 2016 reported that the cut-off value of 3.15 m/s had sensitivity of 97% (95% CI 85% to 100%) and specificity of 62% (95% CI 41% to 80%) (60 participants).

Figure 17. Forest plot of sensitivity and specificity of spleen stiffness measurement (SSM) by point shear wave elastography (pSWE) at any cut-off against hepatic venous pressure gradient (HVPG) reference standard > 12 mmHg (severe portal hypertension; SPH) in 2 studies ordered alphabetically. TP = true positive; FP = false positive; FN = false negative; TN = true negative. Values between brackets are the 95% confidence intervals (CIs) of sensitivity and specificity. The figure shows the estimated sensitivity and specificity of the study (blue square) and its 95% CI (black horizontal line).

Study	TP	FP	FN	TN	SS cut-off value (SPH)	Sensitivity (95% CI)	Specificity (95% CI)
Attia 2015	58	2	4	14	2.53	0.94 [0.84, 0.98]	0.88 [0.62, 0.98]
Takuma 2016	33	10	1	16	3.15	0.97 [0.85, 1.00]	0.62 [0.41, 0.80]



LSM by 2D-shear wave elastography (2D-SWE) for CSPH

Description of included studies

We identified 10 studies with 767 participants that evaluated the diagnostic accuracy of LSM by 2D-SWE for CSPH (Corina 2018; Grgurević 2016; Grgurević 2022; Jansen 2017; Kim 2015; Maruyama 2016; Procopet 2015a; Yoo 2018; Zhu 2019; Zhu 2020). The cut-off values ranged from 11.4 kPa to 29.5 kPa. Only five studies reported the frequency of invaluable or indeterminate results due to technical failure; the failure rates ranged from 0% to 2.6% (Grgurević 2022; Jansen 2017; Kim 2015; Zhu 2019; Zhu 2020).

Meta-analysed results

Figure 18 shows a forest plot of sensitivity and specificity with their 95% CIs, and Figure 19 shows a graphical representation of studies in the ROC space (sensitivity against 1 − specificity). We performed a meta-analysis using the HSROC model, as the primary studies reported diagnostic accuracy estimates of LSM by 2D-SWE using different cut-off values (Figure 19). Three studies evaluated more than one cut-off value (Grgurević 2022; Jansen 2017; Procopet 2015a). In such cases, we extracted the results obtained with the cut-off most frequently used. According to the SROC curve, specificity corresponding to sensitivity of 80% was 83.4% (95% CI 68.2% to 92.2%), while specificity corresponding to sensitivity of 90% could not be calculated by the HSROC model. On the other hand, sensitivities corresponding to specificities of 80% and 90% were 82.2% (95% CI 72.0% to 89.3%) and 73.5% (95% CI 52.6% to 87.4%), respectively. We deemed these estimates at a very low level of certainty, downgrading them by three levels due to risk of bias, inconsistency, and imprecision.

Figure 18. Forest plot of sensitivity and specificity of liver stiffness measurement (LSM) by two-dimensional shear wave elastography (2D-SWE) at any cut-off against hepatic venous pressure gradient (HVPG) reference standard > 10 mmHg (clinically significant portal hypertension; CSPH) in 10 studies ordered alphabetically. TP = true positive; FP = false positive; FN = false negative; TN = true negative. Values between brackets are the 95% confidence intervals (CIs) of sensitivity and specificity. The figure shows the estimated sensitivity and specificity of the study (blue square) and its 95% CI (black horizontal line).

Study	TP	FP	FN	TN	LS cut-off value (CSPH)	Sensitivity (95% CI)	Specificity (95% CI)
Corina 2018	47	2	10	22	11.4	0.82 [0.70, 0.91]	0.92 [0.73, 0.99]
Grgurević 2016	10	1	1	8	12.6	0.91 [0.59, 1.00]	0.89 [0.52, 1.00]
Grgurević 2022	31	2	8	33	20.1	0.79 [0.64, 0.91]	0.94 [0.81, 0.99]
Jansen 2017	71	10	33	41	24.6	0.68 [0.58, 0.77]	0.80 [0.67, 0.90]
Kim 2015	66	3	11	12	15.2	0.86 [0.76, 0.93]	0.80 [0.52, 0.96]
Maruyama 2016	14	1	2	5	15.4	0.88 [0.62, 0.98]	0.83 [0.36, 1.00]
Procopet 2015a	23	5	5	22	17	0.82 [0.63, 0.94]	0.81 [0.62, 0.94]
Yoo 2018	63	9	19	15	12.44	0.77 [0.66, 0.85]	0.63 [0.41, 0.81]
Zhu 2019	66	6	18	14	18.1	0.79 [0.68, 0.87]	0.70 [0.46, 0.88]
Zhu 2020	41	2	6	9	12.86	0.87 [0.74, 0.95]	0.82 [0.48, 0.98]

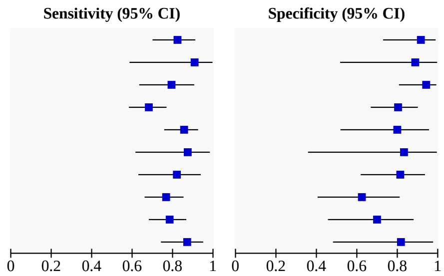
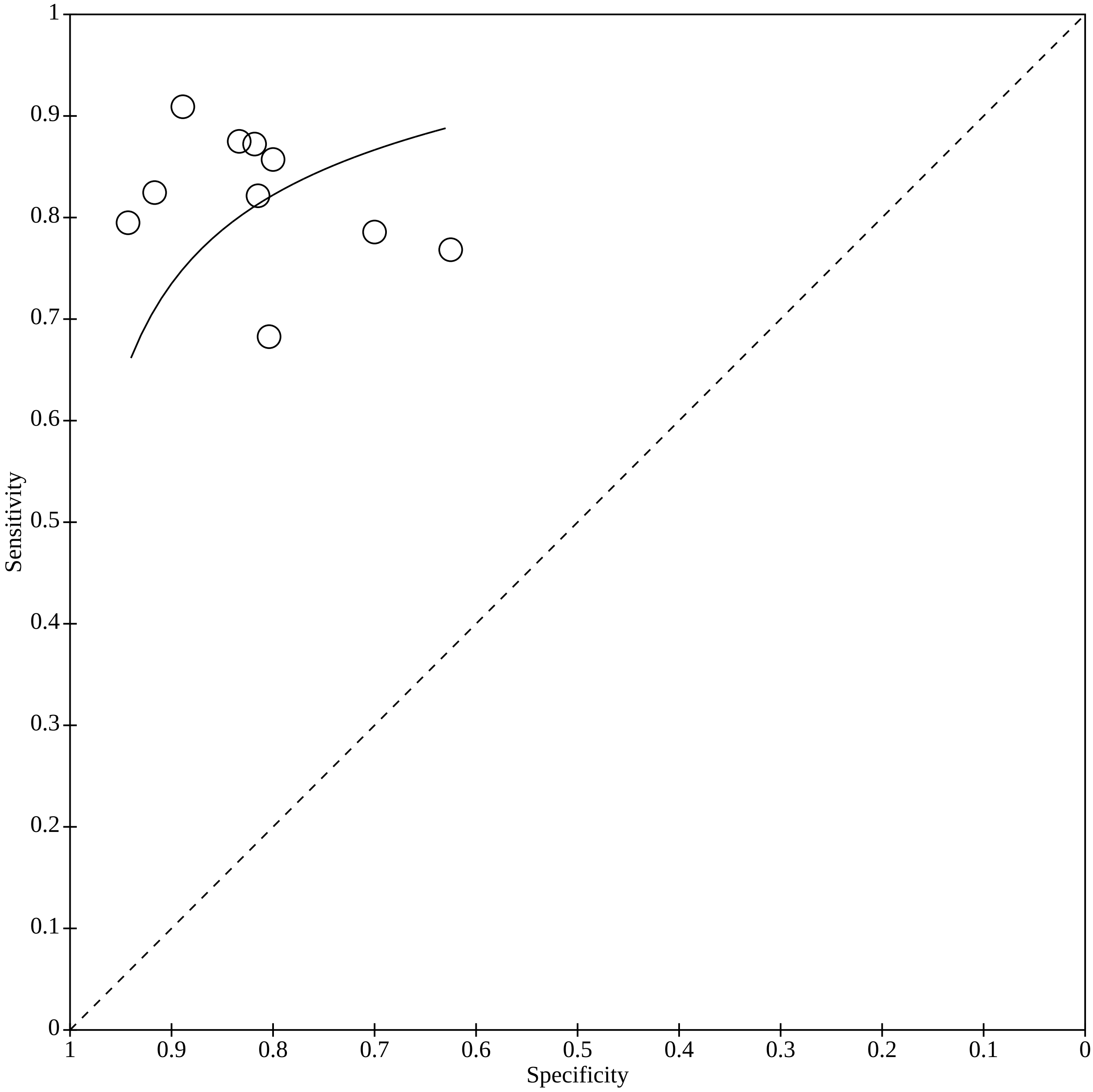


Figure 19. Summary receiver operating characteristic (ROC) comparing liver stiffness measurement (LSM) by two-dimensional shear wave elastography (2D-SWE) with any cut-off value and hepatic venous pressure gradient (HVPG) reference standard > 10 mmHg (clinically significant portal hypertension; CSPH) in 10 studies.



Based on cut-off values used in the included studies, we grouped them into two groups based on a cut-off value range of 3 and 2 kPa (11 to 14 kPa and 15 to 17 kPa). The results of the bivariate analyses for these two subgroups are shown in Table 5. We also identified two studies evaluating cut-off values in the range of 24 to 26 kPa (Jansen 2017; Procopet 2015a). Due to the high risk of obtaining unreliable estimates, we chose not to perform bivariate analysis for this subgroup.

LSM by 2D-SWE for SPH

Description of included studies

We identified seven studies with 492 participants that evaluated the diagnostic accuracy of LSM by 2D-SWE for SPH (Hirooka 2022; Kim 2015; Maruyama 2016; Sohn 2011; Yoo 2018; Zhu 2019; Zhu 2020). The cut-off values ranged from 13.2 kPa to 23.5 kPa, while Hirooka 2022 evaluated the cut-off value in metres per second (2.94 m/s). Three studies reported the frequency of invaluable or indeterminate results due to technical failure; the failure rates ranged from 0% to 1.7% (Kim 2015; Zhu 2019; Zhu 2020).

Meta-analysed results

Figure 20 shows a forest plot of sensitivity and specificity with their 95% CIs, and Figure 21 shows a graphical representation of studies in the ROC space (sensitivity against 1 − specificity). We performed a meta-analysis using the HSROC model, as the primary studies reported diagnostic accuracy estimates of LSM by 2D-SWE using different cut-off values (Figure 21). According to the SROC curve, specificity corresponding to sensitivity of 80% was 79.9% (95% CI 55.8% to 92.6%), while specificity corresponding to 90% could not be calculated by the HSROC model. On the other hand, sensitivities corresponding to 80% and 90% specificities were 79.8% (95% CI 50.4% to 93.9%) and 59.8% (95% CI 19.4% to 90.2%), respectively. We deemed these estimates at a very low level of certainty, downgrading them by three levels due to risk of bias, inconsistency, and imprecision.

Figure 20. Forest plot of sensitivity and specificity of liver stiffness measurement (LSM) by two-dimensional shear wave elastography (2D-SWE) at any cut-off against hepatic venous pressure gradient (HVPG) reference standard > 12 mmHg (severe portal hypertension; SPH) in 7 studies ordered alphabetically. TP = true positive; FP = false positive; FN = false negative; TN = true negative. Values between brackets are the 95% confidence intervals (CIs) of sensitivity and specificity. The figure shows the estimated sensitivity and specificity of the study (blue square) and its 95% CI (black horizontal line).

Study	TP	FP	FN	TN	LS cut-off value (SPH)	Sensitivity (95% CI)	Specificity (95% CI)
Hirooka 2022	20	3	11	47	2.94	0.65 [0.45, 0.81]	0.94 [0.83, 0.99]
Kim 2015	55	5	11	21	21.6	0.83 [0.72, 0.91]	0.81 [0.61, 0.93]
Maruyama 2016	12	1	3	6	17.15	0.80 [0.52, 0.96]	0.86 [0.42, 1.00]
Sohn 2011	15	0	5	12	21.64	0.75 [0.51, 0.91]	1.00 [0.74, 1.00]
Yoo 2018	43	7	32	21	13.21	0.57 [0.45, 0.69]	0.75 [0.55, 0.89]
Zhu 2019	60	6	14	24	23.5	0.81 [0.70, 0.89]	0.80 [0.61, 0.92]
Zhu 2020	20	5	14	19	18.95	0.59 [0.41, 0.75]	0.79 [0.58, 0.93]

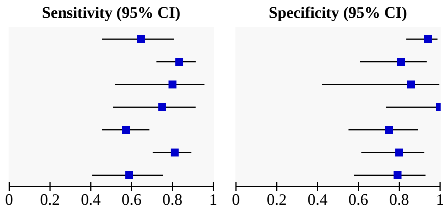
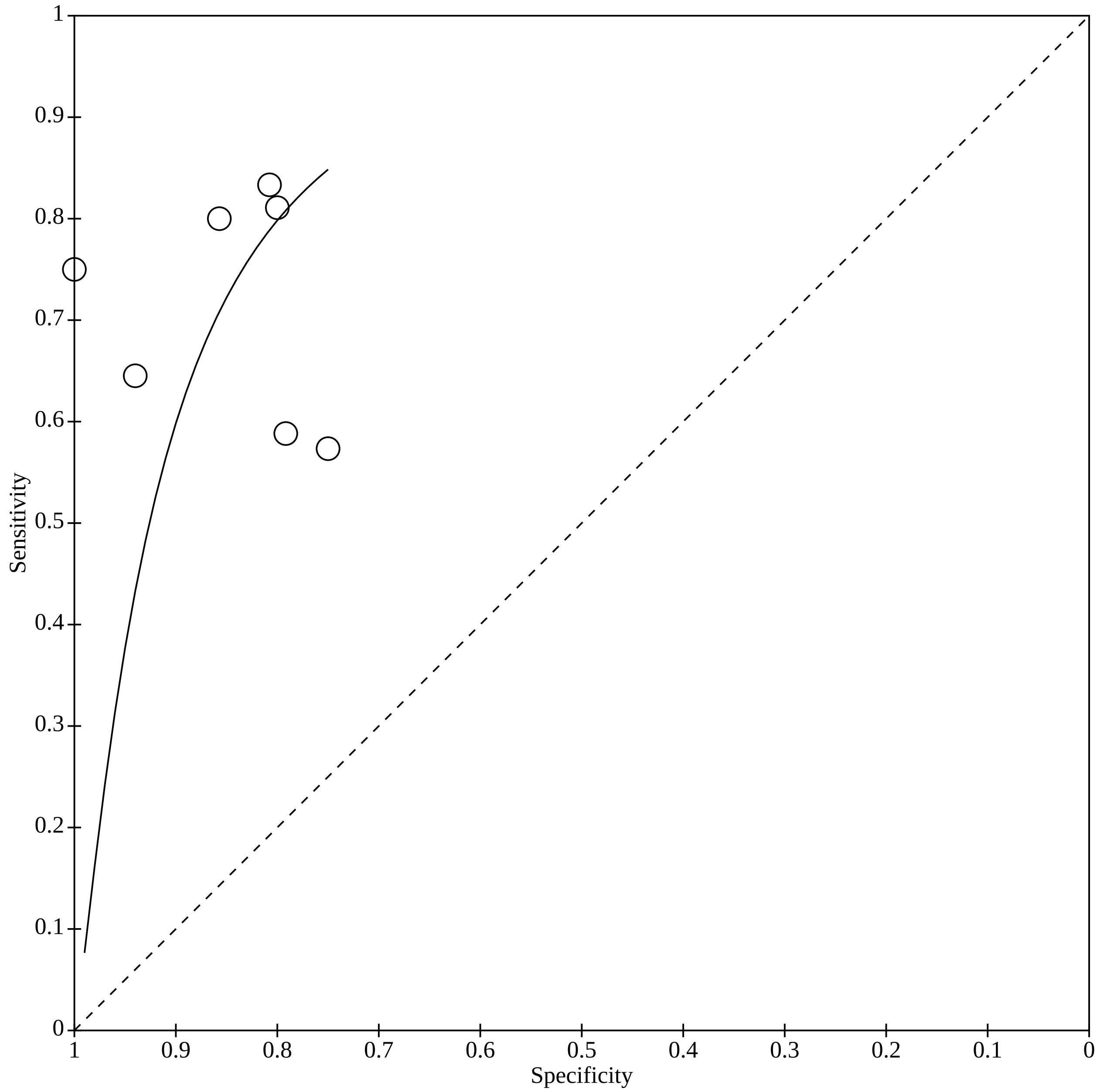


Figure 21. Summary receiver operating characteristic (ROC) comparing liver stiffness measurement (LSM) by two-dimensional shear wave elastography (2D-SWE) with any cut-off value and hepatic venous pressure gradient (HVPG) reference standard > 12 mmHg (severe portal hypertension; SPH) in 7 studies.



Based on cut-off values used in the included studies, we performed meta-analysis including only studies that evaluated cut-off values in the range of 21.5 to 23.5 kPa (the most used cut-off values). The result of the bivariate analyses for this subgroup is shown in Table 5. We also identified a cut-off range of 17 kPa to 19 kPa in two studies (Maruyama 2016; Zhu 2020), and chose not to perform bivariate analysis for this subgroup due to the high risk of unreliable estimates.

SSM by 2D-SWE for CSPH

Description of included studies

We identified six studies with 353 participants that evaluated the diagnostic accuracy of SSM by 2D-SWE for CSPH (Grgurević 2016; Grgurević 2022; Jachs 2023; Jansen 2017; Kennedy 2022; Zhu 2019). The cut-off values ranged from 21.7 kPa to 41.4 kPa. Four studies reported the frequency of invaluable or indeterminate results due to technical failure; the failure rates ranged from 8.3% to 24.7% (Grgurević 2022; Jansen 2017; Kennedy 2022; Zhu 2019).

Meta-analysed results

Figure 22 shows a forest plot of sensitivity and specificity with their 95% CIs, and Figure 23 shows a graphical representation of studies in the ROC space (sensitivity against 1 − specificity). We performed a meta-analysis using the HSROC model, as the primary studies reported diagnostic accuracy estimates of SSM by 2D-SWE using different cut-off values (Figure 23). Two studies evaluated more than one cut-off value (Grgurević 2022; Jansen 2017). In such cases, we extracted the results obtained with the cut-off most frequently used. According to the SROC curve, specificities corresponding to sensitivities of 80% and 90% could not be calculated by a HSROC model. On the other hand, sensitivities corresponding to specificities of 80% and 90% were 81.3% (95% CI 71.6% to 88.2%) and 80.0% (95% CI 59.8% to 91.5%), respectively. We deemed these estimates at a very low level of certainty, downgrading them by three levels due to risk of bias, inconsistency, and imprecision.

Figure 22. Forest plot of sensitivity and specificity of spleen stiffness measurement (SSM) by two-dimensional shear wave elastography (2D-SWE) at any cut-off against hepatic venous pressure gradient (HVPG) reference standard > 10 mmHg (clinically significant portal hypertension; CSPH) in 6 studies ordered alphabetically. TP = true positive; FP = false positive; FN = false negative; TN = true negative. Values between brackets are the 95% confidence intervals (CIs) of sensitivity and specificity. The figure shows the estimated sensitivity and specificity of the study (blue square) and its 95% CI (black horizontal line).

Study	TP	FP	FN	TN	SS cut-off value (CSPH)	Sensitivity (95% CI)	Specificity (95% CI)
Grgurević 2016	7	0	4	9	30.2	0.64 [0.31, 0.89]	1.00 [0.66, 1.00]
Grgurević 2022	30	5	6	22	34.8	0.83 [0.67, 0.94]	0.81 [0.62, 0.94]
Jachs 2023	8	0	2	10	40	0.80 [0.44, 0.97]	1.00 [0.69, 1.00]
Jansen 2017	59	6	15	32	26.3	0.80 [0.69, 0.88]	0.84 [0.69, 0.94]
Kennedy 2022	8	9	3	14	26	0.73 [0.39, 0.94]	0.61 [0.39, 0.80]
Zhu 2019	71	4	13	16	25.3	0.85 [0.75, 0.91]	0.80 [0.56, 0.94]

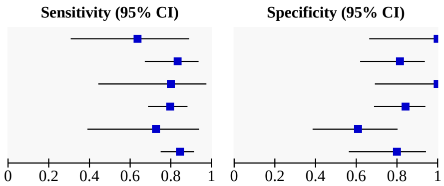
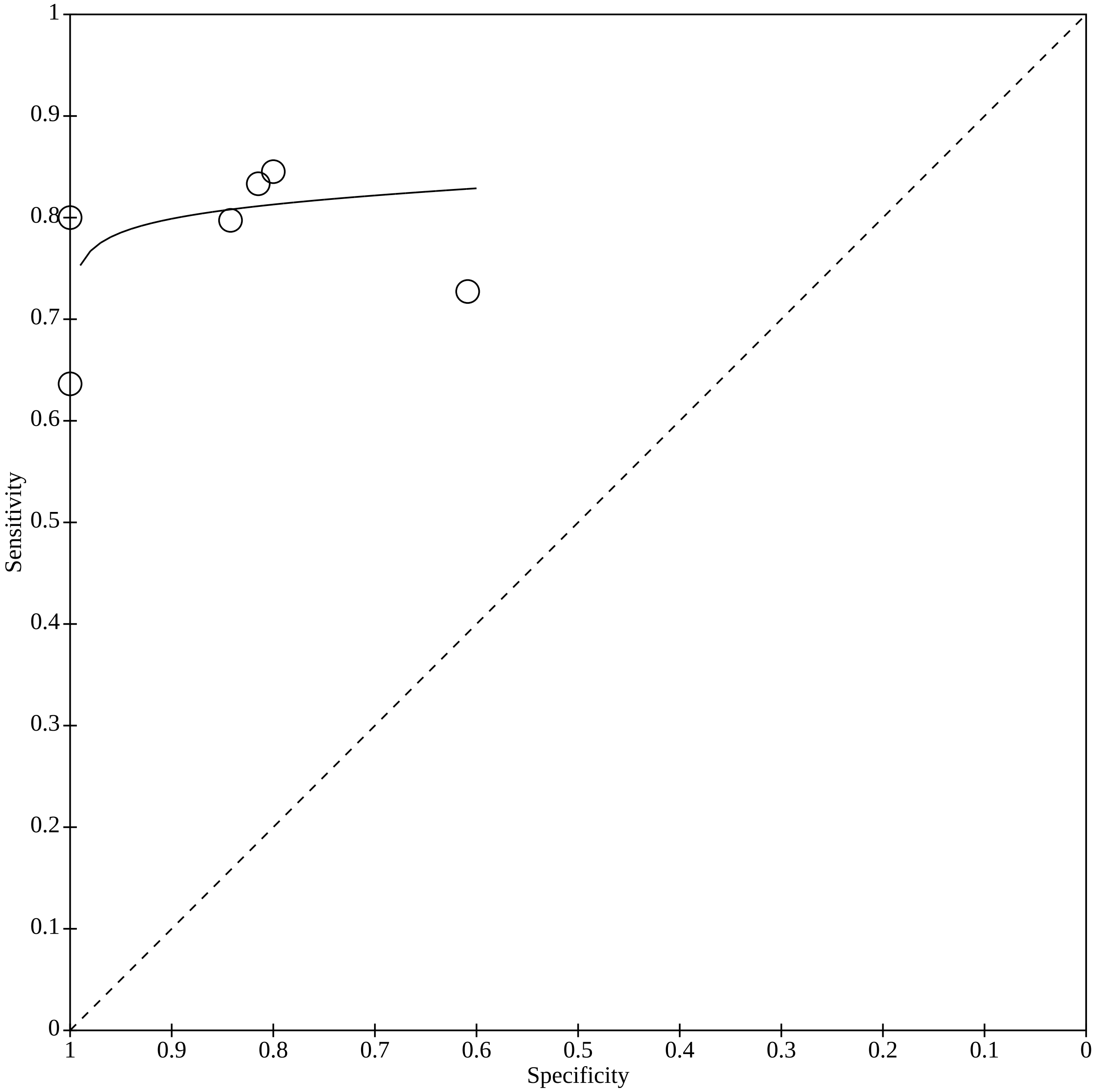


Figure 23. Summary receiver operating characteristic (ROC) comparing spleen stiffness measurement (SSM) by two-dimensional shear wave elastography (2D-SWE) with any cut-off value and hepatic venous pressure gradient (HVPG) reference standard > 10 mmHg (clinically significant portal hypertension; CSPH) in 6 studies.



Based on cut-off values used in the included studies, we performed meta-analysis including only studies that evaluated cut-off values ranging from 25.3 kPa to 26.3 kPa (the most used cut-off values). The result of the bivariate analyses for this subgroup is shown in Table 5. We also identified cut-off ranges of 30 kPa to 31 kPa, 34.8 kPa to 35.8 kPa, and 40 kPa to 42 kPa with only two studies per range, and chose not to perform bivariate analysis for these subgroups due to the high risk of unreliable estimates.

SSM by 2D-SWE for SPH

Description of included studies

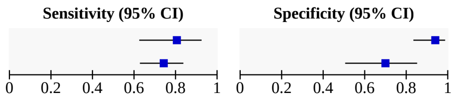
We identified two studies with 185 participants that evaluated the diagnostic accuracy of SSM by 2D-SWE for SPH (Hirooka 2022; Zhu 2019). Zhu 2019 evaluated the cut-off value in kilopascals (33.4 kPa), while Hirooka 2022 evaluated the cut-off value in metres per second (3.30 m/s). One study reported the frequency of invaluable or indeterminate results due to technical failure; the failure rate was 0.7% (Zhu 2019).

Meta-analysed results

Figure 24 shows a forest plot of sensitivity and specificity with their 95% CIs. We could not perform a meta-analysis using the HSROC model, as the primary studies were incomparable due to different units of measurement.

Figure 24. Forest plot of sensitivity and specificity of spleen stiffness measurement (SSM) by two-dimensional shear wave elastography (2D-SWE) at any cut-off against hepatic venous pressure gradient (HVPG) reference standard > 12 mmHg (severe portal hypertension; SPH) in 2 studies ordered alphabetically. TP = true positive; FP = false positive; FN = false negative; TN = true negative. Values between brackets are the 95% confidence intervals (CIs) of sensitivity and specificity. The figure shows the estimated sensitivity and specificity of the study (blue square) and its 95% CI (black horizontal line).

Study	TP	FP	FN	TN	SS cut-off value (SPH)	Sensitivity (95% CI)	Specificity (95% CI)
Hirooka 2022	25	3	6	47	3.3	0.81 [0.63, 0.93]	0.94 [0.83, 0.99]
Zhu 2019	55	9	19	21	33.4	0.74 [0.63, 0.84]	0.70 [0.51, 0.85]



LSM by magnetic resonance elastography (MRE) for CSPH

Description of included studies

We identified no studies that evaluated LSM by MRE for CSPH that matched our inclusion and exclusion criteria.

LSM by MRE for SPH

Description of included studies

We identified no studies that evaluated LSM by MRE for SPH that matched our inclusion and exclusion criteria.

SSM by MRE for CSPH

Description of included studies

We identified only one eligible study that evaluated SSM by MRE for CSPH (Kennedy 2022). SSM was evaluated by 2D-MRE and 3D-MRE. For 2D-MRE with the cut-off value of 8 kPa, the sensitivity was 100% (95% CI 66% to 100%) and specificity was 60% (95% CI 39% to 79%), and for 3D-MRE with the cut-off value of 9.6 kPa, the sensitivity was 100% (95% CI 63% to 100%) and specificity was 77% (95% CI 55% to 92%).

Meta-analysed results

We could not perform a meta-analysis using the HSROC model or bivariate model due to an insufficient number of included studies.

SSM by MRE for SPH

Description of included studies

We identified no studies that evaluated SSM by MRE for SPH that matched our inclusion and exclusion criteria.

Combination of LSM and SSM for CSPH

Combination of LSM and SSM by VCTE for CSPH

Description of included studies

We identified only one eligible study that evaluated the combination of LSM and SSM by VCTE for the diagnosis of CSPH (He 2022). The combination of an LSM cut-off value of 25 kPa and an SSM cut-off value of 50 kPa showed a sensitivity of 46% (95% CI 32% to 61%) and a specificity of 98% (95% CI 90% to 100%).

Meta-analysed results

We could not perform a meta-analysis using the HSROC model or bivariate model due to an insufficient number of included studies.

Combination of LSM and SSM by 2D-SWE for CSPH

Description of included studies

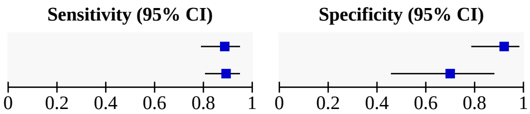
We identified two studies that evaluated the combination of LSM and SSM by 2D-SWE for the diagnosis of CSPH. Jansen 2017 evaluated the combination of LSM cut-off of 38 kPa and SSM cut-off of 27.1 kPa, which had sensitivity 89% (95% CI 79% to 95%) and specificity 92% (95% CI 79% to 98%). Zhu 2019 evaluated different cut-off values (LSM cut-off value of 16.1 kPa and SSM cut-off value of 25.3 kPa), which had sensitivity 89% (95% CI 81% to 95%) and specificity 70% (95% CI 46% to 88%).

Meta-analysed results

Figure 25 shows a forest plot of sensitivity and specificity with their 95% CIs. We could not perform a meta-analysis using the HSROC model, as the primary studies were incomparable because of different cut-off values.

Figure 25. Forest plot of sensitivity and specificity of combination of liver stiffness measurement (LSM) and spleen stiffness measurement (SSM) by two-dimensional shear wave elastography (2D-SWE) at any cut-off against hepatic venous pressure gradient (HVPG) reference standard > 10 mmHg (clinically significant portal hypertension; CSPH) in 2 studies ordered alphabetically. TP = true positive; FP = false positive; FN = false negative; TN = true negative. Values between brackets are the 95% confidence intervals (CIs) of sensitivity and specificity. The figure shows the estimated sensitivity and specificity of the study (blue square) and its 95% CI (black horizontal line).

Study	TP	FP	FN	TN	Sensitivity (95% CI)	Specificity (95% CI)
Jansen 2017	63	3	8	35	0.89 [0.79, 0.95]	0.92 [0.79, 0.98]
Zhu 2019	75	6	9	14	0.89 [0.81, 0.95]	0.70 [0.46, 0.88]



Combination of LSM and SSM for SPH

Combination of LSM and SSM by VCTE for SPH

We identified no studies that evaluated the combination of LSM and SSM by VCTE for SPH that matched our inclusion and exclusion criteria.

Combination of LSM and SSM by 2D-SWE for SPH

Description of included studies

We identified only one eligible study that evaluated the combination of LSM and SSM by 2D-SWE for the diagnosis of SPH (Zhu 2019). The combination of the LSM cut-off value of 23.5 kPa and the SSM cut-off value of 33.4 kPa showed a sensitivity of 84% (95% CI 73% to 91%) and a specificity of 83% (95% CI 65% to 94%).

Meta-analysed results

We could not perform a meta-analysis using the HSROC model or bivariate model due to an insufficient number of included studies.

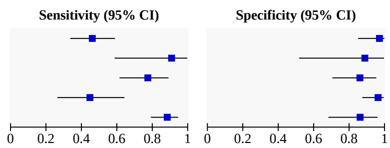
Direct and indirect comparison of LSM by VCTE and SSM by VCTE for CSPH

Five studies directly compared the results of the two index tests: LSM by VCTE and SSM by VCTE, to assess CSPH (Colecchia 2012; Grgurević 2016; Grgurević 2022; Odriozola 2023; Zykus 2015). Due to differing numbers of included participants for each index test across the included studies, a direct comparison was not possible. A graphical presentation of the results of the studies is shown in Figure 26.

Figure 26. Forest plots of sensitivity and specificity of liver stiffness measurement (LSM) and spleen stiffness measurement (SSM) by vibration-controlled transient elastography (VCTE) at any cut-off against hepatic venous pressure gradient (HVPG) reference standard > 10 mmHg (clinically significant portal hypertension; CSPH) in 5 studies ordered alphabetically that evaluated both index tests. TP = true positive; FP = false positive; FN = false negative; TN = true negative. Values between brackets are the 95% confidence intervals (CIs) of sensitivity and specificity. The figure shows the estimated sensitivity and specificity of the study (blue square) and its 95% CI (black horizontal line).

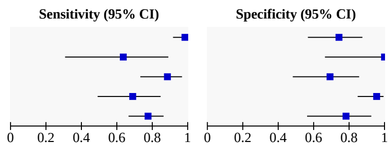
LSM by VCTE (CSPH) "any cut-off"

Study	TP	FP	FN	TN	LS cut-off value (CSPH)	SS cut-off value (CSPH)	Sensitivity (95% CI)	Specificity (95% CI)
Colecchia 2012	30	1	35	34	25		0.46 [0.34, 0.59]	0.97 [0.85, 1.00]
Grgurević 2016	10	1	1	8	10.4		0.91 [0.59, 1.00]	0.89 [0.52, 1.00]
Grgurević 2022	31	5	9	31	20.2		0.78 [0.62, 0.89]	0.86 [0.71, 0.95]
Odriozola 2023	13	2	16	53	27.6		0.45 [0.26, 0.64]	0.96 [0.87, 1.00]
Zykus 2015	69	4	9	25	17.4		0.88 [0.79, 0.95]	0.86 [0.68, 0.96]



SSM by VCTE (CSPH) "any cut-off"

Study	TP	FP	FN	TN	LS cut-off value (CSPH)	SS cut-off value (CSPH)	Sensitivity (95% CI)	Specificity (95% CI)
Colecchia 2012	64	9	1	26		40	0.98 [0.92, 1.00]	0.74 [0.57, 0.88]
Grgurević 2016	7	0	4	9		41.8	0.64 [0.31, 0.89]	1.00 [0.66, 1.00]
Grgurević 2022	31	8	4	18		43.5	0.89 [0.73, 0.97]	0.69 [0.48, 0.86]
Odriozola 2023	20	2	9	43		49.9	0.69 [0.49, 0.85]	0.96 [0.85, 0.99]
Zykus 2015	59	5	17	18		47.6	0.78 [0.67, 0.86]	0.78 [0.56, 0.93]

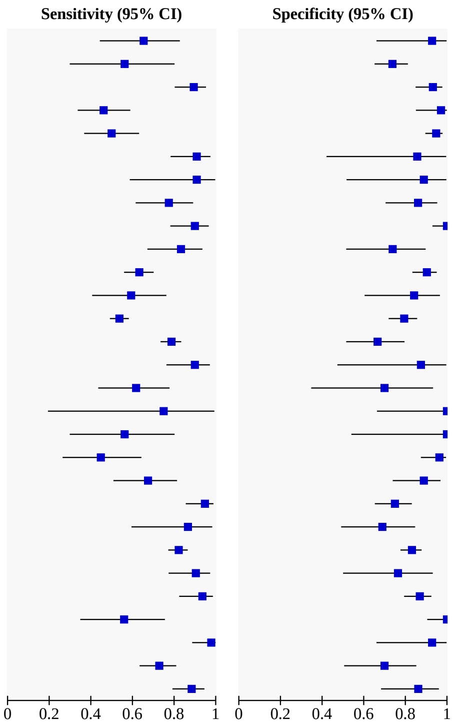


The indirect comparison between 27 studies (29 cohorts, 3818 participants) with LSM by VCTE and six studies (391 participants) with SSM by VCTE each at any cut-off value for CSPH is shown in Figure 27. Due to the considerable variability of cut-off values across both index tests, we opted to compare SROC curves rather than conducting a bivariate analysis for specific cut-off values (Figure 28). We found an overall difference between the two HSROC curves (P < 0.001). However, this result should be interpreted with caution, as the two ROC curves cross.

Figure 27. Forest plots of sensitivity and specificity of liver stiffness measurement (LSM) and spleen stiffness measurement (SSM) by vibration-controlled transient elastography (VCTE) at any cut-off against hepatic venous pressure gradient (HVPG) reference standard > 10 mmHg (clinically significant portal hypertension; CSPH) in 33 studies ordered alphabetically that either evaluated LSM or SSM by VCTE or both. TP = true positive; FP = false positive; FN = false negative; TN = true negative. Values between brackets are the 95% confidence intervals (CIs) of sensitivity and specificity. The figure shows the estimated sensitivity and specificity of the study (blue square) and its 95% CI (black horizontal line).

LSM by VCTE (CSPH) "any cut-off"

Study	TP	FP	FN	TN	Sensitivity (95% CI)	Specificity (95% CI)
Augustin 2015	17	1	9	13	0.65 [0.44, 0.83]	0.93 [0.66, 1.00]
Banini 2022	9	33	7	93	0.56 [0.30, 0.80]	0.74 [0.65, 0.81]
Bureau 2008	68	5	8	69	0.89 [0.80, 0.95]	0.93 [0.85, 0.98]
Colecchia 2012	30	1	35	34	0.46 [0.34, 0.59]	0.97 [0.85, 1.00]
Dajti 2022	30	7	30	128	0.50 [0.37, 0.63]	0.95 [0.90, 0.98]
Gomez-Escolar 2013	40	1	4	6	0.91 [0.78, 0.97]	0.86 [0.42, 1.00]
Grgurević 2016	10	1	1	8	0.91 [0.59, 1.00]	0.89 [0.52, 1.00]
Grgurević 2022	31	5	9	31	0.78 [0.62, 0.89]	0.86 [0.71, 0.95]
He 2022	45	0	5	51	0.90 [0.78, 0.97]	1.00 [0.93, 1.00]
Hong 2013	30	6	6	17	0.83 [0.67, 0.94]	0.74 [0.52, 0.90]
Jachs 2022	119	11	69	103	0.63 [0.56, 0.70]	0.90 [0.83, 0.95]
Jachs 2023	19	3	13	16	0.59 [0.41, 0.76]	0.84 [0.60, 0.97]
Jindal 2022	258	30	222	116	0.54 [0.49, 0.58]	0.79 [0.72, 0.86]
Kumar 2017	219	16	59	32	0.79 [0.73, 0.83]	0.67 [0.52, 0.80]
Lemoine 2008 (alcoholic aetiology)	36	1	4	7	0.90 [0.76, 0.97]	0.88 [0.47, 1.00]
Lemoine 2008 (viral aetiology)	21	3	13	7	0.62 [0.44, 0.78]	0.70 [0.35, 0.93]
Levick 2019	3	0	1	9	0.75 [0.19, 0.99]	1.00 [0.66, 1.00]
Maruyama 2016	9	0	7	6	0.56 [0.30, 0.80]	1.00 [0.54, 1.00]
Odriozola 2023	13	2	16	53	0.45 [0.26, 0.64]	0.96 [0.87, 1.00]
Podrug 2022	27	4	13	32	0.68 [0.51, 0.81]	0.89 [0.74, 0.97]
Procopet 2015b (training group)	55	25	3	75	0.95 [0.86, 0.99]	0.75 [0.65, 0.83]
Procopet 2015b (validation group)	13	9	2	20	0.87 [0.60, 0.98]	0.69 [0.49, 0.85]
Reiberger 2012	227	38	49	188	0.82 [0.77, 0.87]	0.83 [0.78, 0.88]
Salzl 2014	38	4	4	13	0.90 [0.77, 0.97]	0.76 [0.50, 0.93]
Schwabl 2015	44	15	3	100	0.94 [0.82, 0.99]	0.87 [0.79, 0.93]
Taru 2023	14	0	11	37	0.56 [0.35, 0.76]	1.00 [0.91, 1.00]
Vizzutti 2007	46	1	1	13	0.98 [0.89, 1.00]	0.93 [0.66, 1.00]
Zhang 2022	78	9	29	21	0.73 [0.63, 0.81]	0.70 [0.51, 0.85]
Zykus 2015	69	4	9	25	0.88 [0.79, 0.95]	0.86 [0.68, 0.96]



SSM by VCTE (CSPH) "any cut-off"

Study	TP	FP	FN	TN	Sensitivity (95% CI)	Specificity (95% CI)
Colecchia 2012	64	9	1	26	0.98 [0.92, 1.00]	0.74 [0.57, 0.88]
Grgurević 2016	7	0	4	9	0.64 [0.31, 0.89]	1.00 [0.66, 1.00]
Grgurević 2022	31	8	4	18	0.89 [0.73, 0.97]	0.69 [0.48, 0.86]
Odriozola 2023	20	2	9	43	0.69 [0.49, 0.85]	0.96 [0.85, 0.99]
Stefanescu 2013	13	4	1	19	0.93 [0.66, 1.00]	0.83 [0.61, 0.95]
Zykus 2015	59	5	17	18	0.78 [0.67, 0.86]	0.78 [0.56, 0.93]

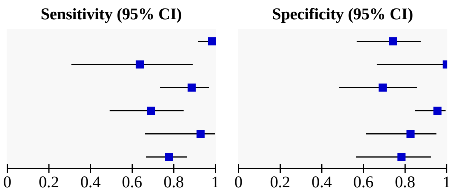
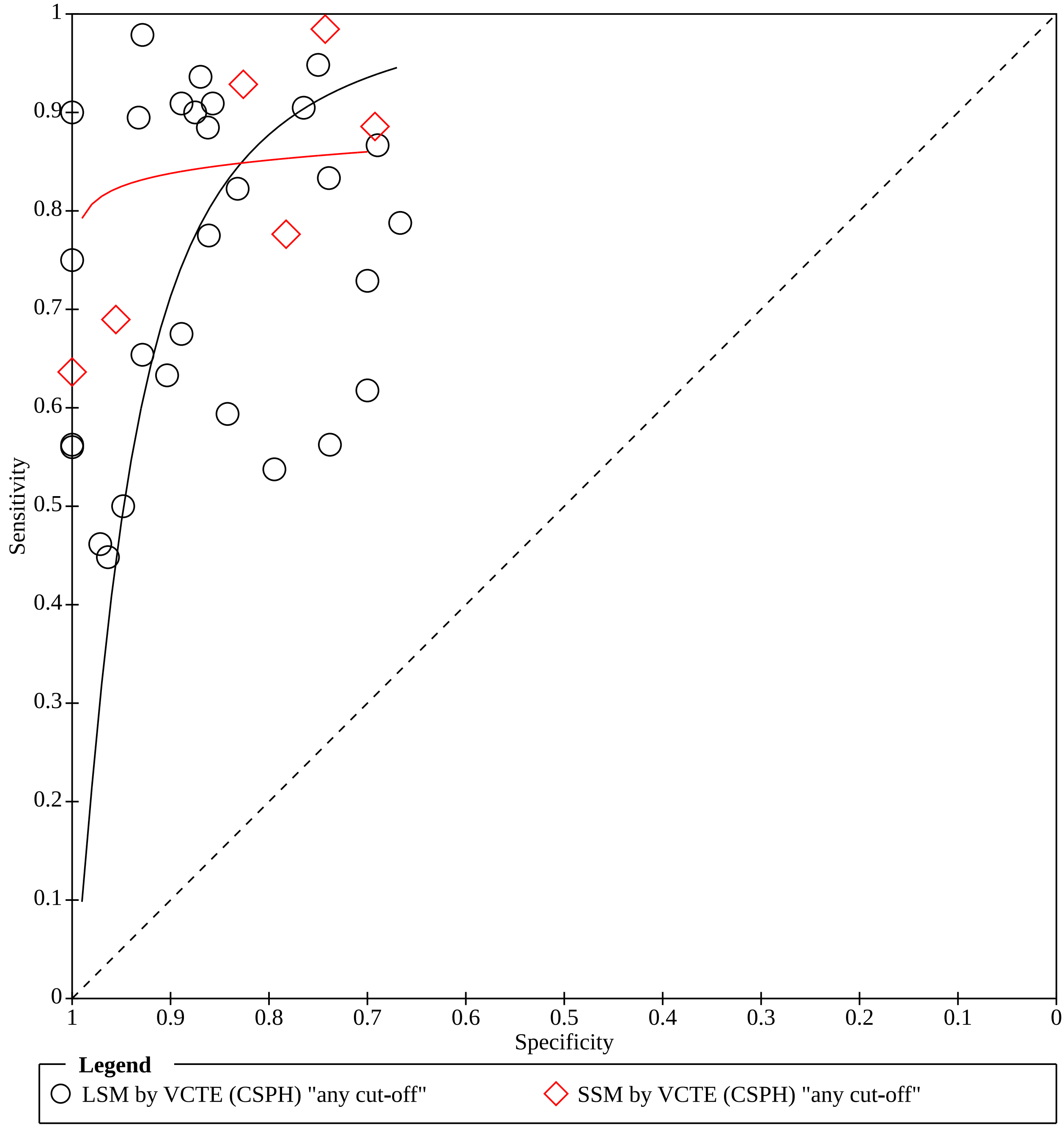


Figure 28. The indirect comparison of summary receiver operating characteristic (ROC) curves of liver stiffness measurement (LSM) by vibration-controlled transient elastography (VCTE) with any cut-off value (27 studies) and spleen stiffness measurement (SSM) by VCTE (6 studies) with any cut-off value. CSPH = clinically significant portal hypertension



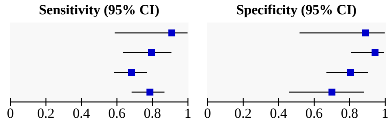
Direct and indirect comparison of LSM by 2D-SWE and SSM by 2D-SWE for CSPH

Four studies directly compared the results of the two index tests: LSM by 2D-SWE and SSM by 2D-SWE to assess CSPH (Grgurević 2016; Grgurević 2022; Jansen 2017; Zhu 2019). Due to differing numbers of included participants for each index test across the included studies, a direct comparison was not possible. A graphical presentation of the results of the studies is shown in Figure 29.

Figure 29. Forest plots of sensitivity and specificity of liver stiffness measurement (LSM) and spleen stiffness measurement (SSM) by two-dimensional shear wave elastography (2D-SWE) at any cut-off against hepatic venous pressure gradient (HVPG) reference standard > 10 mmHg (clinically significant portal hypertension; CSPH) in 4 studies ordered alphabetically that evaluated both index tests. TP = true positive; FP = false positive; FN = false negative; TN = true negative. Values between brackets are the 95% confidence intervals (CIs) of sensitivity and specificity. The figure shows the estimated sensitivity and specificity of the study (blue square) and its 95% CI (black horizontal line).

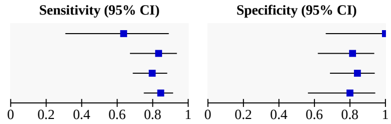
LSM by 2D-SWE (CSPH) "any cut-off"

Study	TP	FP	FN	TN	SS cut-off value (CSPH)	LS cut-off value (CSPH)	Sensitivity (95% CI)	Specificity (95% CI)
Grgurević 2016	10	1	1	8		12.6	0.91 [0.59, 1.00]	0.89 [0.52, 1.00]
Grgurević 2022	31	2	8	33		20.1	0.79 [0.64, 0.91]	0.94 [0.81, 0.99]
Jansen 2017	71	10	33	41		24.6	0.68 [0.58, 0.77]	0.80 [0.67, 0.90]
Zhu 2019	66	6	18	14		18.1	0.79 [0.68, 0.87]	0.70 [0.46, 0.88]



SSM by 2D SWE (CSPH) "any cut-off"

Study	TP	FP	FN	TN	SS cut-off value (CSPH)	LS cut-off value (CSPH)	Sensitivity (95% CI)	Specificity (95% CI)
Grgurević 2016	7	0	4	9	30.2		0.64 [0.31, 0.89]	1.00 [0.66, 1.00]
Grgurević 2022	30	5	6	22	34.8		0.83 [0.67, 0.94]	0.81 [0.62, 0.94]
Jansen 2017	59	6	15	32	26.3		0.80 [0.69, 0.88]	0.84 [0.69, 0.94]
Zhu 2019	71	4	13	16	25.3		0.85 [0.75, 0.91]	0.80 [0.56, 0.94]

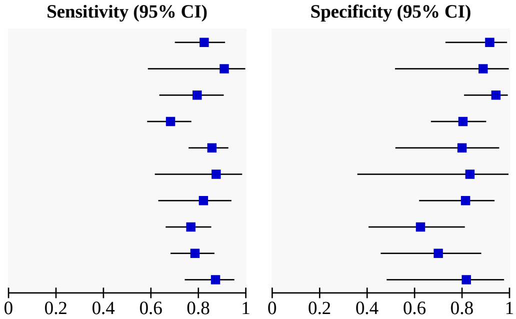


The indirect comparison between 10 studies (767 participants) with LSM by 2D-SWE and six studies (353 participants) with SSM by 2D-SWE each at any cut-off for CSPH is shown in Figure 30. Due to the considerable variability of cut-off values across both index tests, we opted to compare SROC curves rather than conducting a bivariate analysis for specific cut-off values (Figure 31). There was no overall difference between SROC curve estimates (P = 0.62).

Figure 30. Forest plots of sensitivity and specificity of liver stiffness measurement (LSM) and spleen stiffness measurement (SSM) by two-dimensional shear wave elastography (2D-SWE) at any cut-off against hepatic venous pressure gradient (HVPG) reference standard > 10 mmHg (clinically significant portal hypertension; CSPH) in 16 studies ordered alphabetically that evaluated either LSM or SSM by 2D-SWE or both index tests. TP = true positive; FP = false positive; FN = false negative; TN = true negative. Values between brackets are the 95% confidence intervals (CIs) of sensitivity and specificity. The figure shows the estimated sensitivity and specificity of the study (blue square) and its 95% CI (black horizontal line).

LSM by 2D-SWE (CSPH) "any cut-off"

Study	TP	FP	FN	TN	Sensitivity (95% CI)	Specificity (95% CI)
Corina 2018	47	2	10	22	0.82 [0.70, 0.91]	0.92 [0.73, 0.99]
Grgurević 2016	10	1	1	8	0.91 [0.59, 1.00]	0.89 [0.52, 1.00]
Grgurević 2022	31	2	8	33	0.79 [0.64, 0.91]	0.94 [0.81, 0.99]
Jansen 2017	71	10	33	41	0.68 [0.58, 0.77]	0.80 [0.67, 0.90]
Kim 2015	66	3	11	12	0.86 [0.76, 0.93]	0.80 [0.52, 0.96]
Maruyama 2016	14	1	2	5	0.88 [0.62, 0.98]	0.83 [0.36, 1.00]
Procopet 2015a	23	5	5	22	0.82 [0.63, 0.94]	0.81 [0.62, 0.94]
Yoo 2018	63	9	19	15	0.77 [0.66, 0.85]	0.63 [0.41, 0.81]
Zhu 2019	66	6	18	14	0.79 [0.68, 0.87]	0.70 [0.46, 0.88]
Zhu 2020	41	2	6	9	0.87 [0.74, 0.95]	0.82 [0.48, 0.98]



SSM by 2D SWE (CSPH) "any cut-off"

Study	TP	FP	FN	TN	Sensitivity (95% CI)	Specificity (95% CI)
Grgurević 2016	7	0	4	9	0.64 [0.31, 0.89]	1.00 [0.66, 1.00]
Grgurević 2022	30	5	6	22	0.83 [0.67, 0.94]	0.81 [0.62, 0.94]
Jachs 2023	8	0	2	10	0.80 [0.44, 0.97]	1.00 [0.69, 1.00]
Jansen 2017	59	6	15	32	0.80 [0.69, 0.88]	0.84 [0.69, 0.94]
Kennedy 2022	8	9	3	14	0.73 [0.39, 0.94]	0.61 [0.39, 0.80]
Zhu 2019	71	4	13	16	0.85 [0.75, 0.91]	0.80 [0.56, 0.94]

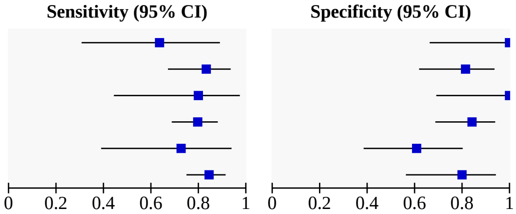
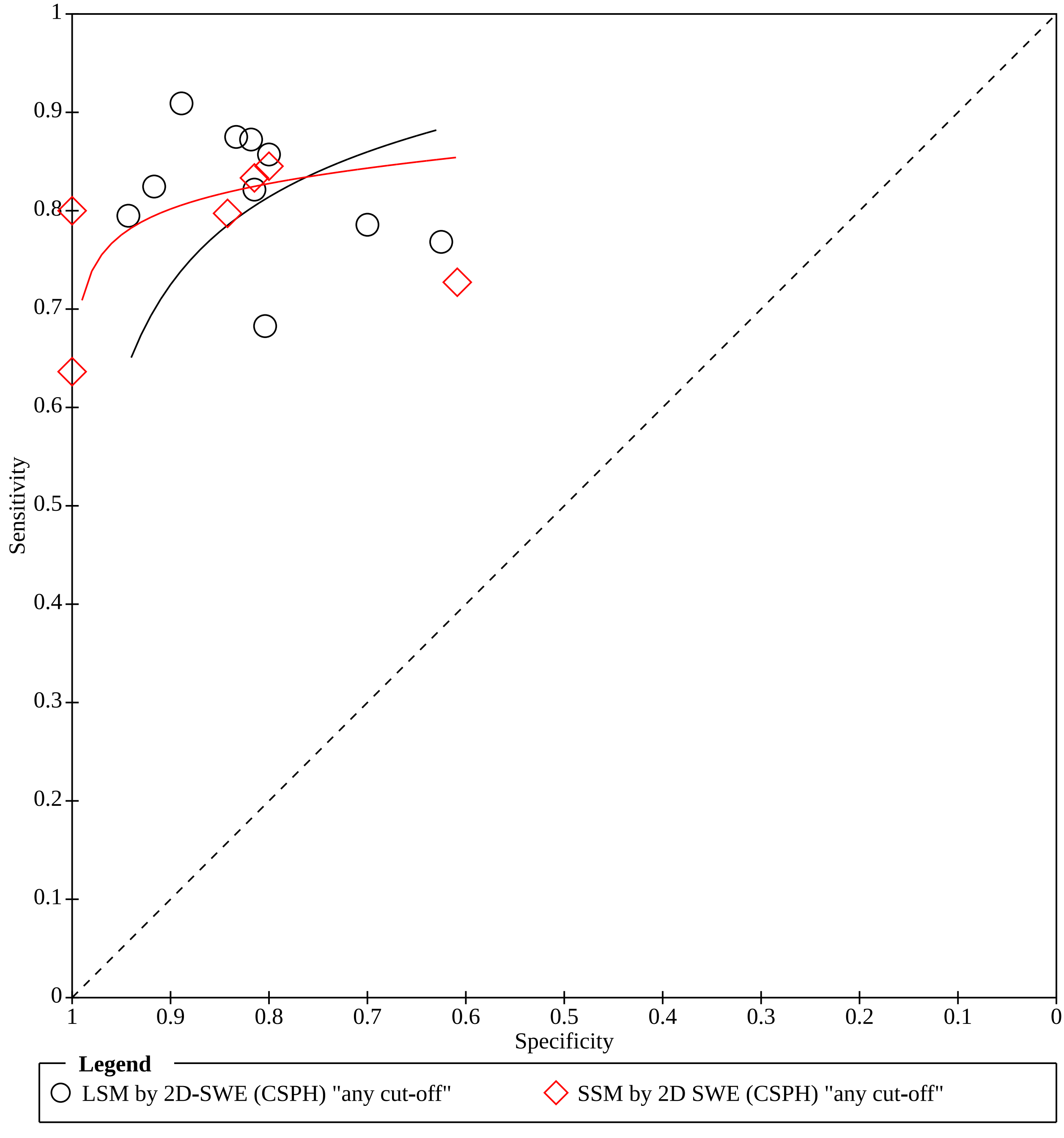


Figure 31. The indirect comparison of summary receiver operating characteristic (ROC) curves of liver stiffness measurement (LSM) by two-dimensional shear wave elastography (2D-SWE) with any cut-off value (10 studies) and spleen stiffness measurement (SSM) by 2D-SWE (6 studies) with any cut-off value. CSPH = clinically significant portal hypertension



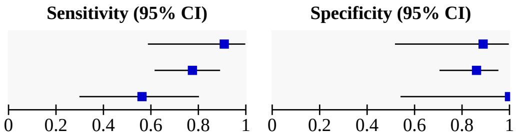
Direct and indirect comparison of LSM by VCTE and LSM by 2D-SWE for CSPH

Three studies compared the results of the two index tests (Grgurević 2016; Grgurević 2022; Maruyama 2016): LSM by VCTE and LSM by 2D-SWE to assess CSPH. Due to differing numbers of included participants for each index test across the included studies, a direct comparison was not possible. A graphical presentation of the results of the studies is shown in Figure 32.

Figure 32. Forest plots of sensitivity and specificity of liver stiffness measurement (LSM) by vibration-controlled transient elastography (VCTE) and LSM by two-dimensional shear wave elastography (2D-SWE) at any cut-off against hepatic venous pressure gradient (HVPG) reference standard > 10 mmHg (clinically significant portal hypertension; CSPH) in 3 studies ordered alphabetically that evaluated both index tests. TP = true positive; FP = false positive; FN = false negative; TN = true negative. Values between brackets are the 95% confidence intervals (CIs) of sensitivity and specificity. The figure shows the estimated sensitivity and specificity of the study (blue square) and its 95% CI (black horizontal line).

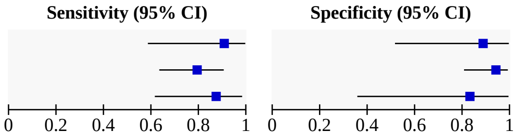
LSM by VCTE (CSPH) "any cut-off"

Study	TP	FP	FN	TN	Sensitivity (95% CI)	Specificity (95% CI)
Grgurević 2016	10	1	1	8	0.91 [0.59, 1.00]	0.89 [0.52, 1.00]
Grgurević 2022	31	5	9	31	0.78 [0.62, 0.89]	0.86 [0.71, 0.95]
Maruyama 2016	9	0	7	6	0.56 [0.30, 0.80]	1.00 [0.54, 1.00]



LSM by 2D-SWE (CSPH) "any cut-off"

Study	TP	FP	FN	TN	Sensitivity (95% CI)	Specificity (95% CI)
Grgurević 2016	10	1	1	8	0.91 [0.59, 1.00]	0.89 [0.52, 1.00]
Grgurević 2022	31	2	8	33	0.79 [0.64, 0.91]	0.94 [0.81, 0.99]
Maruyama 2016	14	1	2	5	0.88 [0.62, 0.98]	0.83 [0.36, 1.00]

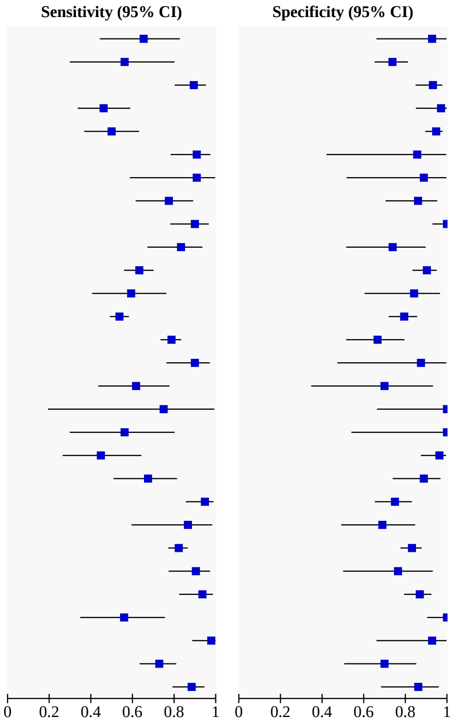


The indirect comparison between 27 studies (29 cohorts, 3818 participants) with LSM by VCTE and 10 studies (767 participants) with LSM by 2D-SWE each at any cut-off for CSPH is shown in Figure 33. Due to the considerable variability of cut-off values across both index tests, we opted to compare SROC curves rather than conducting a bivariate analysis for specific cut-off values (Figure 34). There was no overall difference between SROC curve estimates (P = 0.16).

Figure 33. Forest plots of sensitivity and specificity of liver stiffness measurement (LSM) by vibration-controlled transient elastography (VCTE) and LSM by two-dimensional shear wave elastography (2D-SWE) at any cut-off against hepatic venous pressure gradient (HVPG) reference standard > 10 mmHg (clinically significant portal hypertension; CSPH) in 37 studies ordered alphabetically that evaluated either LSM by VCTE or LSM by 2D-SWE or both index tests. TP = true positive; FP = false positive; FN = false negative; TN = true negative. Values between brackets are the 95% confidence intervals (CIs) of sensitivity and specificity. The figure shows the estimated sensitivity and specificity of the study (blue square) and its 95% CI (black horizontal line).

LSM by VCTE (CSPH) "any cut-off"

Study	TP	FP	FN	TN	Sensitivity (95% CI)	Specificity (95% CI)
Augustin 2015	17	1	9	13	0.65 [0.44, 0.83]	0.93 [0.66, 1.00]
Banini 2022	9	33	7	93	0.56 [0.30, 0.80]	0.74 [0.65, 0.81]
Bureau 2008	68	5	8	69	0.89 [0.80, 0.95]	0.93 [0.85, 0.98]
Colecchia 2012	30	1	35	34	0.46 [0.34, 0.59]	0.97 [0.85, 1.00]
Dajti 2022	30	7	30	128	0.50 [0.37, 0.63]	0.95 [0.90, 0.98]
Gomez-Escolar 2013	40	1	4	6	0.91 [0.78, 0.97]	0.86 [0.42, 1.00]
Grgurević 2016	10	1	1	8	0.91 [0.59, 1.00]	0.89 [0.52, 1.00]
Grgurević 2022	31	5	9	31	0.78 [0.62, 0.89]	0.86 [0.71, 0.95]
He 2022	45	0	5	51	0.90 [0.78, 0.97]	1.00 [0.93, 1.00]
Hong 2013	30	6	6	17	0.83 [0.67, 0.94]	0.74 [0.52, 0.90]
Jachs 2022	119	11	69	103	0.63 [0.56, 0.70]	0.90 [0.83, 0.95]
Jachs 2023	19	3	13	16	0.59 [0.41, 0.76]	0.84 [0.60, 0.97]
Jindal 2022	258	30	222	116	0.54 [0.49, 0.58]	0.79 [0.72, 0.86]
Kumar 2017	219	16	59	32	0.79 [0.73, 0.83]	0.67 [0.52, 0.80]
Lemoine 2008 (alcoholic aetiology)	36	1	4	7	0.90 [0.76, 0.97]	0.88 [0.47, 1.00]
Lemoine 2008 (viral aetiology)	21	3	13	7	0.62 [0.44, 0.78]	0.70 [0.35, 0.93]
Levick 2019	3	0	1	9	0.75 [0.19, 0.99]	1.00 [0.66, 1.00]
Maruyama 2016	9	0	7	6	0.56 [0.30, 0.80]	1.00 [0.54, 1.00]
Odriozola 2023	13	2	16	53	0.45 [0.26, 0.64]	0.96 [0.87, 1.00]
Podrug 2022	27	4	13	32	0.68 [0.51, 0.81]	0.89 [0.74, 0.97]
Procopet 2015b (training group)	55	25	3	75	0.95 [0.86, 0.99]	0.75 [0.65, 0.83]
Procopet 2015b (validation group)	13	9	2	20	0.87 [0.60, 0.98]	0.69 [0.49, 0.85]
Reiberger 2012	227	38	49	188	0.82 [0.77, 0.87]	0.83 [0.78, 0.88]
Salzl 2014	38	4	4	13	0.90 [0.77, 0.97]	0.76 [0.50, 0.93]
Schwabl 2015	44	15	3	100	0.94 [0.82, 0.99]	0.87 [0.79, 0.93]
Taru 2023	14	0	11	37	0.56 [0.35, 0.76]	1.00 [0.91, 1.00]
Vizzutti 2007	46	1	1	13	0.98 [0.89, 1.00]	0.93 [0.66, 1.00]
Zhang 2022	78	9	29	21	0.73 [0.63, 0.81]	0.70 [0.51, 0.85]
Zykus 2015	69	4	9	25	0.88 [0.79, 0.95]	0.86 [0.68, 0.96]



LSM by 2D-SWE (CSPH) "any cut-off"

Study	TP	FP	FN	TN	Sensitivity (95% CI)	Specificity (95% CI)
Corina 2018	47	2	10	22	0.82 [0.70, 0.91]	0.92 [0.73, 0.99]
Grgurević 2016	10	1	1	8	0.91 [0.59, 1.00]	0.89 [0.52, 1.00]
Grgurević 2022	31	2	8	33	0.79 [0.64, 0.91]	0.94 [0.81, 0.99]
Jansen 2017	71	10	33	41	0.68 [0.58, 0.77]	0.80 [0.67, 0.90]
Kim 2015	66	3	11	12	0.86 [0.76, 0.93]	0.80 [0.52, 0.96]
Maruyama 2016	14	1	2	5	0.88 [0.62, 0.98]	0.83 [0.36, 1.00]
Procopet 2015a	23	5	5	22	0.82 [0.63, 0.94]	0.81 [0.62, 0.94]
Yoo 2018	63	9	19	15	0.77 [0.66, 0.85]	0.63 [0.41, 0.81]
Zhu 2019	66	6	18	14	0.79 [0.68, 0.87]	0.70 [0.46, 0.88]
Zhu 2020	41	2	6	9	0.87 [0.74, 0.95]	0.82 [0.48, 0.98]

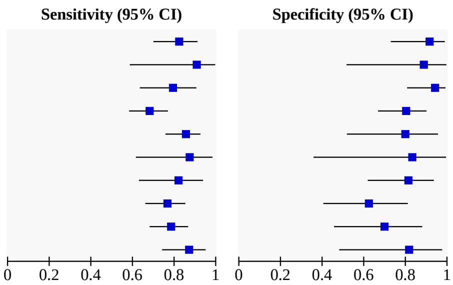
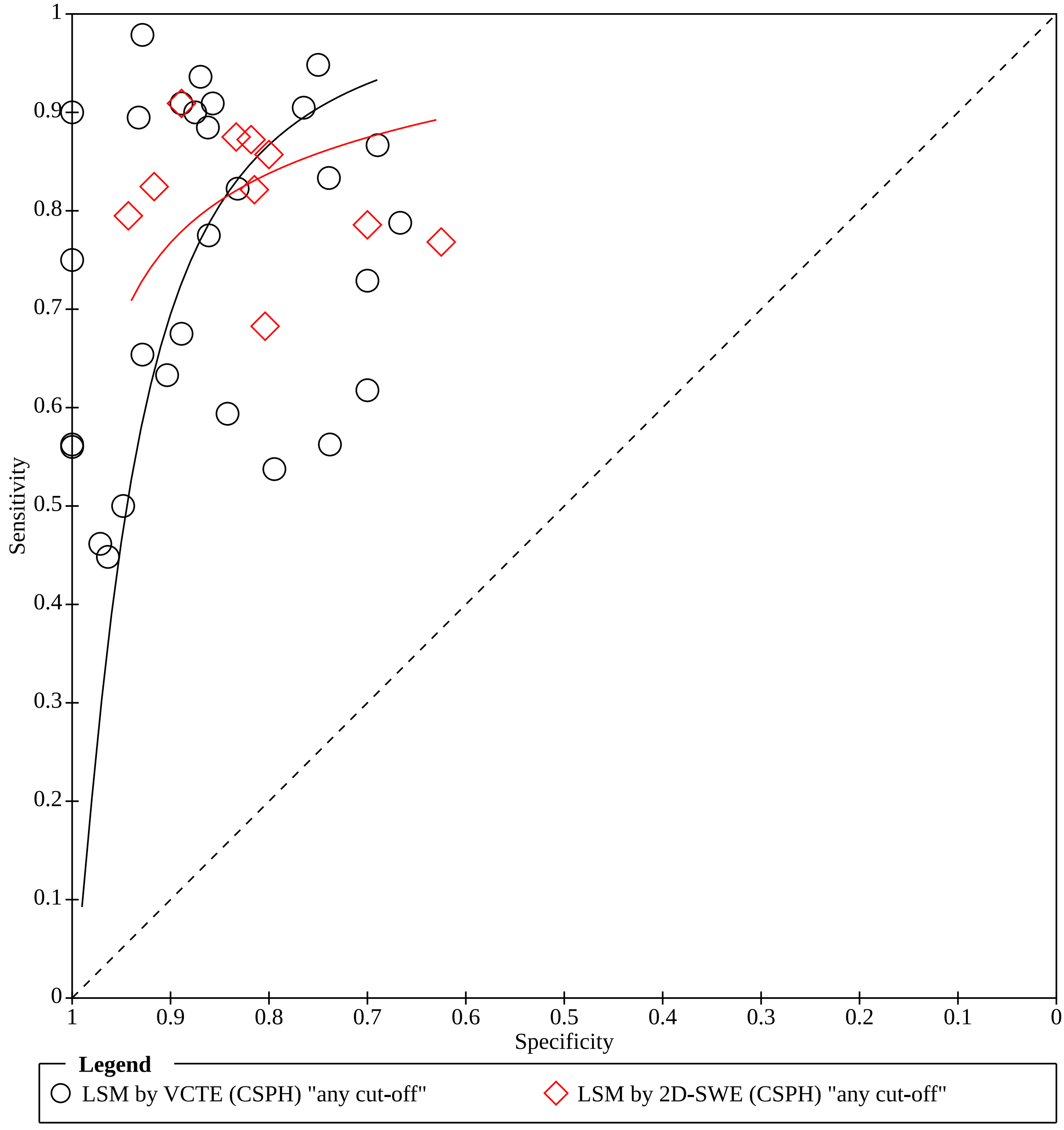


Figure 34. The indirect comparison of summary receiver operating characteristic (ROC) curves of liver stiffness measurement (LSM) by vibration-controlled transient elastography (VCTE) with any cut-off value (27 studies) and LSM by two-dimensional shear wave elastography (2D-SWE) (10 studies) with any cut-off value. CSPH = clinically significant portal hypertension



Direct and indirect comparison of SSM by VCTE and SSM by 2D-SWE for CSPH

Two studies compared the results of the two index tests (Grgurević 2016; Grgurević 2022): SSM by VCTE and SSM by 2D-SWE to assess CSPH. Due to differing numbers of included participants for each index test across the included studies, a direct comparison was not possible. A graphical presentation of the results of the studies is shown in Figure 35.

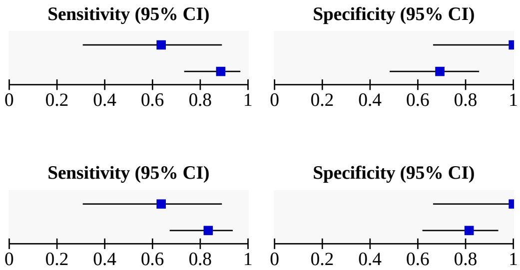
Figure 35. Forest plots of sensitivity and specificity of spleen stiffness measurement (SSM) by vibration-controlled transient elastography (VCTE) and SSM by two-dimensional shear wave elastography (2D-SWE) at any cut-off against hepatic venous pressure gradient (HVPG) reference standard > 10 mmHg (clinically significant portal hypertension; CSPH) in 2 studies ordered alphabetically that evaluated both index tests. TP = true positive; FP = false positive; FN = false negative; TN = true negative. Values between brackets are the 95% confidence intervals (CIs) of sensitivity and specificity. The figure shows the estimated sensitivity and specificity of the study (blue square) and its 95% CI (black horizontal line).

SSM by VCTE (CSPH) "any cut-off"

Study	TP	FP	FN	TN	Sensitivity (95% CI)	Specificity (95% CI)
Grgurević 2016	7	0	4	9	0.64 [0.31, 0.89]	1.00 [0.66, 1.00]
Grgurević 2022	31	8	4	18	0.89 [0.73, 0.97]	0.69 [0.48, 0.86]

SSM by 2D SWE (CSPH) "any cut-off"

Study	TP	FP	FN	TN	Sensitivity (95% CI)	Specificity (95% CI)
Grgurević 2016	7	0	4	9	0.64 [0.31, 0.89]	1.00 [0.66, 1.00]
Grgurević 2022	30	5	6	22	0.83 [0.67, 0.94]	0.81 [0.62, 0.94]



The indirect comparison between six studies (391 participants) with SSM by VCTE and six studies (353 participants) with SSM by 2D-SWE each at any cut-off for CSPH is shown in Figure 36. Due to the considerable variability of cut-off values across both index tests, we opted to compare SROC curves rather than conducting a bivariate analysis for specific cut-off values (Figure 37). There was no overall difference between SROC curve estimates (P = 0.08).

Figure 36. Forest plots of sensitivity and specificity of spleen stiffness measurement (SSM) by vibration-controlled transient elastography (VCTE) and SSM by two-dimensional shear wave elastography (2D-SWE) at any cut-off against hepatic venous pressure gradient (HVPG) reference standard > 10 mmHg (clinically significant portal hypertension; CSPH) in 12 studies ordered alphabetically that evaluated either SSM by VCTE or SSM by 2D-SWE or both index tests. TP = true positive; FP = false positive; FN = false negative; TN = true negative. Values between brackets are the 95% confidence intervals (CIs) of sensitivity and specificity. The figure shows the estimated sensitivity and specificity of the study (blue square) and its 95% CI (black horizontal line).

SSM by VCTE (CSPH) "any cut-off"

Study	TP	FP	FN	TN	Sensitivity (95% CI)	Specificity (95% CI)
Colecchia 2012	64	9	1	26	0.98 [0.92, 1.00]	0.74 [0.57, 0.88]
Grgurević 2016	7	0	4	9	0.64 [0.31, 0.89]	1.00 [0.66, 1.00]
Grgurević 2022	31	8	4	18	0.89 [0.73, 0.97]	0.69 [0.48, 0.86]
Odriozola 2023	20	2	9	43	0.69 [0.49, 0.85]	0.96 [0.85, 0.99]
Stefanescu 2013	13	4	1	19	0.93 [0.66, 1.00]	0.83 [0.61, 0.95]
Zykus 2015	59	5	17	18	0.78 [0.67, 0.86]	0.78 [0.56, 0.93]

SSM by 2D SWE (CSPH) "any cut-off"

Study	TP	FP	FN	TN	Sensitivity (95% CI)	Specificity (95% CI)
Grgurević 2016	7	0	4	9	0.64 [0.31, 0.89]	1.00 [0.66, 1.00]
Grgurević 2022	30	5	6	22	0.83 [0.67, 0.94]	0.81 [0.62, 0.94]
Jachs 2023	8	0	2	10	0.80 [0.44, 0.97]	1.00 [0.69, 1.00]
Jansen 2017	59	6	15	32	0.80 [0.69, 0.88]	0.84 [0.69, 0.94]
Kennedy 2022	8	9	3	14	0.73 [0.39, 0.94]	0.61 [0.39, 0.80]
Zhu 2019	71	4	13	16	0.85 [0.75, 0.91]	0.80 [0.56, 0.94]

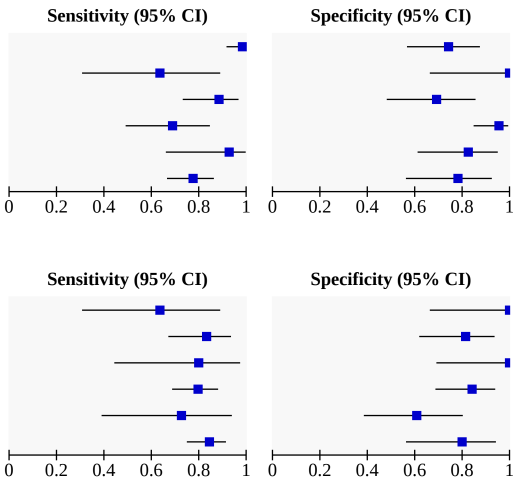
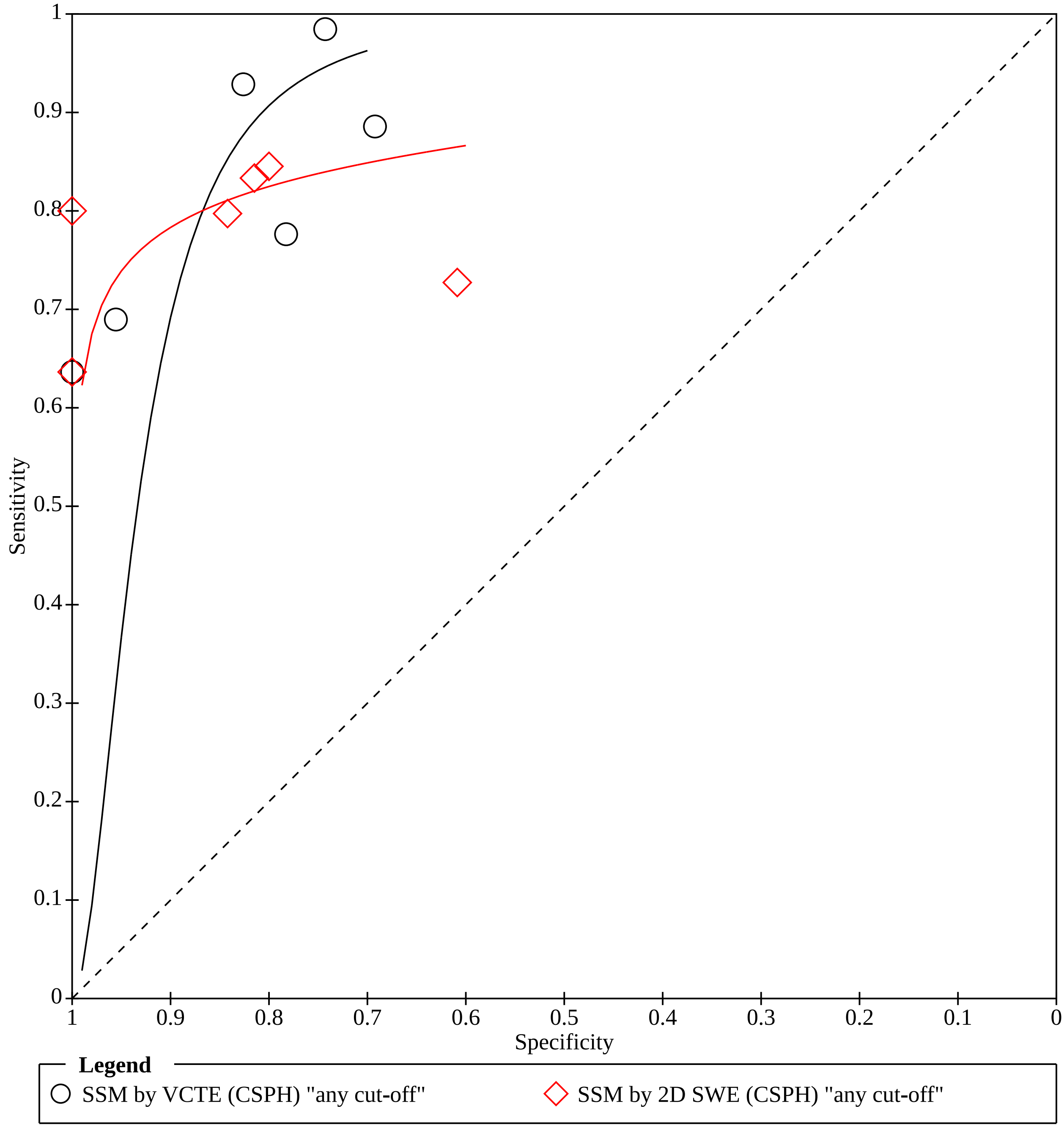


Figure 37. The indirect comparison of summary receiver operating characteristic (ROC) curves of spleen stiffness measurement (SSM) by vibration-controlled transient elastography (VCTE) with any cut-off value (6 studies) and SSM by two-dimensional shear wave elastography (2D-SWE) (6 studies) with any cut-off value. CSPH = clinically significant portal hypertension



DISCUSSION

Summary of main results

The aim of this review was to assess the diagnostic accuracy of liver and spleen stiffness as assessed by VCTE and other elastography methods for diagnosing CSPH in adults with CLD.

We included 47 studies (including 49 cohorts) (Figure 1, Table 6). Our review comprised a total of 41 different analyses which included various cut-off ranges and target conditions (CSPH or SPH) for each evaluated index test and their combinations (Supplementary material 4; Supplementary material 5).

Overall, only two studies including three cohorts were judged at low risk of bias for all domains (Jachs 2022; Procopet 2015b (training group); Procopet 2015b (validation group)). Most studies were deemed at high risk of bias for the index test domain, predominantly because of the lack of predefined cut-off values (Figure 2). Except for the 25 kPa cut-off assessed by LSM-VCTE, the other cut-off values had not been validated in previous studies. Eleven studies assessing LSM by VCTE evaluated more than one cut-off value.

Regarding applicability, almost all studies were judged at low concern for index test and reference standard domains. The main applicability issues were in the patient selection domain (26/47 studies), mainly because of the inclusion of a heterogeneous population that differs from an ideal cACLD population who would benefit the most from possible early treatment with portal pressure-lowering therapy (i.e. beta-blockers). Studies with applicability issues regarding the patient selection domain included participants with decompensated advanced CLD, or up to 20% of participants with hepatocellular carcinoma, cholestatic liver disease, history of NSBB use (not current), or previously known CSPH/SPH or oesophageal varices.

Only a minority of studies reported the frequency of invaluable or indeterminate results due to technical failure, detailing how often the elastography techniques, such as VCTE, 2D-SWE, pSWE, or MRE, were unable to produce reliable measurements. These rates are important for understanding the practical limitations of these diagnostic tools in routine clinical practice.

We summarised the main results of the analyses in Summary of findings 1. We considered the clinical implications of test results as follows.

1. True positives (patients with CSPH/SPH and a positive index test) would receive appropriate treatment with NSBBs.
2. True negatives (patients without CSPH/SPH and a negative index test) would appropriately avoid unnecessary procedures, such as oesophagogastroduodenoscopy, or inappropriate treatment.
3. False negatives (patients with CSPH/SPH but a negative index test) would be misdiagnosed and may miss timely and appropriate treatment.
4. False positives (patients without CSPH/SPH but a positive index test) would be unnecessarily treated with beta-blockers.

We considered sensitivity or specificity to be adequate if ≥ 90%, to effectively rule out or rule in the target condition, respectively.

Due to the wide range of cut-off values reported across studies, we used the HSROC model for each index test to obtain summary estimates. The main results for the detection of CSPH were as follows.

1. LSM by VCTE; 27 studies with 3818 participants
   a. Sensitivity 72.6% (95% CI 60.0% to 82.5%) at fixed specificity of 90%
   b. Specificity 75.9% (95% CI 63.5% to 85.1%) at fixed sensitivity of 90%
2. SSM by VCTE; six studies with 391 participants
   a. Sensitivity 72.9% (95% CI 27.1% to 95.1%) at fixed specificity of 90%
   b. Specificity 80.6% (95% CI 64.1% to 90.7%) at fixed sensitivity of 90%
3. SSM by pSWE; four studies with 248 participants
   a. Sensitivity 84.1% (95% CI 8.4% to 99.7%) at fixed specificity of 90%
   b. Specificity 86.1% (95% CI 40.4% to 98.3%) at fixed sensitivity of 90%
4. LSM by 2D-SWE; 10 studies with 767 participants
   a. Sensitivity 73.5% (95% CI 52.6% to 87.4%) at fixed specificity of 90%
   b. Specificity 83.4% (95% CI 68.2% to 92.2%) at fixed sensitivity of 80% (as estimation at 90% was not feasible)
5. SSM by 2D-SWE; six studies with 353 participants
   a. Sensitivity 80.0% (95% CI 59.8% to 91.5%) at fixed specificity of 90%
   b. Specificity could not be estimated at 80% or 90% sensitivity

We judged all results to be of very low certainty due to high risk of bias, heterogeneity, and imprecision (wide CI estimates). No index test achieved both sensitivity and specificity ≥ 90%, limiting their use for reliably ruling in or ruling out the target condition. Moreover, HSROC model estimations were not feasible for LSM by pSWE and LSM and SSM by MRE due to an insufficient number of studies.

As VCTE was the most reported technique, and 25 kPa the most frequently used and only predefined cut-off value, with the bivariate model in nine studies (1553 participants), we calculated a sensitivity of 62.3% (95% CI 53.0% to 70.7%) and a specificity of 94.1% (95% CI 87.2% to 97.3%), both judged to be of very low certainty. Exploring heterogeneity by prevalence of CSPH (< 50% versus ≥ 50%), we found a relative sensitivity of 0.87 (95% CI 0.67 to 1.14, P = 0.32) and a relative specificity of 0.90 (95% CI 0.83 to 0.97, P = 0.009) (Table 4). These results suggest that specificity, more than sensitivity, may vary by disease prevalence, potentially reflecting differences in case-mix or diagnostic thresholds across study settings. Therefore, LSM by VCTE using the 25 kPa cut-off may be reliably used to rule in CSPH, particularly in populations with lower expected CSPH prevalence (e.g. cACLD), which represents the ideal target population.

For the secondary objective, SPH (portal pressure higher than 12 mmHg), we found the following results.

1. LSM by VCTE (eight studies with 637 participants): sensitivity 67.2% (95% CI 48.8% to 81.4%) at fixed specificity of 90%, while specificity at fixed sensitivity of 90% could not be calculated
2. LSM by 2D-SWE (seven studies with 492 participants): sensitivity 59.8% (95% CI 19.4% to 90.2%) at fixed specificity of 90%, while specificity at fixed sensitivity of 90% could not be calculated

We judged all results to be of very low certainty due to high risk of bias, heterogeneity, and imprecision (wide CI estimates). Due to an insufficient number of studies and data, HSROC model estimations were not feasible for SSM by VCTE and 2D-SWE, and both LSM and SSM for pSWE and MRE.

Moreover, as planned, we grouped the most frequently reported cut-off values of each elastographic technique into ranges that were as narrow as possible. This approach allowed us to perform bivariate analyses and generate meta-analysed estimates, which are reported in Table 5. These data-driven results are of very low certainty due to high risk of bias, heterogeneity, and imprecision.

We intended to perform direct (within-study) comparisons for all possible index test combinations. We expected to find studies comparing LSM and SSM by either VCTE or 2D-SWE, or LSM and SSM by VCTE versus 2D-SWE, as these two index tests had the most identified studies. However, such analyses proved impossible due to the inconsistent numbers of participants included for each index test in the included studies. On the other hand, indirect comparisons showed no overall difference between SROC curves.

Many studies on non-invasive tests for CSPH or SPH, or both, have included non-elastography methods like platelet count, which fall outside the scope of our review. Following the 2015 Baveno VI consensus [4], many studies have validated LSM < 15 kPa combined with platelet count > 150,000 to rule out CSPH, but not specifically LSM and SSM alone. The He 2022 study is the only study that met our criteria for evaluating the combination of LSM and SSM by VCTE. However, it was published only as an abstract with limited information on study design and cohort characteristics. The study assessed SSM > 50 kPa using a dedicated 100 Hz probe together with a standardised probe for LSM at 25 kPa, and showed excellent ability to rule in CSPH (specificity 98%).

Only two studies evaluated LSM and SSM by 2D-SWE; these studies varied in their cut-off values, preventing meaningful meta-analysis (Jansen 2017; Zhu 2019).

No studies evaluated combinations of LSM and SSM by pSWE or MRE. For SPH, only one study evaluated the combination of LSM and SSM by 2D-SWE, showing better accuracy compared to LSM alone (Zhu 2019).

Strengths and weaknesses of the review

Strengths and weaknesses of the included studies

Only two studies including three cohorts were judged at low risk of bias (Jachs 2022; Procopet 2015b (training group); Procopet 2015b (validation group)). Most studies reported the most optimal cut-off values or those deemed best for ruling in or ruling out CSPH or SPH. The lack of predefined cut-off criteria was the most frequent reason for considering studies at high risk of bias. Such post hoc determination of cut-offs tends to overestimate the diagnostic accuracy of the evaluated tests.

We anticipated a limited number of studies evaluating index tests other than VCTE. To address this, we broadened our inclusion criteria to allow up to 20% of participants with decompensated cirrhosis and included studies with single aetiology, hepatocellular carcinoma, cholestatic liver disease, a history of NSBB use (not current), and previously known CSPH/SPH. While this approach expanded the pool of eligible studies and participants, it also impacted the applicability and increased the risk of bias in the patient selection domain of our results.

The prevalence of CSPH and SPH in the included studies was higher than expected (both 62.5%), raising concerns about applicability due to patient selection. Most studies were conducted in tertiary centres, which may not reflect broader clinical settings. To explore the impact of disease prevalence, we performed a subgroup analysis of studies assessing LSM by VCTE with a 25 kPa cut-off, comparing those with CSPH prevalence below versus above 50%.

As anticipated, specificity was significantly higher in populations with lower CSPH prevalence – similar to the populations on which current recommendations are based [4, 28, 37, 68]. On the other hand, the majority of participants were Child-Pugh A (68%), which might indicate the ideal cohort for these index tests. However, we acknowledge that Child-Pugh A class is a suboptimal surrogate for compensated cirrhosis, as it does not exclude patients with prior decompensation who may currently be clinically compensated. Due to incomplete or inconsistent reporting in the included studies, we were unable to reliably extract and analyse the number of participants with past decompensation (currently compensated) separately that were regarded as Child-Pugh A class. Previously decompensated should not be regarded as compensated participants, and does not represent an ideal cohort for portal hypertension estimation, as they most likely already have CSPH or have been treated with NSBB. Thus, we acknowledge that incomplete reporting of these data in the included studies limited our ability to analyse this as a source of heterogeneity. More studies with clearly defined cACLD population (not Child-Pugh A) are needed in future.

We identified a sufficient number of studies evaluating LSM by VCTE. The prespecified cut-off of 25 kPa, validated by the Baveno VI and VII consensus, EASL guidelines, and AASLD practice guidelines, was consistently used across nine studies and over 1500 participants [4, 28, 37, 68]. However, due to the high risk of bias, as well as significant inconsistency and imprecision in the results, the reliability of these findings is very low (very low certainty). Consequently, the results may be inadequate to fully support the recommendations by the Baveno consensus, EASL, and AASLD. Furthermore, only a minority of studies reported the frequency of invaluable or indeterminate results due to technical failure. Finally, many studies did not report all relevant covariates (e.g. viral aetiology), which negatively impacted both their results and ours.

For all other index tests, we found two main issues: varying cut-off values due to the lack of previously evaluated and proposed cut-offs, and a relatively low number of eligible studies. 2D-SWE was the second most evaluated index test, but with cut-off values that varied greatly. Moreover, different devices were used: 2D-SWE by General Electric, Aixplorer Supersonic system, and Toshiba TUS-A500. Although most studies used the Aixplorer device, there is a need for standardisation of both cut-off values and devices.

LSM and SSM by pSWE were evaluated in only a few studies, and no cut-off values could be assessed for either CSPH or SPH. Therefore, more studies are needed to draw definitive conclusions regarding their diagnostic accuracy. A similar situation applies to MRE, for which only one eligible study was included (Kennedy 2022). This study represents one of the few pilot investigations into the diagnostic utility of MRE for CSPH and SPH, underscoring the need for further primary research.

Moreover, since our review focused exclusively on elastography techniques alone (without incorporating other components of composite scores such as platelet count), we excluded studies that evaluated LSM by VCTE in combination with other clinical parameters such as platelet count. This approach prevented us from evaluating the diagnostic performance of currently recommended criteria by the Baveno consensus to rule out CSPH (e.g. LSM < 15 kPa + platelet count > 150,000/mm^3) [28]. This topic warrants further investigation in future systematic reviews, particularly to assess the diagnostic performance of current Baveno recommendations and other validated composite scores that incorporate elastography alongside clinical parameters, as these would be highly relevant for clinical practice.

Finally, we found insufficient data regarding the combination of different index tests and their comparative accuracy. Direct (within-study) comparison between two index tests was impossible due to inconsistent numbers of participants included for each index test in studies that evaluated two or more index tests.

Strengths and weaknesses of the review process

Search strategy

Our strategy allowed us to obtain a substantial number of studies on various index tests conducted across different countries, highlighting a broad interest in incorporating newly evaluated non-invasive diagnostic methods into the care of patients with CLD. To avoid excluding potentially eligible participants from larger studies, we opted to include studies with up to 20% of participants who might otherwise have been excluded (Supplementary material 7). This approach allowed us to incorporate a larger number of studies, but came at the cost of reduced applicability due to a higher risk of bias in patient selection. Nevertheless, we found that, except for LSM by VCTE, relatively few eligible studies were available for other index tests.

Additionally, 34 authors of potentially eligible studies did not respond to our requests for data to complete 2 x 2 tables. This suggests there may be more eligible studies, but we currently lack sufficient data to include them in our review. Nonetheless, we are confident that we have included most, if not all, studies meeting our inclusion and exclusion criteria. Future research should focus on primary studies with prespecified cut-off values to improve the robustness of findings.

Quality assessment and data extraction

We tried to reduce the subjectivity in our attempts to assess the methodological quality of studies by using the objective QUADAS-C tool and GRADE as prespecified in the protocol. Two review authors independently assessed studies by QUADAS-C protocol and compared the results, discussing them in case of disagreement. Most disagreements were present when assessing the patient selection domain (12 studies) and flow and timing domain (7 studies). Moreover, two review authors independently extracted data using a prespecified extraction form, with any uncertainties resolved by discussion or in consultation with a third review author. Most disagreements and discordances were due to simple miscalculations or typos and were easily resolved. Both quality assessment and data extraction processes are a strength of this review, given that every step was done according to the *Cochrane Handbook for Diagnostic Test of Accuracy Reviews* to minimise the possibility of subjectivity and bias [117]. The same review authors assessed the certainty of evidence using the GRADE approach, and the level of agreement was high.

Review analysis

Analyses for most index tests were limited by the low number of identified studies. In addition, substantial heterogeneity between studies, due to varying cut-off values and inconsistent results, complicated the review process and made it challenging to generate meaningful pooled estimates. As planned, we used the HSROC model. We acknowledge that more complex modelling approaches, such as multithreshold methods described in the *Cochrane Handbook for Diagnostic Test of Accuracy Reviews* [117], represent valid analytical options, particularly when studies report multiple thresholds. However, given the limited number of studies reporting multiple cut-off values, inconsistent threshold reporting, and the overall very low certainty of the evidence, we judged that applying such methods would not meaningfully improve the precision or informativeness of our results. We therefore opted to use the HSROC model as the most appropriate synthesis method. We have provided a table with studies that reported multiple thresholds of the same index test for the same target condition and their values (Table 7).

Additionally, we focused on a subgroup of nine studies (1553 participants) that assessed LSM by VCTE using a predefined cut-off of 25 kPa. This allowed us to conduct bivariate meta-analysis and generate pooled estimates at the most commonly used threshold, although based on a reduced number of studies.

Moreover, we grouped the most frequently reported cut-off values into ranges as narrow as possible. This allowed us to perform bivariate analyses and generate meta-analysed estimates, which are reported in Table 5. We acknowledge that these meta-analysed estimates, based on cut-off ranges rather than exact thresholds, may be confusing or prone to misinterpretation in clinical practice, and potentially imprecise. However, this approach was an attempt to manage the considerable heterogeneity in the available data. In some instances, the ranges were relatively narrow (1 kPa to 2 kPa) and supported by a substantial amount of data. These results may have meaningful clinical implications and could help guide the selection of predefined cut-off values in future, in better-designed studies.

Regarding the investigation of sources of heterogeneity and sensitivity analysis, we focused solely on LSM by VCTE with a cut-off value of 25 kPa for CSPH. This cut-off is the only previously validated value and the only single cut-off (not a range) with sufficient study data to analyse. We considered most of the heterogeneity to be related to the use of different cut-off values.

This approach to investigating sources of heterogeneity only on LSM by VCTE prevented us from exploring the type of probe used for SSM by VCTE as a source of heterogeneity. In the majority of earlier studies, SSM by VCTE was usually performed by 50 Hz probe originally designed for LSM; however, recently a spleen-dedicated 100 Hz probe has been introduced (Stefanescu 2020 [217]), but only one study eligible for our review evaluated the diagnostic accuracy of this probe (Odriozola 2023). Since we expected a low number of studies with the spleen-dedicated probe, we decided to investigate SSM by VCTE with two modules combined, as well as whether an M or XL probe was used when LSM 50 Hz probe was used. Our results mostly estimate the use of liver 50 Hz probe (M and XL probe combined) since it was used in all studies except in Odriozola 2023. In future updates of the review, we expect new studies with the novel spleen-dedicated probe and plan to separately assess two different probes, or as a source of heterogeneity.

Among the planned subgroup analyses, we were able to explore the impact of CSPH prevalence (more or less than 50%) and to perform sensitivity analyses considering only full-text studies and only studies in which no participants met our exclusion criteria. These were the only feasible analyses given the limited number of studies and insufficient reporting in some cases (e.g. lack of data or covariates). A lower prevalence of CSPH was associated with higher specificity of the index test (P = 0.009). Ideally, the target population for this index test would consist of individuals with cACLD and a low prevalence of CSPH. Notably, when restricting the analysis to studies that fully adhered to our inclusion criteria (i.e. the ideal cohort), the pooled estimates for sensitivity and specificity were slightly lower than in the overall analysis, but remained broadly comparable. A similar trend was observed when including only full-text publications (see Table 4).

Additionally, further evaluation of elastography methods in specific populations, such as individuals with non-viral aetiologies or exclusively cACLD, is warranted, as originally planned (see Objectives). Assessing the performance of elastography techniques across different liver disease aetiologies would significantly improve the applicability of our findings in diverse clinical settings. For instance, viral hepatitis remains predominant in Asia and many low- and middle-income countries, whereas MASLD and alcohol-related liver disease are more prevalent in Western countries. We were unable to address this issue due to a lack of studies with clearly defined cohorts based on aetiology. Most studies included participants with mixed aetiologies, precluding the generation of pooled estimates for specific subgroups.

Moreover, given that the current guideline recommending LSM by VCTE > 25 kPa to rule in CSPH is applicable only to non-obese MASLD populations, it is critical to investigate obesity as a potential source of heterogeneity in cACLD populations, especially in light of the global obesity epidemic. Unfortunately, the absence of studies specifically enrolling obese participants with cACLD prevented us from exploring this factor.

As specified in our predefined protocol, we did not seek individual participant data, which further limited our ability to assess obesity and other individual-level variables as potential modifiers of diagnostic accuracy. Only aggregate data were extracted. We acknowledge this as a limitation of our review and highlight the evaluation of elastography performance in obese populations as a key area for future research.

Within- and between-study comparisons

We were unable to perform direct comparisons between tests because most studies that evaluated multiple tests reported data for different numbers of participants. As a result, we conducted indirect comparisons using SROC curves. However, due to the variability in cut-off values reported across studies, we could not compare the exact diagnostic accuracies of specific cut-off values for each test.

Comparison with previous research

To our knowledge, this review is the most comprehensive and only systematic review to assess the diagnostic accuracy of all elastography methods for detecting CSPH and SPH.

Several previous reviews have explored similar topics but with notable differences. For instance, You and colleagues published a systematic review in 2017 that included 11 studies with 1399 participants evaluating LSM by VCTE [99]. Their review, which included a broader range of studies including those with liver tumours, transplanted patients, HIV-positive patients, and patients with known varices – groups excluded by the exclusion criteria in our review – reported cut-off ranges of 13.6 kPa to 18 kPa and 21 kPa to 25 kPa with sensitivities of 92% and 71%, and specificities of 81% and 91%, respectively [99]. In contrast, our review included a larger number of studies with more narrowly defined populations, allowing for reduced clinical heterogeneity and more targeted subgroup analyses. Additionally, we used narrower cut-off ranges to better reflect clinical decision-making and provide more precise, applicable estimates of test performance.

Another meta-analysis published in 2017 by Kim and colleagues had substantial overlap with You's review and calculated summary sensitivity and specificity, without providing any cut-off-specific estimates [102]. Similarly, Song and colleagues in 2018 evaluated LSM by VCTE specifically in patients with viral aetiology, reporting sensitivities of 96%, 85%, and 74% for cut-offs around 13.6 kPa, 17.6 kPa, and 21.5 kPa, and corresponding specificities of 60%, 80%, and 94% [98]. Despite methodological differences, their results are broadly consistent with ours. A more recent meta-analysis by Kumar and colleagues included 26 studies evaluating LSM by VCTE for CSPH, with a total of 4337 participants, applying broader inclusion criteria than our review. Remarkably, most studies in that analysis were judged to be at low risk of bias in all domains, in contrast to our findings, where only two out of 47 studies were assessed as having a low risk of bias [106]. They reported an overall summary sensitivity of 79% and specificity of 88%, and a weighted mean of optimal cut-off value of 22.8 kPa – results that are broadly in line with ours, though based on more heterogeneously defined populations. The most recent meta-analysis, published by Rockey and colleagues in 2024, included only nine studies [105], a lower number than in our review, and not directly comparable.

The first systematic review on spleen stiffness was published by Song and colleagues in 2018. It included nine studies assessing SSM by various elastographic techniques with variable cut-off values. Their analysis demonstrated a correlation between SSM values and HVPG and a summary sensitivity of 88% (95% CI 70% to 96%) and a summary specificity of 84% (95% CI 72% to 92%) for CSPH [97]. Using a similar approach, Hu and colleagues in 2021 conducted a broader meta-analysis that included 32 studies. They reported a summary sensitivity of 85% (95% CI 69% to 93%) and a summary specificity of 86% (95% CI 74% to 93%) for CSPH [95]. This approach does not provide specific information about individual methods or cut-off values and is less applicable for clinical practice due to its overall pooled estimates.

Dajti and colleagues conducted a systematic review and individual patient data meta-analysis in 2023 focusing on SSM for diagnosing CSPH [100]. They found that combining the Baveno VII criteria for LSM (25 kPa) with either SSM > 40 kPa or platelet count < 150,000 significantly improved diagnostic accuracy, reducing the grey area of indeterminate findings [100]. These findings are consistent with those of our review, particularly regarding the high specificity of the 25 kPa cut-off for LSM.

Regarding 2D-SWE, Thiele and colleagues performed an individual patient data meta-analysis with 328 participants from five studies in 2020 [101]. They reported a cut-off of 14 kPa with a summary sensitivity of 91% and specificity of 37%, showing high false-negative rates, likely due to the inclusion of decompensated cirrhosis patients.

A previous review evaluating pSWE and 2D-SWE together reported substantial heterogeneity due to varying devices, techniques, and cut-off values, limiting the applicability to clinical practice [46]. To our knowledge, no systematic review has yet focused exclusively on pSWE for diagnosing CSPH or SPH.

In the case of MRE, previous reviews faced challenges due to limited studies and the use of suboptimal reference standards, such as the presence of splenomegaly, varices, or ascites [96, 103]. As a result, these reviews were unable to perform a robust meta-analysis.

Finally, Hai and colleagues published a network meta-analysis comparing various elastography methods, contrast-enhanced ultrasound (CEUS), and abdominal ultrasound in 2022. They found that pSWE and CEUS had the highest diagnostic accuracy, followed by VCTE, though substantial heterogeneity among the included studies limits the applicability of their results [104].

Comparison with current clinical guidelines

The use of various elastography techniques has been recommended in several clinical guidelines [28, 37, 68, 216, 218]. Our results suggest that LSM by VCTE > 25 kPa may have a specificity greater than 90%, allowing CSPH to be ruled in with reasonable confidence. Accordingly, the 'rule-of-five' recommended by Baveno consensus, and adopted by other guidelines, recommends to rule in CSPH when > 25 kPa in virus- and/or alcohol-related cACLD and non-obese (BMI < 30 kg/m^2) MASLD-related cACLD [28, 37, 68, 216, 218]. Due to the limited number of studies, we were unable to provide estimates stratified by aetiology or by obesity status. Future updates of this review will aim to address these knowledge gaps through subgroup analyses, in the case of sufficient data. Additionally, our findings show that an LSM cut-off in the range of 11.4 kPa to 14.4 kPa by VCTE may have a sensitivity above 90%, supporting its use to rule out CSPH. This supports the AASLD recommendation to rule out CSPH if LSM by VCTE < 15 kPa and raises the question of whether platelet count > 150,000/mm^3, as required by the Baveno consensus for ruling out CSPH, is truly necessary in combination with LSM < 15 kPa [28, 68, 216].

Although our findings do not allow us to define a clear role for SSM by VCTE in ruling in or ruling out CSPH, the Baveno consensus recommends ruling in CSPH if SSM by VCTE exceeds 50 kPa, at least in patients with viral aetiology [28]. The AASLD guideline states that SSM by VCTE > 40 kPa may be used to rule in CSPH, without specifying the underlying aetiology [68]. The EASL guideline acknowledges the utility of SSM by VCTE as an additional non-invasive tool, but does not provide specific cut-off values [37]. These guideline recommendations, as well as our own findings, are primarily based on studies that assessed SSM using the liver (50 Hz) probe. In fact, all studies included in our review used the 50 Hz probe, except for Odriozola 2023, which evaluated the spleen-dedicated 100 Hz probe. A growing number of studies are now investigating the spleen-specific 100 Hz probe in cACLD cohorts – although, to date, only one has focused on CSPH. As recommended by the Baveno guidelines, further research on the performance of the spleen-dedicated probe is warranted. This will be addressed in future updates of our review, provided a sufficient number of studies become available [28].

According to the 'rule-of-five', approximately 40% to 50% of patients have intermediate LSM values (15 kPa to 25 kPa) and fall into the so-called 'grey zone'. In these cases, SSM by VCTE, with cut-offs between 40 kPa and 50 kPa, has been proposed as a decisive tool for confirming CSPH or excluding high-risk varices [28, 37, 68, 100, 216]. We were unable to identify a sufficient number of studies evaluating the combination of LSM and SSM by VCTE. However, He 2022 showed that combination of LSM > 25 kPa and SSM > 50 kPa could be used to rule in CSPH, achieving a specificity of 98%. Most available studies instead assess Baveno consensus algorithms, which include platelet count < 150,000/mm^3 in addition to LSM and SSM. Therefore, systematic reviews aiming to evaluate such composite scores are warranted.

The results of our review were at a very low level of certainty due to high risk of bias, inconsistency, and imprecision in almost all included studies. Therefore, our estimates should be interpreted with caution. Similarly, current guideline recommendations are mostly based on the same studies at high risk of bias that we included in our review.

In contrast to other guidelines that do not yet recommend shear wave elastography methods for routine clinical use, the World Federation for Ultrasound in Medicine and Biology (WFUMB) guidelines suggest the 'rule of four' for LSM by pSWE and 2D-SWE [216]. According to WFUMB, an LSM cut-off of > 21 kPa (for 2D-SWE) or 2.6 m/s (for pSWE) strongly suggests CSPH. Some data suggest that an LSM cut-off of > 25 kPa by 2D-SWE might be adequate to rule in CSPH [Grgurević 2022, 218]. However, our review could not support these recommendations due to an insufficient number of eligible studies. Furthermore, no guidelines currently recommend using SSM by any shear wave elastography technique. Although interest in shear wave elastography is increasing, its role in routine clinical practice remains controversial.

There is a lack of studies evaluating MRE for CSPH, and there are no recommendations for its use in guidelines for this purpose [28, 37, 68, 216, 218].

None of the guidelines recommend any elastography technique for SPH [28, 37, 68, 216, 218].

Applicability of findings to the review question

The review question addressed a broad range of index tests for different target conditions across heterogeneous cohorts. Due to the limited number of studies, most sources of heterogeneity could not be thoroughly investigated, highlighting the need for further primary research for many of the index tests. Despite these challenges, we used the QUADAS-C tool to evaluate the applicability of our findings to routine clinical practice.

Applicability concerns

Patient selection domain: many studies had high applicability concerns due to issues such as insufficient reporting of the cohort characteristics or inclusion of participants with single aetiology, hepatocellular carcinoma, cholestatic liver disease, history of NSBB use (not current), or previously known CSPH/SPH or oesophageal varices. To expand our pool of included studies, we allowed up to 20% of such participants. Despite this, we found a median CSPH prevalence of 62.5% (IQR 51% to 76.7%), similar to the 66% found in a large cross-sectional study [17] with participants with compensated CLD, which aligns with our review's clinical question. However, we also observed an unexpectedly high prevalence of SPH (62.5%; IQR 54.0% to 71.7%), even though only 50% to 60% of people with CSPH are expected to have oesophageal varices [2, 25]. This discrepancy suggests the inclusion of more severe patients, at least in studies that targeted SPH.

Index test domain: most studies were judged as having a low concern for applicability regarding the index tests. Nearly all studies, except for one (Zhang 2022), used the FibroScan device for LSM or SSM by VCTE, which is the most validated method for assessing CSPH/SPH and other conditions such as liver fibrosis. Similarly, pSWE and 2D-SWE were conducted using standardised methods across studies, though the range of devices used (e.g. Siemens ACUSON S2000, Philips ElastPQ, and Aixplorer Supersonic system) indicates the need for future standardisation. While most studies used the Aixplorer Supersonic system for 2D-SWE, pSWE was performed with both Siemens and Philips devices. Future research should consider standardising devices to enhance comparability.

Reference standard domain: all studies were deemed low concern for applicability regarding the reference standards, as HVPG was performed in a standardised manner as specified in the methods.

AUTHORS' CONCLUSIONS

Implications for practice

Clinically significant portal hypertension (CSPH) in chronic liver disease is a key driver of decompensation, with severe portal hypertension (SPH) leading to complications such as oesophageal bleeding. Early detection is essential to initiate timely and appropriate treatment. While hepatic venous pressure gradient measurement remains the gold standard for assessing portal pressure, its invasiveness and cost have prompted a growing interest in non-invasive alternatives such as liver and spleen stiffness measurements. These tests offer indirect estimates of portal pressure and may serve as substitutes for invasive procedures in selected settings.

This review examined a broad range of elastographic techniques for two target conditions (CSPH and SPH) across heterogeneous chronic liver disease populations. Valid results were obtained only for liver stiffness measurement (LSM) by vibration-controlled transient elastography (VCTE) and 2D-shear wave elastography (2D-SWE). The accuracy of other techniques could not be assessed due to insufficient data. No index test demonstrated both sensitivity and specificity ≥ 90%, limiting their ability to reliably confirm or exclude CSPH or SPH. Most studies were at high risk of bias due to patient selection, post hoc determination of cut-off values, and incomplete reporting of test failures, rendering the evaluation of diagnostic accuracy unreliable. Applicability is also limited, as many studies included patients with decompensated cirrhosis. Furthermore, only a minority of studies reported test failure rates – an important consideration for routine clinical implementation.

Using LSM by VCTE with the most frequently employed predefined cut-off value of 25 kPa, we estimated that 38% of people with CSPH would be missed (false negatives), and 6% of people without CSPH would be unnecessarily treated (false positives). These results are based on very low-certainty evidence.

The current guideline recommendation of using LSM by VCTE > 25 kPa to rule in CSPH appears to be based on very low-certainty evidence. Our findings are consistent with this threshold and are also based on studies with high risk of bias and very low-certainty evidence. This threshold should therefore be applied with caution in clinical practice and ideally interpreted in conjunction with other clinical and diagnostic information.

The limited number of studies hindered accuracy assessment of most other index tests or their combinations and prevented making any meaningful comparisons.

Implications for research

Our review highlights that the studies assessing the accuracy of various elastography techniques for diagnosing CSPH and SPH are unreliable due to high risk of bias, inconsistency, and imprecision (wide confidence interval estimates). Despite widespread use and guideline recommendations, there are insufficient data on their diagnostic performance, including comparisons, combinations, and technical failure rates, to fully support their use in clinical practice. This suggests that many of these tests were introduced without a clear definition of their characteristics, such as cut-off values, or their role in the clinical pathway.

There is a critical need for more high-quality primary studies with predefined cut-off values and comprehensive reporting on test failures and patient selection. Future research should focus on the standardisation of elastography devices and methods to reduce heterogeneity. Studies should consider specific subpopulations, such as those with compensated advanced chronic liver disease, and different aetiologies of liver disease, to enable more personalised and clinically relevant conclusions. Further research should evaluate the combination of different non-invasive tests, mainly with conditionally independent tests (such as platelet count or other biomarkers), to improve diagnostic accuracy and applicability in clinical practice. Since most studies lacked predefined cut-off criteria, and thus by reporting the most optimal cut-offs overestimated the diagnotic accuracies of tests, we encourage authors of future diagnostic test accuracy studies to use cut-off values and ranges evaluated in our review as their predefined criteria for more objective estimation of diagnostic accuracies of elastography techniques.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: 10.1002/14651858.CD015415.pub2.

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Analyses

Supplementary material 5 Data package

Supplementary material 6 QUADAS-2 and QUADAS-C methodology assessment tools

Supplementary material 7 Differences between protocol and review

Supplementary material 8 Search strategy for other resources

ADDITIONAL INFORMATION

Acknowledgements

Cochrane Review Group funding acknowledgement: the Danish State is the largest single funder of the Cochrane Hepato-Biliary

Group (CHBG) through its investment in the Copenhagen Trial Unit, Centre for Clinical Intervention Research, the Capital Region of Denmark, Rigshospitalet, Copenhagen, Denmark. Disclaimer: the views and opinions expressed in this review are those of the authors and do not necessarily reflect those of the Danish State or the Copenhagen Trial Unit.

Cochrane Hepato-Biliary and Cochrane Diagnostic Test Accuracy Editorial Teams supported the authors in the development of this review.

The following people from the Cochrane Hepato-Biliary (CHB) Editorial Team conducted the editorial process for this article:

- Sign-off Editor (final editorial decision): Christian Gluud, Co-ordinating Editor CHBG, Denmark;
- Contact Editor (provided editorial decision): Chavdar Pavlov, Russia;
- Managing Editor (selected editors, provided editorial guidance to authors, edited the article): Dimitrinka Nikolova, Denmark;
- Information Specialist (database searches): Sarah Louise Klingenberg, Denmark.

Agostino Colli, Mirella Fraquelli, and Giovanni Casazza are editors within the CHBG but were not involved with the editorial process of this review.

Editorial and peer-reviewer contributions

The following people conducted the editorial process for this article:

- DTA Sign-off Editor (final editorial decision): Tanya Walsh, The University of Manchester;
- Clinical Sign-off Editor (final editorial decision): Emmanuel Tsochatzis, UCL Institute for Liver and Digestive Health, London, UK;
- Managing Editor (selected peer reviewers, provided editorial guidance to authors, edited the article): Liz Bickerdike, Cochrane Central Editorial Service;
- Editorial Assistant (conducted editorial policy checks, collated peer-reviewer comments, and supported the editorial team): Justin Mann, Cochrane Central Editorial Service;
- Copy Editor (copy editing and production): Lisa Winer, Cochrane Central Production Service;
- Peer reviewers (provided comments and recommended an editorial decision): Elton Dajti, Department of Medical and Surgical Sciences, University of Bologna, Bologna, Italy (clinical review), Tanya Walsh, The University of Manchester (statistical review), Dr Robert Meertens, University of Exeter (methods review), Julie Glanville, Glanville.info, York, UK (search review).

Contributions of authors

LV: wrote the protocol; assessed studies for inclusion; extracted data; assessed risk of bias; interpreted the analysis; and wrote the final review.

TN: wrote the protocol; acted as arbiter if review authors could not reach a consensus; extracted data; assessed risk of bias; interpreted the analysis; and wrote the final review.

DŠ: commented on the protocol, and critically commented on the final review.

MF: co-ordinated protocol design; critically commented on the protocol; interpreted the analysis; and critically commented on the final review.

CM: commented on the protocol; assessed studies for inclusion; and critically commented on the final review.

GC: critically commented on the protocol; co-ordinated protocol design; carried out and interpreted the analysis; and critically commented on the final review.

AC: wrote the protocol; critically commented on the protocol; co-ordinated protocol design; interpreted the analysis; and wrote the final review.

All authors approved the publication of the protocol and final review.

Declarations of interest

LV: none known

TN: none known

DŠ: none known

MF: none known

CM: none known

GC: none known

AC: AbbVie (Independent Contractor – Consultant), and HavasLife Milan (Independent Contractor – Consultant)

Sources of support

Internal sources

- No sources of support, Other

External sources

- New Source of support, Other

  The Cochrane Hepato-Biliary Group supported the development of the review.

Registration and protocol

Vranić L, Nadarevic T, Štimac D, Fraquelli M, Manzotti C, Casazza G, Colli A. Liver and spleen stiffness as assessed by vibration controlled transient elastography for diagnosing clinically significant portal hypertension in comparison with other elastography-based techniques in adults with chronic liver disease. Cochrane Database of Systematic Reviews 2022, Issue 8. Art. No.: CD015415. DOI: 10.1002/14651858.CD015415.

Data, code and other materials

Supplementary material 4; Supplementary material 5

History

Protocol first published: Issue 8, 2022

REFERENCES

1. Reiberger T. The value of liver and spleen stiffness for evaluation of portal hypertension in compensated cirrhosis. Hepatology Communications 2022;**6**(5):950-64. [DOI: 10.1002/hep4.1855]

2. Garcia-Tsao G, Abraldes JG, Berzigotti A, Bosch J. Portal hypertensive bleeding in cirrhosis: risk stratification, diagnosis, and management: 2016 practice guidance by the American Association for the Study of Liver Diseases. *Hepatology (Baltimore, Md.)* 2017;**65**:310–35.

3. Bosch J, Abraldes JG, Berzigotti A, García-Pagan JC. The clinical use of HVPG measurements in chronic liver disease. *Nature Reviews. Gastroenterology & Hepatology* 2009;**6**(10):573-82.

4. de Franchis R, Baveno VI Faculty. Expanding consensus in portal hypertension: Report of the Baveno VI Consensus Workshop: Stratifying risk and individualizing care for portal hypertension. *Journal of Hepatology* 2015;**63**(3):743-52.

5. Geraci G, Pisello F, Modica G, Li Volsi F, Arnone E, Sciumè C. Complications of elective esophago-gastro-duodenoscopy (EGDS). Personal experience and literature review. *Il Giornale di Chirurgia* 2009;**30**:502-6.

6. Leffler DA, Kheraj R, Garud S, Neeman N, Nathanson LA, Kelly CP, et al. The incidence and cost of unexpected hospital use after scheduled outpatient endoscopy. *Archives of Internal Medicine* 2010;**170**:1752-7.

7. D'Amico G, Garcia-Tsao G, Pagliaro L. Natural history and prognostic indicators of survival in cirrhosis: a systematic review of 118 studies. *Journal of Hepatology* 2006;**44**(1):217-31.

8. D'Amico G, Pasta L, Morabito A, D'Amico M, Caltagirone M, Malizia G, et al. Competing risks and prognostic stages of cirrhosis: a 25-year inception cohort study of 494 patients. *Alimentary Pharmacology & Therapeutics* 2014;**39**(10):1180-93.

9. Samonakis DN, Koulentaki M, Coucoutsi C, Augoustaki A, Baritaki C, Digenakis E, et al. Clinical outcomes of compensated and decompensated cirrhosis: a long term study. *World Journal of Hepatology* 2014;**6**(7):504-12.

10. Groszmann RJ, Garcia-Tsao G, Bosch J, Grace ND, Burroughs AK, Planas R, et al. Beta-blockers to prevent gastroesophageal varices in patients with cirrhosis. *New England Journal of Medicine* 2005;**353**:2254-61.

11. Ripoll C, Groszmann R, Garcia-Tsao G, Grace N, Burroughs A, Planas R, et al. Hepatic venous pressure gradient predicts clinical decompensation in patients with compensated cirrhosis. *Gastroenterology* 2007;**133**(2):481-8.

12. Ripoll C, Groszmann RJ, Garcia-Tsao G, Bosch J, Grace N, Burroughs A, et al. Hepatic venous pressure gradient predicts development of hepatocellular carcinoma independently of severity of cirrhosis. *Journal of Hepatology* 2009;**50**(5):923-8.

13. Amitrano L, Guardascione MA, Manguso F, Bennato R, Bove A, DeNucci C, et al. The effectiveness of current acute variceal bleed treatments in unselected cirrhotic patients: refining short-term prognosis and risk factors. *American Journal of Gastroenterology* 2012;**107**(12):1872-8.

14. Fortune BE, Garcia-Tsao G, Ciarleglio M, Deng Y, Fallon MB, Sigal S, et al. Child-Turcotte-Pugh class is best at stratifying risk in variceal hemorrhage: analysis of a US multicenter prospective study. *Journal of Clinical Gastroenterology* 2017;**51**(5):446-53.

15. Reverter E, Tandon P, Augustin S, Turon F, Casu S, Bastiampillai R, et al. A MELD-based model to determine risk of mortality among patients with acute variceal bleeding. *Gastroenterology* 2014;**146**(2):412-9.

16. Turco L, Garcia-Tsao G. Portal hypertension: pathogenesis and diagnosis. *Clinics in Liver Disease* 2019;**23**(4):573-87.

17. Abraldes JG, Bureau C, Stefanescu H, Augustin S, Ney M, Blasco H, et al. Noninvasive tools and risk of clinically significant portal hypertension and varices in compensated cirrhosis: The "Anticipate" study. *Hepatology (Baltimore, Md.)* 2016;**64**(6):2173-84.

18. Bosch J, Iwakiri Y. The portal hypertension syndrome: etiology, classification, relevance, and animal models. *Hepatology International* 2018;**12**:1-10.

19. Abraldes JG, Tarantino I, Turnes J, Garcia-Pagan JC, Rodes J, Bosch J. Hemodynamic response to pharmacological treatment of portal hypertension and long-term prognosis of cirrhosis. *Hepatology (Baltimore, Md.)* 2003;**37**(4):902-8.

20. D'Amico G, Garcia-Pagan JC, Luca A, Bosch J. HVPG reduction and prevention of variceal bleeding in cirrhosis: a systematic review. *Gastroenterology* 2006;**131**:1611-24.

21. Sanyal AJ, Harrison SA, Ratziu V, Abdelmalek MF, Diehl AM, Caldwell S, et al. The natural history of advanced fibrosis due to nonalcoholic steatohepatitis: data from the simtuzumab trials. *Hepatology (Baltimore, Md.)* 2019;**70**(6):1913-27.

22. Villanueva C, Aracil C, Colomo A, Hernandez-Gea V, Lopez-Balaguer JM, Alvarez-Urturi C, et al. Acute hemodynamic response to beta-blockers and prediction of long-term outcome in primary prophylaxis of variceal bleeding. *Gastroenterology* 2009;**137**(1):119-28.

23. Garcia-Tsao G, Groszmann RJ, Fisher RL, Conn HO, Atterbury CE, Glickman M. Portal pressure, presence of gastroesophageal varices and variceal bleeding. *Hepatology (Baltimore, Md.)* 1985;**5**(3):419-24.

24. Lebrec D, De Fleury P, Rueff B, Nahum H, Benhamou JP. Portal hypertension, size of esophageal varices, and risk of gastrointestinal bleeding in alcoholic cirrhosis. *Gastroenterology* 1980;**79**:1139-44.

25. Kibrit J, Khan R, Jung BH, Koppe S. Clinical assessment and management of portal hypertension. *Seminars in Interventional Radiology* 2018;**35**(3):153-9.

26. Villanueva C, Albillos A, Genescà J, Abraldes JG, Calleja JL, Aracil C, et al. Development of hyperdynamic circulation and response to β-blockers in compensated cirrhosis with portal hypertension. *Hepatology (Baltimore, Md.)* 2016;**63**(1):197-206.

27. Villanueva C, Albillos A, Genescà J, Garcia-Pagan JC, Calleja JL, Aracil C, et al. B blockers to prevent decompensation of cirrhosis in patients with clinically significant portal hypertension (PREDESCI): a randomised, double-blind, placebo-controlled, multicentre trial. *Lancet* 2019;**3933**(10181):1597-608.

28. de Franchis R, Bosch J, Garcia-Tsao G, Reiberger T, Ripoll C; Baveno VII Faculty. Baveno VII - Renewing consensus in portal hypertension. *Journal of Hepatology* 2022;**76**(4):959-74. [DOI: 10.1016/j.jhep.2021.12.022]

29. Sigrist RM, Liau J, Kaffas AE, Chammas MC, Willmann JK. Ultrasound elastography: review of techniques and clinical applications. *Theranostics* 2017;**7**(5):1303-29.

30. Shi KQ, Fan YC, Pan ZZ, Lin XF, Liu WY, Chen YP, et al. Transient elastography: a meta-analysis of diagnostic accuracy in evaluation of portal hypertension in chronic liver disease. *Liver International* 2013;**33**(1):62-71.

31. Stefanescu H, Marasco G, Calès P, Fraquelli M, Rosselli M, Ganne-Carriè N, et al. A novel spleen-dedicated stiffness measurement by FibroScan® improves the screening of high-risk oesophageal varices. *Liver International* 2020;**40**(1):175-85.

32. Fraquelli M, Rigamonti C, Casazza G, Conte D, Donato MF, Ronchi G, et al. Reproducibility of transient elastography in the evaluation of liver fibrosis in patients with chronic liver disease. *Gut* 2007;**56**:968-73.

33. European Association for the Study of the Liver; Asociacion Latinoamericana para el Estudio del Higado. EASL-ALEH Clinical Practice Guidelines: Non-invasive tests for evaluation of liver disease severity and prognosis. *Journal of Hepatology* 2015;**63**(1):237-64.

34. Castéra L, Foucher J, Bernard PH, Carvalho F, Allaix D, Merrouche W, et al. Pitfalls of liver stiffness measurement: a 5-year prospective study of 13,369 examinations. *Hepatology (Baltimore, Md.)* 2010;**51**(3):828-35.

35. Castera L, Friedrich-Rust M, Loomba R. Noninvasive assessment of liver disease in patients with nonalcoholic fatty liver disease. *Gastroenterology* 2019;**156**(5):1264-81.

36. Vuppalanchi R, Siddiqui MS, Van Natta ML, Hallinan E, Brandman D, Kowdley K, et al. Performance characteristics of vibration-controlled transient elastography for evaluation of nonalcoholic fatty liver disease. *Hepatology (Baltimore, Md.)* 2018;**67**(1):134–44.

37. European Association for the Study of the Liver. EASL Clinical Practice Guidelines on non-invasive tests for evaluation of liver disease severity and prognosis - 2021 update. *Journal of Hepatology* 2021;**75**(3):659-89.

38. Pavlov CS, Casazza G, Nikolova D, Tsochatzis E, Burroughs AK, Ivashkin VT, et al. Transient elastography for diagnosis of stages of hepatic fibrosis and cirrhosis in people with alcoholic liver disease. *Cochrane Database of Systematic Reviews* 2015, Issue 1. Art. No: CD010542. [DOI: 10.1002/14651858.CD010542.pub2]

39. Colecchia A, Montrone L, Scaioli E, Bacchi-Reggiani ML, Colli A, Casazza G, et al. Measurement of spleen stiffness to evaluate portal hypertension and the presence of esophageal varices in patients with HCV-related cirrhosis. *Gastroenterology* 2012;**143**(3):646-54.

40. Fraquelli M, Giunta M, Pozzi R, Rigamonti C, Della Valle S, Massironi S, et al. Feasibility and reproducibility of spleen transient elastography and its role in combination with liver transient elastography for predicting the severity of chronic viral hepatitis. *Journal of Viral Hepatitis* 2014;**21**(2):90-8.

41. Zykus R, Jonaitis L, Petrenkienė V, Pranculis A, Kupčinskas L. Liver and spleen transient elastography predicts portal hypertension in patients with chronic liver disease: a prospective cohort study. *BMC Gastroenterology* 24;**15**:183.

42. Woo H, Lee JY, Yoon JH, Kim W, Cho B, Choi BI. Comparison of the reliability of acoustic radiation force impulse imaging and supersonic shear imaging in measurement of liver stiffness. *Radiology* 2015;**277**(3):881–6.

43. Cabassa P, Ravanelli M, Rossini A, Contessi G, Almajdalawi R, Maroldi R. Acoustic radiation force impulse quantification of spleen elasticity for assessing liver fibrosis. *Abdominal Imaging* 2015;**40**:738–44.

44. Ferraioli G, Tinelli C, Lissandrin R, Zicchetti M, Bernuzzi S, Salvaneschi L, et al. Ultrasound point shear wave elastography assessment of liver and spleen stiffness: effect of training on repeatability of measurements. *European Radiology* 2014;**24**:1283–9.

45. Cassinotto C, Charrie A, Mouries A, Lapuyade B, Hiriart JB, Vergniol J, et al. Liver and spleen elastography using supersonic shear imaging for the non-invasive diagnosis of cirrhosis severity and oesophageal varices. *Digestive and Liver Disease* 2015;**47**(8):695-701.

46. Suh CH, Kim KW, Park SH, Lee SS, Kim HS, Tirumani SH, et al. Shear wave elastography as a quantitative biomarker of clinically significant portal hypertension: a systematic review and meta-analysis. *AJR. American Journal of Roentgenology* 2018;**210**(5):185-95.

47. Salzl P, Reiberger T, Ferlitsch M, Payer B, Schwengerer B, Trauner M, et al. Evaluation of portal hypertension and varices by acoustic radiation force impulse imaging of the liver compared to transient elastography and AST to platelet ratio index. *Ultraschall in der Medizin* 2014;**35**:528-33.

48. Mandorfer M, Hernández-Gea V, García-Pagán JC, Reiberger T. Noninvasive diagnostics for portal hypertension: a comprehensive review. *Seminars in Liver Disease* 2020;**40**:240-55.

49. Dietrich CF, Bamber J, Berzigotti A, Bota S, Cantisani V, Castera L, et al. EFSUMB guidelines and recommendations on the clinical use of liver ultrasound elastography, update 2017 (long version). *Ultraschall in der Medizin* 2017;**38**(4):16-47.

50. Elkrief L, Rautou PE, Ronot M, Lambert S, Dioguardi Burgio M, Francoz C, et al. Prospective comparison of spleen and liver stiffness by using shear-wave and transient elastography for detection of portal hypertension in cirrhosis. *Radiology* 2015;**275**(2):589-98.

51. Jansen C, Bogs C, Verlinden W, Thiele M, Möller P, Görtzen J, et al. Shear-wave elastography of the liver and spleen identifies clinically significant portal hypertension: a prospective multicentre study. *Liver International* 2017;**37**(3):396-405.

52. Kim TY, Jeong WK, Sohn JH, Kim J, Kim MY, Kim Y. Evaluation of portal hypertension by real-time shear wave elastography in cirrhotic patients. *Liver International* 2015;**35**(11):2416-24.

53. Procopet B, Berzigotti A, Abraldes JG, Turon F, Hernandez-Gea V, García-Pagán JC, et al. Real-time shear-wave elastography: applicability, reliability and accuracy for clinically significant portal hypertension. *Journal of Hepatology* 2015;**62**:1068-75.

54. Zhu YL, Ding H, Fu TT, Peng SY, Chen SY, Luo JJ, et al. Portal hypertension in hepatitis B-related cirrhosis: diagnostic accuracy of liver and spleen stiffness by 2-D shear-wave elastography. *Hepatology Research* 2019;**49**(5):540-9.

55. Elkrief L, Ronot M, Andrade F, Dioguardi Burgio M, Issoufaly T, Zappa M, et al. Non-invasive evaluation of portal hypertension using shear-wave elastography: analysis of two algorithms combining liver and spleen stiffness in 191 patients with cirrhosis. *Alimentary Pharmacology & Therapeutics* 2018;**47**(5):621-30.

56. Bamber J, Cosgrove D, Dietrich CF, Fromageau J, Bojunga J, Calliada F, et al. EFSUMB guidelines and recommendations on the clinical use of ultrasound elastography. Part 1: basic principles and technology. *Ultraschall in der Medizin* 2013;**34**(2):169-84.

57. Attia D, Schoenemeier B, Rodt T, Negm AA, Lenzen H, Lankisch TO, et al. Evaluation of liver and spleen stiffness with acoustic radiation force impulse quantification elastography for diagnosing clinically significant portal hypertension. *Ultraschall in der Medizin* 2015;**36**(6):603-10.

58. Takuma Y, Nouso K, Morimoto Y, Tomokuni J, Sahara A, Takabatake H, et al. Portal hypertension in patients with liver cirrhosis: diagnostic accuracy of spleen stiffness. *Radiology* 2016;**279**(2):609-19.

59. Guglielmo FF, Venkatesh SK, Mitchell DG. Liver MR elastography technique and image interpretation: pearls and pitfalls. *Radiographics* 2019;**39**(7):1983-2002.

60. Yin M, Glaser KJ, Talwalkar JA, Chen J, Manduca A, Ehman RL. Hepatic MR elastography: clinical performance in a series of 1377 consecutive examinations. *Radiology* 2016;**278**(1):114–24.

61. Lee YJ, Lee JM, Lee JE, Lee KB, Lee ES, Yoon J-H, et al. MR elastography for noninvasive assessment of hepatic fibrosis: reproducibility of the examination and reproducibility and repeatability of the liver stiffness value measurement. *Magnetic Resonance Imaging* 2014;**39**(2):326–31.

62. Cheng YW, Chang YC, Chen YL, Chen RC, Chou CT. Feasibility of measuring spleen stiffness with MR elastography and splenic volume to predict hepatic fibrosis stage. *PloS One* 2019;**14**(5):e0217876.

63. Talwalkar JA, Yin M, Venkatesh S, Rossman PJ, Grimm RC, Manduca A, et al. Feasibility of in vivo MR elastographic splenic stiffness measurements in the assessment of portal hypertension. *AJR. American Journal of Roentgenology* 2009;**193**(1):122-7.

64. Singh S, Venkatesh SK, Loomba R, Wang Z, Sirlin C, Chen J, et al. Magnetic resonance elastography for staging liver fibrosis in non-alcoholic fatty liver disease: a diagnostic accuracy systematic review and individual participant data pooled analysis. *European Radiology* 2016;**26**(5):1431-40.

65. Kennedy P, Stocker D, Carbonell G, Said D, Bane O, Hectors S, et al. MR elastography outperforms shear wave elastography for the diagnosis of clinically significant portal hypertension. *European Radiology* 2022;**32**(12):8339-49. [DOI: 10.1007/s00330-022-08935-9]

66. Ronot M, Lambert S, Elkrief L, Doblas S, Rautou PE, Castera L, et al. Assessment of portal hypertension and high-risk oesophageal varices with liver and spleen three-dimensional multifrequency MR elastography in liver cirrhosis. *European Radiology* 2014;**24**:1394–402.

67. Mandorfer M, Kozbial K, Schwabl P, Freissmuth C, Schwarzer R, Stern R, et al. Sustained virologic response to interferon-free therapies ameliorates HCV-induced portal hypertension. *Journal of Hepatology* 2016;**65**(4):692-9.

68. Sterling RK, Asrani SK, Levine D, Duarte-Rojo A, Patel K, Fiel MI, et al. AASLD Practice Guideline on noninvasive liver disease assessment of portal hypertension. *Hepatology (Baltimore, Md.)* 2025;**81**(3):1060-85.

69. Colli A, Gana JC, Yap J, Adams-Webber T, Rashkovan N, Ling SC, et al. Platelet count, spleen length, and platelet count-to-spleen length ratio for the diagnosis of oesophageal varices in people with chronic liver disease or portal vein thrombosis. *Cochrane Database of Systematic Reviews* 2017, Issue 4. Art. No: CD008759. [DOI: 10.1002/14651858.CD008759.pub2]

70. Leeming DJ, Karsdal MA, Byrjalsen I, Bendtsen F, Trebicka J, Nielsen MJ, et al. Novel serological neo-epitope markers of extracellular matrix proteins for the detection of portal hypertension. *Alimentary Pharmacology & Therapeutics* 2013;**38**:1086-96.

71. Ferlitsch M, Reiberger T, Hoke M, Salzl P, Schwengerer B, Ulbrich G, et al. Von Willebrand factor as new noninvasive predictor of portal hypertension, decompensation and mortality inpatients with liver cirrhosis. *Hepatology (Baltimore, Md.)* 2012;**56**:1439-47.

72. La Mura V, Reverter JC, Flores-Arroyo A, Raffa S, Reverter E, Seijo S, et al. Von Willebrand factor levels predict clinical outcome in patients with cirrhosis and portal hypertension. *Gut* 2011;**60**:1133-8.

73. Hametner S, Ferlitsch A, Ferlitsch M, Etschmaier A, Schöfl R, Ziachehabi A, et al. The VITRO score (von Willebrand factor antigen/thrombocyte ratio) as a new marker for clinically significant portal hypertension in comparison to other non-invasive parameters of fibrosis including ELF test. *PloS One* 2016;**11**:e0149230. [DOI: 10.1371/journal.pone.0149230]

74. Berzigotti A, Seijo S, Arena U, Abraldes JG, Vizzutti F, García–Pagán JC, et al. Elastography, spleen size, and platelet count identify portal hypertension in patients with compensated cirrhosis. *Gastroenterology* 2013;**144**:102-11.

75. Simbrunner B, Marculescu R, Scheiner B, Schwabl P, Bucsics T, Stadlmann A, et al. Non-invasive detection of portal hypertension by enhanced liver fibrosis score in patients with different aetiologies of advanced chronic liver disease. *Liver International* 2020;**40**:1713-24.

76. Lisotti A, Azzaroli F, Buonfiglioli F, Montagnani M, Cecinato P, Turco L, et al. Indocyanine green retention test as a noninvasive marker of portal hypertension and esophageal varices in compensated liver cirrhosis. *Hepatology (Baltimore, Md.)* 2014;**59**:643-50.

77. Lisotti A, Azzaroli F, Cucchetti A, Buonfiglioli F, Cecinato P, Calvanese C, et al. Relationship between indocyanine green retention test, decompensation and survival in patients with Child-Pugh A cirrhosis and portal hypertension. *Liver International* 2016;**36**:1313-21.

78. Amat-Roldan I, Berzigotti A, Gilabert R, Bosch J. Assessment of hepatic vascular network connectivity with automated graph analysis of dynamic contrast-enhanced US to evaluate portal hypertension in patients with cirrhosis: a pilot study. *Radiology* 2015;**277**:268–76.

79. Berzigotti A, Piscaglia F, Amat-Roldan I, Gilabert R, Procopet B, Stefanescu H, et al. Non-invasive measurement of HVPG using graph analysis of dynamic contrast-enhanced ultrasound: the CLEVER study. *Journal of Hepatology* 2018;**68**:76-7.

80. Eisenbrey JR, Dave JK, Halldorsdottir VG, Merton DA, Miller C, Gonzalez JM, et al. Chronic liver disease: noninvasive subharmonic aided pressure estimation of hepatic venous pressure gradient. *Radiology* 2013;**268**(2):581-8.

81. Kim MY, Suk KT, Baik SK, Kim HA, Cha SH, Kwak HR, et al. Hepatic vein arrival time as assessed by contrast-enhanced ultrasonography is useful for the assessment of portal hypertension in compensated cirrhosis. *Hepatology (Baltimore, Md.)* 2012;**56**:1053–62.

82. Annet L, Materne R, Danse E, Jamart J, Horsmans Y, Van Beers BE. Hepatic flow parameters measured with MR imaging and Doppler US: correlations with degree of cirrhosis and portal hypertension. *Radiology* 2003;**229**:409–14.

83. Asenbaum U, Ba-Ssalamah A, Mandorfer M, Nolz R, Furtner J, Reiberger T, et al. Effects of portal hypertension on gadoxetic acid-enhanced liver magnetic resonance: diagnostic and prognostic implications. *Investigative Radiology* 2017;**52**(8):462-9.

84. Gouya H, Grabar S, Vignaux O, Saade A, Pol S, Legmann P, et al. Portal hypertension in patients with cirrhosis: indirect assessment of hepatic venous pressure gradient by measuring azygos flow with 2D-cine phase-contrast magnetic resonance imaging. *European Radiology* 2016;**26**(7):1981-90.

85. Iranmanesh P, Vazquez O, Terraz S, Majno P, Spahr L, Poncet A, et al. Accurate computed tomography-based portal pressure assessment in patients with hepatocellular carcinoma. *Journal of Hepatology* 2014;**60**(5):969-74.

86. Liu F, Ning Z, Liu Y, Liu D, Tian J, Luo H, et al. Development and validation of a radiomics signature for clinically significant portal hypertension in cirrhosis (CHESS1701): a prospective multicenter study. *EBioMedicine* 2018;**36**:151-8.

87. Qi X, An W, Liu F, Qi R, Wang L, Liu Y. Virtual hepatic venous pressure gradient with CT angiography (CHESS 1601): a prospective multicenter study for the noninvasive diagnosis of portal hypertension. *Radiology* 2019;**290**(2):370-7.

88. Cho EJ, Kim MY, Lee J-H, Lee IY, Lim YL, Choi DH, et al. Diagnostic and prognostic values of noninvasive predictors of portal hypertension in patients with alcoholic cirrhosis. *PloS One* 2015;**10**:e0133935. [DOI: 10.1371/journal.pone.0133935]

89. Kirnake V, Arora A, Sharma P, Goyal M, Chawlani R, Toshniwal J, et al. Non-invasive aspartate aminotransferase to platelet ratio index correlates well with invasive hepatic venous pressure gradient in cirrhosis. *Indian Journal of Gastroenterology* 2018;**37**:335-41.

90. Procopet B, Cristea VM, Robic MA, Grigorescu M, Agachi PS, Metivier S, et al. Serum tests, liver stiffness and artificial neural networks for diagnosing cirrhosis and portal hypertension. *Digestive and Liver Disease* 2015;**47**:411-6.

91. Verma V, Sarin SK, Sharma P, Kumar A. Correlation of aspartate aminotransferase/platelet ratio index with hepatic venous pressure gradient in cirrhosis. *United European Gastroenterology Journal* 2014;**2**:226-31.

92. Kim BK, Han KH, Park JY, Ahn SH, Kim JK, Paik YH, et al. A liver stiffness measurement-based, noninvasive prediction model for high-risk esophageal varices in B-viral liver cirrhosis. *American Journal of Gastroenterology* 2010;**105**(6):1382-90.

93. Hong WK, Kim MY, Baik SK, Shin SY, Kim JM, Kang YS, et al. The usefulness of non-invasive liver stiffness measurements in predicting clinically significant portal hypertension in cirrhotic patients: Korean data. *Clinical and Molecular Hepatology* 2013;**19**:370-5.

94. Reiberger T, Ferlitsch A, Payer BA, Pinter M, Schwabl P, Stift J, et al. Noninvasive screening for liver fibrosis and portal hypertension by transient elastography – a large single center experience. *Wiener Klinische Wochenschrift* 2012;**124**:395-402.

95. Hu X, Huang X, Hou J, Ding L, Su C, Meng F. Diagnostic accuracy of spleen stiffness to evaluate portal hypertension and esophageal varices in chronic liver disease: a systematic review and meta-analysis. *European Radiology* 2021;**31**(4):2392-404.

96. Singh R, Wilson MP, Katlariwala P, Murad MH, McInnes MD, Low G. Accuracy of liver and spleen stiffness on magnetic resonance elastography for detecting portal hypertension: a systematic review and meta-analysis. *European Journal of Gastroenterology & Hepatology* 2021;**32**(2):237-45.

97. Song J, Huang J, Huang H, Liu S, Luo Y. Performance of spleen stiffness measurement in prediction of clinical significant portal hypertension: a meta-analysis. *Clinics and Research in Hepatology and Gastroenterology* 2018;**42**(3):216-26.

98. Song J, Ma Z, Huang J, Liu S, Luo Y, Lu Q, et al. Comparison of three cut-offs to diagnose clinically significant portal hypertension by liver stiffness in chronic viral liver diseases: a meta-analysis. *European Radiology* 2018;**28**(12):5221-30.

99. You MW, Kim KW, Pyo J, Huh J, Kim HJ, Lee SJ, et al. A meta-analysis for the diagnostic performance of transient elastography for clinically significant portal hypertension. *Ultrasound in Medicine & Biology* 2017;**43**(1):59-68.

100. Dajti E, Ravaioli F, Zykus R, Rautou PE, Elkrief L, Grgurevic I, et al; Spleen Stiffness—IPD-MA Study Group. Accuracy of spleen stiffness measurement for the diagnosis of clinically significant portal hypertension in patients with compensated advanced chronic liver disease: a systematic review and individual patient data meta-analysis. *Lancet Gastroenterology and Hepatology* 2023;**8**(9):816-28.

101. Thiele M, Hugger MB, Kim Y, Rautou PE, Elkrief L, Jansen C, et al. 2D shear wave liver elastography by Aixplorer to detect portal hypertension in cirrhosis: an individual patient data meta-analysis. *Liver International* 2020;**40**(6):1435-46.

102. Kim G, Kim MY, Baik SK. Transient elastography versus hepatic venous pressure gradient for diagnosing portal hypertension: a systematic review and meta-analysis. *Clinical and Molecular Hepatology* 2017;**23**(1):34-41.

103. Yiyi M, Xiaoqin Q, Lei Z. Spleen stiffness on magnetic resonance elastography for the detection of portal hypertension: a systematic review and meta-analysis. *Iranian Journal of Public Health* 2022;**51**(9):1925-35.

104. Hai Y, Chong W, Eisenbrey JR, Forsberg F. Network meta-analysis: noninvasive imaging modalities for identifying clinically significant portal hypertension. *Digestive Diseases and Sciences* 2022;**67**(7):3313-26.

105. Rockey DC, Alsawas M, Duarte-Rojo A, Patel K, Levine D, Asrani SK, et al. Non-invasive liver disease assessment to identify portal hypertension: a systematic review supporting the AASLD Practice Guideline. Hepatology 2024 Mar 15 [Epub ahead of print]. [DOI: 10.1097/HEP.0000000000000841]

106. Kumar A, Maruyama H, Arora A, Sharma P, Anikhindi SA, Bansal N, et al. Diagnostic accuracy of transient elastography in diagnosing clinically significant portal hypertension in patients with chronic liver disease: a systematic review and meta-analysis. *Journal of Medical Ultrasonics (2001)* 2022;**49**(3):333-46.

107. Colli A, Gana JC, Turner D, Yap J, Adams-Webber T, Ling SC, et al. Capsule endoscopy for the diagnosis of oesophageal varices in people with chronic liver disease or portal vein thrombosis. *Cochrane Database of Systematic Reviews* 2014, Issue 10. Art. No: CD008760. [DOI: 10.1002/14651858.CD008760.pub2]

108. Wong RJ, Aguilar M, Cheung R, Perumpail RB, Harrison SA, Younossi ZM, et al. Nonalcoholic steatohepatitis is the second leading etiology of liver disease among adults awaiting liver transplantation in the United States. *Gastroenterology* 2015;**148**(3):547-55.

109. Colli A, Fraquelli M, Casazza G, Conte D, Nikolova D, Duca P, et al. The architecture of diagnostic research: from bench to bedside – research guidelines using liver stiffness as an example. *Hepatology (Baltimore, Md.)* 2014;**60**(1):408-18.

110. Whiting PF, Rutjes AW, Westwood ME, Mallett S, Deeks JJ, Reitsma JB, et al. QUADAS-2: a revised tool for the quality assessment of diagnostic accuracy studies. *Annals of Internal Medicine* 2011;**155**(8):529-36.

111. Sourianarayanane A, Talluri J, Humar A, McCullough AJ. Stage of fibrosis and portal pressure correlation in nonalcoholic steatohepatitis. *European Journal of Gastroenterology & Hepatology* 2017;**29**(5):516-23.

112. Ferrusquía-Acosta J, Bassegoda O, Turco L, Reverter E, Pellone M, Bianchini M, et al. Agreement between wedged hepatic venous pressure and portal pressure in non-alcoholic steatohepatitis-related cirrhosis. *Journal of Hepatology* 2021;**74**(4):811-8.

113. Bettinger D, Berzigotti A, Mandorfer M, Ripoll C, Labenz C, Zizer E, et al. Transjugular diagnostic procedures in hepatology: indications, techniques and interpretation. *JHEP Reports* 2025;**7**(8):101437. [DOI: 10.1016/j.jhepr.2025.101437]

114. Spijker R, Dinnes J, Glanville J, Eisinga A. Chapter 6: Searching for and selecting studies. In: Deeks JJ, Bossuyt PM, Leeflang MM, Takwoingi Y (editors). *Cochrane Handbook for Systematic Reviews of Diagnostic Test Accuracy* Version 2.0 (updated July 2023). The Cochrane Collaboration, 2023. Available from https://www.cochrane.org/authors/handbooks-and-manuals/handbook-systematic-reviews-diagnostic-test-accuracy.

115. Rethlefsen ML, Kirtley S, Waffenschmidt S, Ayala AP, Moher D, Page MJ, et al; PRISMA-S Group. PRISMA-S: an extension to the PRISMA statement for reporting literature searches in systematic reviews. *Systematic Reviews* 2021;**10**(1):39.

116. Lefebvre C, Glanville J, Briscoe S, Littlewood A, Marshall C, Metzendorf M-I, et al. Technical supplement to Chapter 4: Searching for and selecting studies. In: Higgins JP, Thomas J, Chandler J, Cumpston MS, Li T, Page MJ, Welch VA, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.3 (updated February 2022). Cochrane, 2022. Available from https://www.cochrane.org/authors/handbooks-and-manuals/handbook/archive/v6.3.

117. Deeks JJ, Bossuyt PM, Leeflang MM, Takwoingi Y, editor(s). *Cochrane Handbook for Systematic Reviews of Diagnostic Test Accuracy* Version 2.0. The Cochrane Collaboration, 2023. Available from https://www.cochrane.org/authors/handbooks-and-manuals/handbook-systematic-reviews-diagnostic-test-accuracy.

118. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ (Clinical Research Ed.)* 2021;**372**:n71.

119. Page MJ, Moher D, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. PRISMA 2020 explanation and elaboration: updated guidance and exemplars for reporting systematic reviews. *BMJ (Clinical Research Ed.)* 2021;**372**:n160.

120. Colecchia A, Montrone L, Scaioli E, Bacchi-Reggiani ML, Colli A, Casazza G, et al. Measurement of spleen stiffness to evaluate portal hypertension and the presence of esophageal varices in patients with HCV-related cirrhosis. *Gastroenterology* 2012;**143**(3):646-54.

121. Review Manager (RevMan). Version 8.5.1. The Cochrane Collaboration, 2024. Available at https://revman.cochrane.org.

122. Yang B, Mallett S, Takwoingi Y, Davenport CF, Hyde CJ, Whiting PF, et al; QUADAS-C Group. QUADAS-C: a tool for assessing risk of bias in comparative diagnostic accuracy studies. *Annals of Internal Medicine* 2021;**174**(11):1592-9.

123. Takwoingi Y, Leeflang MM, Deeks JJ. Empirical evidence of the importance of comparative studies of diagnostic test accuracy. *Annals of Internal Medicine* 2013;**158**(7):544-54.

124. Takwoingi Y, Deeks J. MetaDAS: A SAS macro for meta-analysis of diagnostic accuracy studies. User Guide Version 1.3. https://methods.cochrane.org/sdt/sites/methods.cochrane.org.sdt/files/uploads/MetaDAS%20Quick%20Reference%20v1.3%20May%202012.pdf (accessed 12 September 2024).

125. Balshem H, Helfand M, Schunemann HJ, Oxman AD, Kunz R, Brozek J, et al. GRADE guidelines: 3. Rating the quality of evidence. *Journal of Clinical Epidemiology* 2011;**64**(4):401-6.

126. GRADEpro GDT. Version accessed 12 September 2024. Hamilton (ON): McMaster University (developed by Evidence Prime), 2024. Available at https://www.gradepro.org.

127. Schünemann HJ, Mustafa R, Brozek J, Steingart KR, Leeflang M, Murad MH, et al. GRADE guidelines: 21 part 1. Study design, risk of bias and indirectness in rating the certainty across a body of evidence for test accuracy. *Journal of Clinical Epidemiology* 2020;**122**:129-41.

128. Schünemann HJ, Mustafa R, Brozek J, Steingart KR, Leeflang M, Murad MH, et al. GRADE guidelines: 21 part 2. Inconsistency, Imprecision, publication bias and other domains for rating the certainty of evidence for test accuracy and presenting it in evidence profiles and summary of findings tables. *Journal of Clinical Epidemiology* 2020;**122**:142-52.

129. Attia D, Schoenemeier B, Rodt T, Negm AA, Lenzen H, Lankisch TO, et al. Evaluation of liver and spleen stiffness with acoustic radiation force impulse quantification elastography for diagnosing clinically significant portal hypertension. *Ultraschall in der Medizin* 2015;**36**(6):603-10. [DOI: 10.1055/s-0041-107971]

130. Augustin S, Millán L, González A, Martell M, Gelabert A, Segarra A, et al. Detection of early portal hypertension with routine data and liver stiffness in patients with asymptomatic liver disease: a prospective study. *Journal of Hepatology* 2014;**60**(3):561-9.

131. Banini BA, Patel S, Yu JW, Kang L, Bailey C, Strife BJ, et al. Derivation and validation of a model to predict clinically significant portal hypertension using transient elastography and FIB-4. *Journal of Clinical Gastroenterology* 2023;**57**(2):189-97. [DOI: 10.1097/MCG.0000000000001664]

132. Borghi A, Venerandi L, Palamà C, Salvatore V, Schiumerini R, Renzulli M, et al. Splenic stiffness assessed by ARFI (acoustic radiation force impulse) correlates with portal pressure in liver cirrhosis. *Journal of Hepatology* 2012;**44**:21-2. [DOI: 10.1016/S1590-8658(12)60060-7]

133. Bureau C, Metivier S, Peron JM, Selves J, Robic MA, Gourraud PA, et al. Transient elastography accurately predicts presence of significant portal hypertension in patients with chronic liver disease. *Alimentary Pharmacology & Therapeutics* 2008;**27**(12):1261-8. [DOI: 10.1111/j.1365-2036.2008.03701.x]

134. Chandan KN, Kumar A, Rastogi A, Bihari C, Sarin S. Transient Elastography (TE) correlates well with Hepatic Venous Pressure Gradient (HVPG) and liver fibrosis score on liver biopsy in patients with portal hypertension. *Hepatology International* 2012;**6**:304-5.

135. Corina R, Procopet B, Platon M, Maniu A, Petra F, Grigoras C, et al. Liver stiffness measurement by 2D SWE from General Electric is similar with Transient Elastography in clinical significant portal hypertension diagnosis. *Journal of Hepatology* 2018;**68**(1):S637. [DOI: 10.1016/S0168-8278(18)31531-9]

136. Dajti E, Ravaioli F, Marasco G, Alemanni LV, Colecchia L, Ferrarese A, et al. A combined Baveno VII and spleen stiffness algorithm to improve the noninvasive diagnosis of clinically significant portal hypertension in patients with compensated advanced chronic liver disease. *American Journal of Gastroenterology* 2022;**117**(11):1825-33. [DOI: 10.14309/ajg.0000000000001887]

137. Gomez-Escolar L, Sanchez Paya J, Garcıa Herola A, Palazon Azorin JM. A prospective study: transient elastography predicts portal hypertension and complications in patients with chronic liver disease. *Journal of Hepatology* 2013;**58**:S292.

138. Grgurević I, Cvetko D, Bokun T, Mustapić S, Lukić A, Banić M, et al. Liver and spleen stiffness assessed by transient elastography and real-time 2D shear wave elastography and ELFTM test have very good performance for non-invasive detection and estimating severity of portal hypertension. *United European Gastroenterology Journal* 2016;**5**(Suppl 1):A361.

139. Grgurevic I, Madir A, Trkulja V, Bozin T, Aralica G, Podrug K, et al. Assessment of clinically significant portal hypertension by two-dimensional shear wave elastography. *European Journal of Clinical Investigation* 2022;**52**(6):e13750. [DOI: 10.1111/eci.13750]

140. He R, Liu C, Jie L, Ma J, Guo Y, Guo X, et al. Sequential algorithm of spleen stiffness measured by a dedicated 100 Hz examination and Baveno VII criteria for clinically significant portal hypertension in compensated cirrhosis. *Journal of Hepatology* 2022;**77**:636.

141. Hirooka M, Koizumi Y, Nakamura Y, Yano R, Okazaki Y, Sunago K, et al. Spleen stiffness in patients with chronic liver disease evaluated by 2-D shear wave elastography with ultrasound multiparametric imaging. *Hepatology Research* 2023;**53**(2):93-103.

142. Hong WK, Kim MY, Baik SK, Shin SY, Kim JM, Kang YS, et al. The usefulness of non-invasive liver stiffness measurements in predicting clinically significant portal hypertension in cirrhotic patients: Korean data. *Clinical and Molecular Hepatology* 2013;**19**(4):370-5.

143. Jachs M, Hartl L, Simbrunner B, Bauer D, Paternostro R, Scheiner B, et al. The sequential application of Baveno VII criteria and VITRO score improves diagnosis of clinically significant portal hypertension. *Clinical Gastroenterology and Hepatology* 2022;**21**(7):1854-63. [DOI: 10.1016/j.cgh.2022.09.032]

144. Jachs M, Sandmann L, Hartl L, Tergast T, Schwarz M, Bauer DJ, et al. Validation of Baveno VII criteria and other non-invasive diagnostic algorithms for clinically significant portal hypertension in hepatitis delta. *Journal of Hepatology* 2024;**81**(2):248-57. [DOI: 10.1016/j.jhep.2024.03.005]

145. Jansen C, Bogs C, Verlinden W, Thiele M, Möller P, Görtzen J, et al. Shear-wave elastography of the liver and spleen identifies clinically significant portal hypertension: A prospective multicentre study. *Liver International* 2017;**37**(3):396-405.

146. Jindal A, Agarwal S, Sharma S, Kumar M, Saraya A, Sarin SK. Assessment of the performance of non-invasive criteria for the evaluation of clinically significant portal hypertension in patients with compensated advanced chronic liver disease. *Digestive Diseases and Sciences* 2023;**68**(5):2149-57.

147. Kennedy P, Stocker D, Carbonell G, Said D, Bane O, Hectors S, et al. MR elastography outperforms shear wave elastography for the diagnosis of clinically significant portal hypertension. *European Radiology* 2022;**32**(12):8339-49.

148. Kim TY, Jeong WK, Sohn JH, Kim J, Kim MY, Kim Y. Evaluation of portal hypertension by real-time shear wave elastography in cirrhotic patients. *Liver International* 2015;**35**(11):2416-24.

149. Kumar A, Khan NM, Anikhindi SA, Sharma P, Bansal N, Singla V, et al. Correlation of transient elastography with hepatic venous pressure gradient in patients with cirrhotic portal hypertension: A study of 326 patients from India. *World Journal of Gastroenterology* 2017;**23**(4):687-96.

150. Lemoine M, Katsahian S, Ziol M, Nahon P, Ganne-Carrie N, Kazemi F, et al. Liver stiffness measurement as a predictive tool of clinically significant portal hypertension in patients with compensated hepatitis C virus or alcohol-related cirrhosis. *Alimentary Pharmacology & Therapeutics* 2008;**28**(9):1102-10.

151. Lemoine M, Katsahian S, Ziol M, Nahon P, Ganne-Carrie N, Kazemi F, et al. Liver stiffness measurement as a predictive tool of clinically significant portal hypertension in patients with compensated hepatitis C virus or alcohol-related cirrhosis. *Alimentary Pharmacology & Therapeutics* 2008;**28**(9):1102-10.

152. Levick C, Phillips-Hughes J, Collier J, Banerjee R, Cobbold JF, Wang LM, et al. Non-invasive assessment of portal hypertension by multi-parametric magnetic resonance imaging of the spleen: A proof of concept study. *PloS One* 2019;**14**(8):e0221066.

153. Maruyama H, Kobayashi K, Kiyono S, Sekimoto T, Kanda T, Yokosuka O. Two-dimensional shear wave elastography with propagation-based reliability assessment for grading hepatic fibrosis and portal hypertension. *Journal of Hepato-Biliary-Pancreatic Sciences* 2016;**23**(9):595-602.

154. Odriozola A, Puente Á, Cuadrado A, Iruzubieta P, Arias-Loste MT, Redondo C, et al. High accuracy of spleen stiffness measurement in diagnosing clinically significant portal hypertension in metabolic-associated fatty liver disease. *Liver International* 2023;**43**(7):1446-57.

155. Podrug K, Trkulja V, Zelenika M, Bokun T, Madir A, Kanizaj TF, et al. Validation of the new diagnostic criteria for clinically significant portal hypertension by platelets and elastography. *Digestive Diseases and Sciences* 2022;**67**(7):3327-32.

156. Procopet B, Berzigotti A, Abraldes JG, Turon F, Hernandez-Gea V, García-Pagán JC, et al. Real-time shear-wave elastography: applicability, reliability and accuracy for clinically significant portal hypertension. *Journal of Hepatology* 2015;**62**(5):1068-75.

157. Procopet B, Cristea VM, Robic MA, Grigorescu M, Agachi PS, Metivier S, et al. Serum tests, liver stiffness and artificial neural networks for diagnosing cirrhosis and portal hypertension. *Digestive and Liver Disease* 2015;**47**(5):411-6.

158. Procopet B, Cristea VM, Robic MA, Grigorescu M, Agachi PS, Metivier S, et al. Serum tests, liver stiffness and artificial neural networks for diagnosing cirrhosis and portal hypertension. *Digestive and Liver Disease* 2015;**47**(5):411-6.

159. Reiberger T, Ferlitsch A, Payer BA, Pinter M, Schwabl P, Stift J, et al. Noninvasive screening for liver fibrosis and portal hypertension by transient plastography – a large single center experience. *Wiener Klinische Wochenschrift* 2012;**124**(11-12):395-402.

160. Rosselli M, Roccarina D, Patch D, Sekhar M, Majumdar A, O'Beirne J, et al. Point shear wave elastography (ElastPQ) is a reliable non-invasive tool for the diagnosis and characterisation of portal hypertension. *Journal of Hepatology* 2017;**66**(1):S667. [DOI: 10.1016/S0168-8278(17)31803-2]

161. Salzl P, Reiberger T, Ferlitsch M, Payer BA, Schwengerer B, Trauner M, et al. Evaluation of portal hypertension and varices by acoustic radiation force impulse imaging of the liver compared to transient elastography and AST to platelet ratio index. *Ultraschall in der Medizin* 2014;**35**(6):528-33.

162. Schonemeier B, Attia D, Dettmer A, Rifai K, Pischke S, Lenzen H, et al. Value of liver elastography using Acoustic Radiation Force Impulse Imaging (ARFI) for the diagnosis of significant portal hypertension and liver fibrosis in patients with chronic liver disease. *Ultraschall in der Medizin* 2013;**34**:WS_SL1_02. [DOI: 10.1055/s-0033-1354800]

163. Schwabl P, Bota S, Salzl P, Mandorfer M, Payer BA, Ferlitsch A, et al. New reliability criteria for transient elastography increase the number of accurate measurements for screening of cirrhosis and portal hypertension. *Liver International* 2015;**35**(2):381-90.

164. Sohn JH, Kim TY, Jeong WK, Jeong JY, Jun DW. Liver stiffness measurement by shear wave elastography is very useful and superior to damping index by Doppler ultrasonography for predicting severe portal hypertension. *Hepatology (Baltimore, Md.)* 2011;**54**:952A.

165. Stefanescu H, Procopet B, Platon-Lupsor M, Bureau C. Is there any place for spleen stiffness measurement in portal hypertension? American Journal of Gastroenterology 2013;**108**(10):1660-1.

166. Takuma Y, Nouso K, Morimoto Y, Tomokuni J, Sahara A, et al. Portal hypertension in patients with liver cirrhosis: diagnostic accuracy of spleen stiffness. *Radiology* 2016;**279**(2):609-19.

167. Taru V, Taru MG, Petrushev B, Ioana R, Stefanescu H, Fodor A, et al. Recently validated non-invasive tests for liver fibrosis assessment have great performance in identifying NASH patients at risk for decompensation. *Journal of Hepatology* 2023;**78**:S295-S296.

168. Tseng Y, Li F, Wang J, Chen S, Jiang W, Shen X, et al. Spleen and liver stiffness for noninvasive assessment of portal hypertension in cirrhotic patients with large esophageal varices. *Journal of Clinical Ultrasound* 2018;**46**(7):442-9.

169. Vizzutti F, Arena U, Romanelli RG, Rega L, Foschi M, Colagrande S, et al. Liver stiffness measurement predicts severe portal hypertension in patients with HCV-related cirrhosis. *Hepatology (Baltimore, Md.)* 2007;**45**(5):1290-7.

170. Wang Y, Zhang G, Mu X, Ma H, Wang M. Diagnostic value of magnetic resonance elastography in cirrhotic portal hypertension. *Journal of Hepatology* 2020;**73**:S532-S533.

171. Yoo HW, Kim YS, Kim SG, Lee JM, Yoo J, Jang JY, et al. The association of hepatic venous pressure gradient with liver stiffness assessed by two-dimensional shear wave elastography and transient elastography in patients with liver cirrhosis. *Hepatology International* 2018;**12**(Suppl 2):S373. [DOI: 10.1007/s12072-018-9852-3]

172. Zhang M, Niu X, Zhao D, Qi R, Qi X, Dong J, et al. Limited diagnostic value of liver stiffness for clinically significant portal hypertension in HBV-related cirrhosis. *Abdominal Radiology (New York)* 2022;**47**(11):3712-23.

173. Zhu YL, Ding H, Fu TT, Peng SY, Chen SY, Luo JJ, et al. Portal hypertension in hepatitis B-related cirrhosis: Diagnostic accuracy of liver and spleen stiffness by 2-D shear-wave elastography. *Hepatology Research* 2019;**49**(5):540-9.

174. Zhu YL, Ding H, Fu TT, Xu ZT, Xue LY, Chen SY, et al. Diagnostic accuracy of liver and spleen stiffness by two dimensional shear wave elastography for portal hypertension in hepatitis B-related cirrhosis. *Zhonghua Yi Xue za Zhi* 2020;**100**(21):1654-7. [DOI: 10.3760/cma.j.cn112137-20191029-02340]

175. Zocco MA, Cintoni M, Ainora ME, Garcovich M, Lupascu A, Iezzi R, et al. Noninvasive evaluation of clinically significant portal hypertension in patients with liver cirrhosis: the role of contrast-enhanced ultrasound perfusion imaging and elastography. *Ultraschall in der Medizin* 2022;**44**(4):428-35. [DOI: 10.1055/a-1933-2847]

176. Zykus R, Jonaitis L, Petrenkienė V, Pranculis A, Kupčinskas L. Liver and spleen transient elastography predicts portal hypertension in patients with chronic liver disease: a prospective cohort study. *BMC Gastroenterology* 2015;**15**:183. [DOI: 10.1186/s12876-015-0414-z]

177. Lyusina EO, Pavlov CS, Ivashkin VT. Elastography in the diagnosis of portal hypertension in patients with liver cirrhosis [Elastometriya v diagnostike portal'noi gipertenzii u patsientov s tsirrozom pecheni]. *Terapevticheskii Arkhiv* 2017;**89**(2):33-7.

178. Perez L, Pincelli V, Pittamiglio L, Rodríguez Pezzino M, Santos D, Saralegui M, et al. Fibroscan as a diagnosis of portal hypertension in cirrhotic patients [Fibroscan como diagnóstico de hipertensión portal en pacientes cirróticos]. *Rev. Urug. Med. Int.* 2019;**4**:26-34.

179. Baik SK, Cho YZ, Kim MY, Cho MY, Hwang HW, Kwon SO, et al. Non-invasive prediction of clinically significant portal hypertension using transient elastography and spleen diameter/platelet count ratio in b-viral and alcohol cirrhosis: external validation study. In: Journal of Hepatology. Vol. 60. 2014:S241. [DOI: 10.1016/S0168-8278(14)60677-2]

180. Benmassaoud A, Zhao X, Kablawi D, Valenti D, Bessissow A, Chen T, et al. Comparison of non-invasive tests to diagnose portal hypertension, and predict hepatic decompensation: which one should we choose ? In: Hepatology (Baltimore, Md.). Vol. 74. 2021:1210A-1211A.

181. Bota S, Purevsambuu T, Reiberger T, Mandorfer M, Schwabl P, Sieghart W, et al. The value of different non-invasive tests for predicting the presence of clinically significant portal hypertension and esophageal varices in cirrhotic patients. In: Journal of Hepatology. Vol. 62. 2015:S200.

182. Buck M, Garcia-Tsao G, Groszmann RJ, Stalling C, Grace ND, Burroughs AK, et al. Novel inflammatory biomarkers of portal pressure in compensated cirrhosis patients. *Hepatology (Baltimore, Md.)* 2014;**59**(3):1052-9.

183. Chang JP, Low AsC, Ooi CC, Cheng LtE, Tay MF, Lau DS. Prospective comparison of spleen stiffness measured by magnetic resonance elastography versus shear-wave elastography for non-invasive assessment of clinically significant and severe portal hypertension. In: Hepatology (Baltimore, Md.). 2019:495A-496A.

184. Cho EJ, Lee JH, Kim MY, Yoo JJ, Choi WM, Cho YY, et al. Which noninvasive hepatic fibrosis test most effectively predicts portal pressure and survival in alcoholic liver disease patients? In: Hepatology (Baltimore, Md.). 2013:987A. [DOI: 10.1002/hep.26873]

185. De Filippi F, Facciotto C, Camera A, Gattoni F, Rossi S, Chiesa A, et al. Liver stiffness measurement as non-invasive tool for the assessment of portal hypertension in cirrhotic patients. In: Digestive and Liver Disease. Vol. 43. 2011:S71.

186. Hametner S, Ferlitsch A, Ferlitsch M, Etschmaier A, Schöfl R, Ziachehabi A, et al. The VITRO score (von willebrand factor antigen/thrombocyte ratio) as a new marker for clinically significant portal hypertension in comparison to other non-invasive parameters of fibrosis including ELF test. *PloS One* 2016;**11**(2):e0149230.

187. Han X, Yang D, Xu H, Wang Y, Yin H, Yang Z. Superiority of spleen stiffness on two-dimensional magnetic resonance elastography over liver stiffness and serum tests in assessing portal hypertension in chronic liver disease. *Quantitative Imaging in Medicine and Surgery* 2024;**14**(2):1429-40.

188. Herran AO, Fortea JI, Puente A, Cuadrado A, Fabrega E, Arias-Loste MT et al. Evaluation of clinically significant portal hypertension with a novel spleen-dedicated probe using transient elastography. In: Journal of Hepatology. Vol. 77. 2022:S624. [DOI: 10.1016/S0168-8278(22)01566-5]

189. Hirooka M, Ochi H, Koizumi Y, Kisaka Y, Abe M, Ikeda Y, et al. Splenic elasticity measured with real-time tissue elastography is a marker of portal hypertension. *Radiology* 2011;**261**(3):960-8.

190. Hirooka M, Ochi H, Koizumi Y, Abe M, Ikeda Y, Matsuura B, et al. Prediction of gastroesophageal varices using splenic elasticity for portal hypertension (SEP) score. In: Journal of Hepatology. 2012:S247-S248. [DOI: 10.1016/S0168-8278%2812%2960638-2]

191. Iacob S, Iacob AR, Gheorghe C, Gheorghe L. Shear wave elastography use for screening of clinically significant portal hypertension after hcv cure. In: Gastroenterology. Vol. 156. 2019:S1052. [DOI: 10.1016/S0016-5085(19)39587-3]

192. Ilan Y, Harrison SA, Traber PG, Chalasani NP, Garcia-Tsao G. 13C-Methacetin Breath Test accurately assesses clinically significant portal hypertension in patients with NASH cirrhosis. *Journal of Hepatology* 2017;**66**(1):S100.

193. Jachs M, Fischer P, Odriozola A, Moga L, Kwanten WJ, Saltini D, et al. Spleen stiffness measurement by a dedicated 100Hz vibration-controlled transient elastography probe improves the noninvasive diagnosis of clinically significant portal hypertension in a prospective multicenter study. In: Hepatology (Baltimore, Md.). 2023:S212-S214.

194. Kedarisetty CK, Bhardwaj A, Vashishta C, Sharma MK, Bhadoria AS, Sarin SK. Hepatic venous pressure gradient: bio-data of a cirrhotic. In: Hepatology International. 2014:S391-S392.

195. Liu C, Chen S, Yan X, Xiang Y, Hui J, Liu Z, et al. Diagnostic value of portal vein velocity for portal hypertension in patients with hepatitis b virus-related cirrhosis. *Current Medical Imaging* 2021;**17**(11):1363-8.

196. Liu F, Ning Z, Liu Y, Liu D, Tian J, Luo H, et al. Development and validation of a radiomics signature for clinically significant portal hypertension in cirrhosis (CHESS1701): a prospective multicenter study. *EBioMedicine* 2018;**36**:151-8.

197. Ortiz GA, Palumbo C, Sayyar MW, Balakrishnan M, Mohanty A, Garcia-Tsao G. Spleen stiffness measurements (SSM) in the identification of clinically-significant portal hypertension (CSPH) in patients with compensated cirrhosis. In: Gastroenterology. 2020:1385.

198. Ozkaya E, Bane O, Geahchan A, Kennedy P, Hectors S, Fischman A, et al. Noninvasive diagnosis of clinically significant portal hypertension with MR elastography, T1, and T1rho mapping of the liver and spleen. *Hepatology (Baltimore, Md.)* 2023;**78**(1):1431-2.

199. Park SH, Park TE, Kim YM, Kim SJ, Baik GH, Kim JB, et al. Non-invasive model predicting clinically-significant portal hypertension in patients with advanced fibrosis. *Journal of Gastroenterology and Hepatology* 2009;**24**(7):1289-93.

200. Park SY, Kim MY, Baik SK, Hong JH, Kwon SO, Cheon GJ, et al. The usefulness of measurement of liver stiffness interval change in prediction of the therapeutic response of propranolol in clinical significant portal hypertension. In: Journal of Hepatology. Vol. 56. 2012:S253.

201. Radu C, Stancu O, Sav R, Bugariu A, Suciu A, Grigoras C, et al. Liver stiffness better predicts portal hypertension after HCV eradication. Journal of Gastrointestinal and Liver Diseases 2018;**27**(2):204.

202. Reiberger T, Ferlitsch A, Pinter M, Homoncik M, Ulbrich G, Peck-Radosavljevic M. The correlation of liver stiffness with portal pressure and fibrosis stage is influenced by vasoactive treatment and etiology of liver disease. In: Gastroenterology. 2010:S815.

203. Ryu SR, Yoo JJ, Kang SH, Jeong SW, Kim MY, Cho YK, et al. The cut-off value of transient elastography to the value of hepatic venous pressure gradient in alcoholic cirrhosis. *Clinical and Molecular Hepatology* 2021;**27**(1):197-206.

204. Sav R, Farcau O, Fischer P, Stancu O, Horhat A, Procopet B, et al. HCV eradication: a game-changer in the relationship between liver stiffness and portal pressure in compensated advanced liver disease. In: United European Gastroenterology Journal. 2018:A558.

205. Semmler G, Hartl L, Mendoza YP, Simbrunner B, Jachs M, Balcar L, et al. Simple blood tests to diagnose compensated advanced chronic liver disease and stratify the risk of clinically significant portal hypertension. Cochrane

Database of Systematic Reviews 2024. [DOI: 10.1097/HEP.0000000000000829]

206. Sharma P, Kirnake V, Tyagi P, Bansal N, Singla V, Kumar A, et al. Spleen stiffness in patients with cirrhosis in predicting esophageal varices. *American Journal of Gastroenterology* 2013;**108**(7):1101-7.

207. Simbrunner B, Marculescu R, Scheiner B, Schwabl P, Bucsics T, Stadlmann A, et al. Non-invasive detection of portal hypertension by enhanced liver fibrosis score in patients with different aetiologies of advanced chronic liver disease. *Liver International* 2020;**40**(7):1713-24.

208. Stefanescu H, Allegretti G, Serra C, Marasco G, Gamal N, Conti F, et al. Impulsional, point and bidimensional shear wave elastometry for portal hypertension: same stiffness thresholds? In: Journal of Hepatology. 2015:S355.

209. Vonghia L, Verlinden W, Vanwolleghem T, Michielsen P, Francque S. Measurement of spleen and liver stiffness by shear wave elastography to noninvasively evaluate hepatic venous pressure gradient and portal hypertension. In: Journal of Hepatology. 2015:S372.

210. Vuppalanchi R, Lawitz E, Abdelmalek MF, Rinella ME, Harrison SA, Ryan MJ, et al. Liver stiffness measured by vibration controlled transient elastography is an excellent surrogate for identifying clinically significant portal hypertension in patients with compensated NASH cirrhosis. In: Hepatology (Baltimore, Md.). Vol. 66. 2017:245A.

211. Wang P, Hu X, Xie F. Predictive value of liver and spleen stiffness measurement based on two-dimensional shear wave elastography for the portal vein pressure in patients with compensatory viral cirrhosis. *PeerJ* 2023;**11**:e15956.

212. Zhu YL, Hong D, Fu TT, Chen SY, Luo JJ, Xu C, et al. Evaluation of liver fibrosis and portal hypertension in chronic hepatitis b patients by acoustic radiation force impulse elastograph. *Iranian Journal of Radiology* 2020;**17**(1):e82396. [DOI: 10.5812/iranjradiol.82396]

213. Berzigotti A, Abraldes JG, Llop E, Erice E, Gilabert R, Garcia-Pagan JC, et al. Comparison of ultrasonographic evaluation of liver surface and transient elastography in the diagnosis of cirrhosis and portal hypertension. In: Journal of Hepatology. 2009:S74--S75. [DOI: 10.1016/S0168-8278(09)60180-X]

214. Berzigotti A, Seijo S, Arena U, Abraldes JG, Vizzutti F, García-Pagán JC, et al. Elastography, spleen size, and platelet count identify portal hypertension in patients with compensated cirrhosis. *Gastroenterology* 2013;**144**(1):102-11.

215. Sperl J, Frankova S, Jirsa M. Liver stiffness measurement is superior to plasma markers in the non-invasive assessment of clinically significant portal hypertension. In: Hepatology (Baltimore, Md.). 2019:503A.

216. Ferraioli G, Barr RG, Berzigotti A, Sporea I, Wong VW, Reiberger T, et al. WFUMB Guideline/Guidance on Liver Multiparametric Ultrasound: part 1. Update to 2018 Guidelines on Liver Ultrasound Elastography. *Ultrasound in Medicine & Biology* 2024;**50**(8):1071-87.

217. Stefanescu H, Marasco G, Calès P, Fraquelli M, Rosselli M, Ganne-Carriè N, et al. A novel spleen-dedicated stiffness measurement by FibroScan® improves the screening of high-risk oesophageal varices. *Liver International* 2020;**40**(1):175-85.

218. Kaplan DE, Ripoll C, Thiele M, Fortune BE, Simonetto DA, Garcia-Tsao G, et al. AASLD Practice Guidance on risk stratification and management of portal hypertension and varices in cirrhosis. *Hepatology (Baltimore, Md.)* 2024;**79**(5):1180-211.

ADDITIONAL TABLES

Table 1. List of abbreviations

cACLD	compensated advanced chronic liver disease
CLD	chronic liver disease
CHBG	Cochrane Hepato-Biliary Group
CSPH	clinically significant portal hypertension
HBV	hepatitis B virus
HCV	hepatitis C virus
HDV	hepatitis D virus
HVPG	hepatic venous pressure gradient
LSM	liver stiffness measurement

Table 1. List of abbreviations *(Continued)*

MASLD	metabolic dysfunction-associated steatotic liver disease
MRE	magnetic resonance elastography
NAFLD	non-alcoholic fatty liver disease
NASH	non-alcoholic steatohepatitis
NSBB	non-selective beta blockers
pSWE	point shear wave elastography
SPH	severe portal hypertension
SSM	spleen stiffness measurement
VCTE	vibration-controlled transient elastography
2D-SWE	two-dimensional shear wave elastography

Table 2. Possible advantages and disadvantages of different index tests

	Vibration-controlled transient elastography (VCTE)	Two-dimensional shear wave elastography (2D-SWE)	Point shear wave elastography (pSWE)	Magnetic resonance elastography (MRE)
Advantages	• Most validated and widely used method • Very user-friendly (rapid, easy to learn, performed at bedside) • High inter- and intraobserver agreement • Established in clinical practice for both LSM and SSM • Recommended in current guidelines to stage liver fibrosis and portal hypertension	• Can be implemented in most ultrasound machines • 'Real-time' imaging • Larger region of interest than VCTE and pSWE • High intra- and inter-operator reproducibility • Supposed low failure rate for LSM and SSM	• Can be implemented in most ultrasound machines • 'Real-time' imaging • High intra- and interoperator reproducibility • Supposed low failure rate for LSM and SSM	• Can be implemented in most magnetic resonance imaging machines • Evaluation of the entire liver and spleen parenchyma • High repeatability • Supposed low failure rate for LSM • Almost 100% feasible for SSM
Disadvantages	• Requires a dedicated device • Impaired applicability in the case of obesity and ascites • Inability to choose the scanned region • A reported failure rate ranging from 15% to 20% for both LSM and SSM • Need for ultrasound-guided	• Not yet established in clinical practice • Quality criteria not defined • LSM and SSM cut-offs for CSPH/SPH vary substantially between studies • Interference in the case of elevated	• Not yet established in clinical practice • Smaller region of interest than 2D-SWE • Quality criteria not defined • LSM and SSM cut-offs for CSPH/SPH vary substantially between studies	• Not yet established in clinical practice • Limited availability • High cost • Time-consuming • Quality criteria not defined

Table 2. Possible advantages and disadvantages of different index tests *(Continued)*

	• Interference in the case of elevated transaminases, acute hepatitis, extrahepatic cholestasis, liver congestion, recent food intake, and excessive alcohol consumption	transaminases, acute hepatitis, extrahepatic cholestasis, liver congestion, food intake, and excessive alcohol consumption	• Interference in the case of elevated transaminases, acute hepatitis, extrahepatic cholestasis, liver congestion, food intake, and excessive alcohol consumption

2D-SWE = 2D-shear wave elastography; CSPH = clinically significant portal hypertension; LSM = liver stiffness measurement; MRE = magnetic resonance elastography; pSWE = point shear wave elastography; SPH = severe portal hypertension; SSM = spleen stiffness measurement; VCTE = vibration-controlled transient elastography

Table 3. Overview of Syntheses and Included Studies table*

Study ID Country of study	Study design Type of study report	Index test(s)	Population (Number of participants)	Target condition	Time interval index test – reference standard	Overall risk of bias (study level)
Attia 2015 Germany	Prospective Full text	LSM by pSWE, SSM by pSWE	Chronic liver disease (78)	CSPH SPH	Same day	High risk
Augustin 2015 Spain	Prospective Full text	LSM by VCTE	Chronic liver disease (40)	CSPH	Within 1 month	High risk
Banini 2022 USA	Retrospective Full text	LSM by VCTE	Chronic liver disease (142)	CSPH	No data	High risk
Borghi 2012 Italy	Retrospective Abstract	SSM by pSWE	Liver cirrhosis (40)	CSPH	No data	High risk
Bureau 2008 France	Prospective Full text	LSM by VCTE	Chronic liver disease (144)	CSPH	Same day	High risk
Chandan 2012 India	Retrospective Abstract	LSM by VCTE	Chronic liver disease (108)	SPH	No data	High risk
Colecchia 2012 Italy	Prospective Full text	LSM by VCTE, SSM by VCTE	HCV-related chronic liver disease (100)	CSPH SPH	1 day	High risk
Corina 2018 Romania	Retrospective Abstract	LSM by 2D-SWE	Advanced liver disease (81)	CSPH	No data	High risk
Dajti 2022 Italy	Retrospective Full text	LSM by VCTE	Compensated advanced chronic liver disease (195)	CSPH	Within 6 months	High risk

Table 3. Overview of Syntheses and Included Studies table* *(Continued)*

Gomez-Escolar 2013 Spain	Prospective Abstract	LSM by VCTE	Chronic liver disease (51)	CSPH	No data	High risk
Grgurević 2016 Croatia	Retrospective Abstract	LSM by VCTE, SSM by VCTE, LSM by 2D-SWE, SSM by 2D-SWE	Chronic liver disease (20)	CSPH	No data	High risk
Grgurević 2022 Croatia	Prospective Full text	LSM by VCTE, SSM by VCTE, LSM by 2D-SWE, SSM by 2D-SWE	Chronic liver disease (76)	CSPH	Same day	High risk
He 2022 China	Prospective Abstract	LSM by VCTE LSM+SSM by VCTE (combination)	Compensated liver cirrhosis (101)	CSPH	No data	High risk
Hirooka 2022 Japan	Retrospective Full text	LSM by VCTE, SSM by VCTE, LSM by 2D-SWE, SSM by 2D-SWE	Chronic liver disease (81)	SPH	Within 1 week	High risk
Hong 2013 South Korea	Retrospective Full text	LSM by VCTE	Liver cirrhosis (59)	CSPH SPH	No data	High risk
Jachs 2022 Austria	Retrospective Full text	LSM by VCTE	Compensated advanced chronic liver disease (302)	CSPH	Same day	Low risk
Jachs 2023 Austria/Germany	Retrospective Full text	LSM by VCTE, SSM by 2D-SWE	HBV/HDV-related compensated advanced chronic liver disease (51)	CSPH	Within 3 months	High risk
Jansen 2017 Belgium/Denmark/Germany	Prospective Full text	LSM by 2D-SWE, SSM by 2D-SWE, LSM+SSM by 2D-SWE (combination)	Chronic liver disease (155)	CSPH	Within 2 days	High risk
Jindal 2022 India	Retrospective Full text	LSM by VCTE	Compensated advanced chronic liver disease (626)	CSPH	No data	High risk

Table 3. Overview of Syntheses and Included Studies table* *(Continued)*

Kennedy 2022 USA	Prospective Full text	SSM by 2D-SWE, SSM by MRE	Chronic liver disease (33)	CSPH	Within 6 months	High risk
Kim 2015 South Korea	Prospective Full text	LSM by 2D-SWE	Liver cirrhosis (92)	CSPH SPH	Same day	High risk
Kumar 2017 India	Retrospective Full text	LSM by VCTE	Liver cirrhosis (326)	CSPH	Within 1 week	High risk
Lemoine 2008 France	Retrospective Full text	LSM by VCTE	1. cohort: HCV-related compensated liver cirrhosis (44) 2. cohort: alcohol-related compensated liver cirrhosis (48)	CSPH	Same day	High risk
Levick 2019 UK	Prospective Full text	LSM by VCTE	Suspected liver cirrhosis (13)	CSPH	No data	High risk
Maruyama 2016 Japan	Prospective Full text	LSM by VCTE, LSM by 2D-SWE	Chronic liver disease (25)	CSPH SPH	Within 170 days	High risk
Odriozola 2023 Spain	Retrospective Full text	LSM by VCTE, SSM by VCTE	Chronic liver disease (84)	CSPH	Within 1 year	High risk
Podrug 2022 Croatia	Retrospective Full text	LSM by VCTE	Compensated advanced chronic liver disease (76)	CSPH	Within 1 month	High risk
Procopet 2015 Spain	Prospective Full text	LSM by 2D-SWE	Compensated chronic liver disease (55)	CSPH	Same day	High risk
Procopet 2015 (2) France	Prospective Full text	LSM by VCTE	Compensated chronic liver disease (198)	CSPH	Same day	Low risk
Reiberger 2012 Austria	Prospective Full text	LSM by VCTE	Chronic liver disease (502)	CSPH	Within 3 days	High risk
Roselli 2017 UK	Retrospective Abstract	SSM by pSWE	Chronic liver disease (70)	CSPH	Same day	High risk
Salzl 2014 Austria	Prospective Full text	LSM by VCTE, LSM by pSWE	Liver cirrhosis (59)	CSPH	No data	High risk
Schonemeier 2013 Germany	Prospective Abstract	LSM by pSWE	Chronic liver disease (45)	CSPH	No data	High risk

Table 3. Overview of Syntheses and Included Studies table* *(Continued)*

Schwabl 2015 Austria	Retrospective Full text	LSM by VCTE	Chronic liver disease (162)	CSPH	Same day	High risk
Sohn 2011 South Korea	Retrospective Abstract	LSM by 2D-SWE	Liver cirrhosis (32)	SPH	Same day	High risk
Stefanescu 2013 France	Retrospective Letter	SSM by VCTE	Liver cirrhosis (37)	CSPH	No data	High risk
Takuma 2016 Japan	Prospective Full text	SSM by pSWE	Liver cirrhosis (60)	CSPH SPH	Within 1 week	High risk
Taru 2023 Romania	Retrospective Abstract	LSM by VCTE	NAFLD/NASH-related chronic liver disease (62)	CSPH	No data	High risk
Tseng 2018 China	Prospective Full text	LSM by VCTE, SSM by VCTE	Liver cirrhosis (99)	SPH	1 day	High risk
Vizzutii 2008 Italy	Prospective Full text	LSM by VCTE	HCV-related liver cirrhosis (61)	CSPH SPH	1 day	High risk
Wang 2020 China	Retrospective Abstract	SSM by MRE	Chronic liver disease (27)	SPH	Within 1 week	High risk
Yoo 2018 South Korea	Retrospective Abstract	LSM by 2D-SWE	Advanced chronic liver disease (103)	CSPH SPH	Within 6 months	High risk
Zhang 2022 China	Retrospective Full text	LSM by VCTE	Liver cirrhosis (137)	CSPH	Within 1 week	High risk
Zhu 2019 China	Retrospective Full text	LSM by 2D-SWE, SSM by 2D-SWE, LSM + SSM by 2D-SWE (combination)	HBV-related liver cirrhosis (104)	CSPH SPH	Within 1 week	High risk
Zhu 2020 China	Retrospective Full text	LSM by 2D-SWE	HBV-related liver cirrhosis (58)	CSPH SPH	Within 1 week	High risk
Zocco 2022 Italy	Prospective Full text	LSM by pSWE	Liver cirrhosis (46)	CSPH SPH	Same day	High risk
Zykus 2015	Prospective	LSM by VCTE,	Chronic liver disease (107)	CSPH	1 day	High risk

Table 3. Overview of Syntheses and Included Studies table* *(Continued)*

Lithuania	Full text	SSM by VCTE	SPH

*All studies used hepatic venous pressure gradient (HVPG) as the reference standard. All studies have a cross-sectional design.
2D-SWE = 2D-shear wave elastography; CSPH = clinically significant portal hypertension; HBV = hepatitis B virus; HCV = hepatitis C virus; HDV = hepatitis D virus; LSM = liver stiffness measurement; MRE = magnetic resonance elastography; NAFLD = non-alcoholic fatty liver disease; NASH = non-alcoholic steatohepatitis; pSWE = point shear wave elastography; SPH = severe portal hypertension; SSM = spleen stiffness measurement; VCTE = vibration-controlled transient elastography

Table 4. Heterogeneity and sensitivity analysis (LSM by VCTE for CSPH 25 kPa)

Subgroup	No. of studies	Sensitivity (95% CI)	Specificity (95% CI)	P value
All	9	62.3% (53% to 70.7%)	94.1% (87.2% to 97.3%)	-
> 80% viral aetiology	2	Not estimable	Not estimable	-
< 80% viral aetiology	4			
Not reported	3			
> 50% CSPH	6	59.6% (49.2% to 69.2%)	88.3% (79.8% to 93.6%)	0.021
< 50% CSPH	3	68.1% (52.8% to 80.3%)	98.2% (92.8% to 99.6%)	
> 75% Child-Pugh A	2	Not estimable	Not estimable	-
< 75% Child-Pugh A	1			
Not reported	6			
> 20% indeterminate data	1	Not estimable	Not estimable	-
< 20% indeterminate data	3			
Not reported	5			
Low risk of bias	1	Not estimable	Not estimable	-
Blinding for reference standard results	2	Not estimable	Not estimable	-
Time interval between the index test and the reference standard (< 1 week)	2	Not estimable	Not estimable	-
Full text	7	56.8% (51.3% to 62.1%)	90.3% (84.4% to 94.2%)	-
No exclusion criteria met	5	55.2% (49% to 61.3%)	91.4% (83.7% to 95.7%)	-

CI = confidence interval; CSPH = clinically significant portal hypertension; LSM = liver stiffness measurement; VCTE = vibration-controlled transient elastography

Table 5. Summary of diagnostic accuracy estimates across various cut-off ranges

Index test for CSPH (cut-off range)	Number of participants (number of studies)	Sensitivity (95% CI)	Specificity (95% CI)	Certainty of evidence

Table 5. Summary of diagnostic accuracy estimates across various cut-off ranges *(Continued)*

LSM by VCTE (11.4 to 14.4 kPa)	757 participants (6 studies, 7 cohorts)	97.2% (84.5% to 100%)	63.1% (34.5% to 84.8%)	Very low[a] ⊕○○○
LSM by VCTE (15 to 18 kPa)	1257 (9 studies)	89.4% (85.7% to 92.9%)	80% (72.9% to 85.6%)	Very low[a] ⊕○○○
LSM by VCTE (20 to 23 kPa)	1246 (11 studies)	79% (74.4% to 83.7%)	85.3% (77.7% to 90.6%)	Very low[a] ⊕○○○
LSM by VCTE (25 to 28 kPa)	1726 (12 studies)	60.7% (52.7% to 68.1%)	94.6% (89.6% to 97.2%)	Very low[a] ⊕○○○
SSM by VCTE (40 to 44 kPa)	218 (4 studies)	91% (74.7% to 97.3%)	79.5% (66.6% to 88.4%)	Very low[b] ⊕○○○
SSM by VCTE (46 to 50 kPa)	334 (4 studies)	73.6% (58.6% to 84.6%)	90.9% (83.3% to 95.2%)	Very low[b] ⊕○○○
LSM by 2D-SWE (11 to 14 kPa)	394 (6 studies)	85.7% (78.6% to 90.7%)	75.3% (63.5% to 84.2%)	Very low[a] ⊕○○○
LSM by 2D-SWE (15 to 17 kPa)	428 (5 studies)	85.4% (79.5% to 89.8%)	74.5% (65.2% to 82.1%)	Very low[a] ⊕○○○
SSM by 2D-SWE (25.3 to 26.3 kPa)	250 (3 studies)	81.5% (74.4% to 87%)	76.5% (63.1% to 86.1%)	Very low[b] ⊕○○○

Index test for SPH (cut-off range)	**Number of participants (number of studies)**	**Sensitivity (95% CI)**	**Specificity (95% CI)**	**Certainty of evidence**
LSM by VCTE (16 to 18 kPa)	258 (3 studies)	93.4% (77.2% to 98.3%)	68.1% (54% to 79.5%)	Very low[b] ⊕○○○
LSM by VCTE (20 to 22 kPa)	315 (3 studies)	81.3% (73.4% to 87.3%)	75.7% (65.3% to 83.7%)	Very low[b] ⊕○○○
LSM by VCTE (24 to 26 kPa)	240 (3 studies)	71% (54.4% to 83.2%)	87.4% (66% to 96.1%)	Very low[b] ⊕○○○
LSM by 2D-SWE (21.5 to 23.5 kPa)	228 (3 studies)	81.2% (75.1% to 86.2%)	83.8% (73.3% to 90.8%)	Very low[b] ⊕○○○

Table 5. Summary of diagnostic accuracy estimates across various cut-off ranges *(Continued)*

Abbreviations: 2D-SWE = 2D-shear wave elastography; CI = confidence interval; CSPH = clinically significant portal hypertension; LSM = liver stiffness measurement; SPH = severe portal hypertension; SSM = spleen stiffness measurement; VCTE = vibration-controlled transient elastography

GRADE certainty of the evidence

High: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

[a]We deemed these estimates to be at a very low level of certainty, downgrading them by three levels due to risk of bias, inconsistency, and imprecision. Risk of bias was downgraded by one level because almost all studies were judged at high risk of bias; inconsistency was downgraded by one level due to wide range of cut-off values or different definitions of populations in studies; imprecision was downgraded by one level due to wide CIs of estimates.
[b]We deemed these estimates to be at a very low level of certainty, downgrading them by one level due to risk of bias and by two levels due to imprecision. Risk of bias was downgraded by one level because all studies were judged at high risk of bias; imprecision was downgraded by two levels due to generally low number of participants per study, low number of studies, and wide CIs of estimates.

Table 6. List of overall number of studies and participants by each index test and specific cut-off

Test	No. of studies	No. of participants
All tests	47	7817
1 LSM by VCTE (CSPH) "any cut-off"	29	3818
2 LSM by VCTE (CSPH) cut-off 11.4 to 14.4 kPa	7	757
3 LSM by VCTE (CSPH) cut-off 15 to 18 kPa	9	1257
4 LSM by VCTE (CSPH) cut-off 20 to 23 kPa	11	1246
5 LSM by VCTE (CSPH) cut-off 25 to 28 kPa	12	1726
6 LSM by VCTE (CSPH) cut-off 25 kPa	9	1553
7 LSM by VCTE (SPH) "any cut-off"	8	637
8 LSM by VCTE (SPH) cut-off 16 to 18 kPa	3	258
9 LSM by VCTE (SPH) cut-off 20 to 22 kPa	3	315
10 LSM by VCTE (SPH) cut-off 24 to 26 kPa	3	240
11 SSM by VCTE (CSPH) "any cut-off"	6	391
12 SSM by VCTE (CSPH) cut-off 40 to 44 kPa	4	218
13 SSM by VCTE (CSPH) cut-off 46 to 50 kPa	4	334
14 SSM by VCTE (SPH) "any cut-off"	4	379
15 SSM by VCTE (SPH) cut-off 47 to 51 kPa	4	379

Table 6. List of overall number of studies and participants by each index test and specific cut-off *(Continued)*

16 LSM by pSWE (CSPH) "any cut-off"	4	228
17 LSM by pSWE (CSPH) cut-off 2.50 to 2.60 m/s	2	104
18 LSM by pSWE (SPH) "any cut-off"	2	124
19 SSM by pSWE (CSPH) "any cut-off"	4	248
20 SSM by pSWE (CSPH) cut-off 3.10 to 3.30 m/s	2	100
21 SSM by pSWE (SPH) "any cut-off"	2	138
22 LSM by 2D-SWE (CSPH) "any cut-off"	10	767
23 LSM by 2D-SWE (CSPH) cut-off 11 to 14 kPa	6	394
24 LSM by 2D-SWE (CSPH) cut-off 15 to 17 kPa	5	428
25 LSM by 2D-SWE (CSPH) cut-off 24 to 26 kPa	2	210
26 LSM by 2D SWE (SPH) "any cut-off"	7	492
27 LSM by 2D SWE (SPH) cut-off 17 to 19 kPa	2	80
28 LSM by 2D SWE (SPH) cut-off 21.5 to 23.5 kPa	3	228
29 SSM by 2D SWE (CSPH) "any cut-off"	6	353
30 SSM by 2D SWE (CSPH) cut-off 25.3 to 26.3 kPa	3	250
31 SSM by 2D SWE (CSPH) cut-off 30 to 31 kPa	2	83
32 SSM by 2D SWE (CSPH) cut-off 34.8 to 35.8 kPa	2	175
33 SSM by 2D SWE (CSPH) cut-off 40 to 42 kPa	2	83
34 SSM by 2D SWE (SPH) "any cut-off"	2	185
35 LSM by MRE (CSPH) "any cut-off"	0	0
36 LSM by MRE (SPH) "any cut-off"	0	0
37 SSM by MRE (CSPH) "any cut-off"	1	30
38 SSM by MRE (SPH) "any cut-off"	1	27
39 LSM + SSM combination by VCTE (CSPH) "any cut-off"	1	101
40 LSM + SSM combination by 2D-SWE (CSPH) "any cut-off"	2	213
41 LSM + SSM combination by 2D-SWE (SPH) "any cut-off"	1	104

2D-SWE = 2D-shear wave elastography; CSPH = clinically significant portal hypertension; LSM = liver stiffness measurement; MRE = magnetic resonance elastography; pSWE = point shear wave elastography; SPH = severe portal hypertension; SSM = spleen stiffness measurement; VCTE = vibration-controlled transient elastography

Table 7. Studies with multiple thresholds reported for single index test and same target condition

Study	Index test (target condition)	Thresholds
Augustin 2015	LSM by VCTE (CSPH)	21.1 kPa 25 kPa
Banini 2022	LSM by VCTE (CSPH)	8.5 kPa 21.3 kPa
Bureau 2008	LSM by VCTE (CSPH)	11.7 kPa 13 kPa 17 kPa 21 kPa
Colecchia 2012 (1)	LSM by VCTE (CSPH)	13 kPa 16 kPa 21 kPa 24.2 kPa 25 kPa
Colecchia 2012 (2)	LSM by VCTE (SPH)	16.4 kPa 21 kPa 25 kPa
Colecchia 2012 (3)	SSM by VCTE (CSPH)	40 kPa 48 kPa 52.8 kPa
Colecchia 2012 (4)	SSM by VCTE (SPH)	41.3 kPa 49 kPa 55 kPa
Grgurević 2022 (1)	LSM by VCTE (CSPH)	15 kPa 20.2 kPa 27.7 kPa
Grgurević 2022 (2)	SSM by VCTE (CSPH)	29.3 kPa 43.5 kPa 46.7 kPa
Grgurević 2022 (3)	LSM by 2D-SWE (CSPH)	13.5 kPa 20.1 kPa

Table 7. Studies with multiple thresholds reported for single index test and same target condition *(Continued)*

Grgurević 2022 (4)	SSM by 2D-SWE (CSPH)	30.1 kPa 34.8 kPa 41.4 kPa
Jachs 2023	LSM by VCTE (CSPH)	15 kPa 25 kPa
Jansen 2017 (1)	LSM by 2D-SWE (CSPH)	16 kPa 24.6 kPa 29.5 kPa
Jansen 2017 (2)	SSM by 2D-SWE (CSPH)	21.7 kPa 26.3 kPa 35.6 kPa
Odriozola 2023 (1)	LSM by VCTE (CSPH)	10.6 kPa 27.6 kPa
Odriozola 2023 (2)	SSM by VCTE (CSPH)	37 kPa 49.9 kPa
Procopet 2015a	LSM by 2D-SWE (CSPH)	14.1 kPa 17 kPa 25.8 kPa
Procopet 2015b (training group)	LSM by VCTE (CSPH)	13.6 kPa 21.1 kPa
Procopet 2015b (validation group)	LSM by VCTE (CSPH)	13.6 kPa 21.1 kPa
Zykus 2015	LSM by VCTE (CSPH)	11.4 kPa 17.4 kPa 21.9 kPa
Zykus 2015	LSM by VCTE (SPH)	12.1 kPa 20.6 kPa 35 kPa

2D-SWE = 2D-shear wave elastography; CSPH = clinically significant portal hypertension; LSM = liver stiffness measurement; SPH = severe portal hypertension; SSM = spleen stiffness measurement; VCTE = vibration-controlled transient elastography